

SPIN AND ORBITAL ANGULAR MOMENTUM

The modern study of optical angular momentum can be said to have started with the paper of Allen *et al.* (2.1, **Paper 2.1**). This work showed that any beam with the amplitude distribution $u(r, \phi, z) = u_0(r, z) \exp i l \phi$, carried angular momentum about the beam axis. Moreover, this angular momentum could be separated into orbital and spin components. The orbital contribution is determined solely by the azimuthal phase dependence and is equivalent to $l\hbar$ per photon. The spin angular momentum is determined by the polarisation and has the value $\sigma_z \hbar$ per photon, where $-1 \leq \sigma_z \leq 1$, with the extremal values corresponding to pure circular polarisation. A physically realisable example of light with this phase distribution is a Laguerre-Gaussian beam, familiar from paraxial optics (2.2).

The identification of an orbital angular momentum is strongly suggested by a powerful analogy between paraxial optics and quantum mechanics (2.3). Here the Schrödinger wave equation is identical to the paraxial form of the wave equation with t replaced by z , while the operator corresponding to the z -component of orbital angular momentum can be represented in the form $L_z = -i\hbar \partial/\partial \phi$. The analogy allows much of paraxial optics, including orbital angular momentum, to be studied using the formalism of quantum mechanics. This is has been amply demonstrated by van Enk and Nienhuis (2.4, **Paper 2.2**) who introduce an eigenfunction description of laser beams and by Nienhuis and Allen (2.5, **Paper 2.3**) who compare paraxial modes and their properties to those of the quantum harmonic oscillator. The propagation of spin and orbital angular momenta through optical elements can usefully be described without reference to the spatial form of the light beam, by a generalisation of the Jones matrix description of optical polarisation (2.6, **Paper 2.4**). This abstraction is analogous to separating the angular and radial wavefunctions in quantum mechanics.

The separation of optical angular momentum into spin and orbital parts is not, however, as straightforward as the analogy with quantum mechanics might suggest. There are two principal difficulties. The first arises from problems in distinguishing between the concepts of spin and orbital angular momentum and of the idea of intrinsic and extrinsic angular momentum. The second concerns the difficulty in identifying meaningful spin and orbital angular momenta within full Maxwellian electromagnetism. A number of publications have dealt with each of these problems.

It is helpful, in approaching the meaning and interpretation of optical orbital and spin angular momenta, to keep in mind the corresponding problem in the mechanics of a set of material bodies. These might be a set of electrons and nuclei, a gas of molecules or even a

system of planets and a star. Each body may have its own spin and orbital angular momentum, the latter arising as a consequence of its linear momentum. For such a system of bodies, labelled by the subscript α , we may write the total angular momentum as

$$\mathbf{J} = \sum_{\alpha} [(\mathbf{r}_{\alpha} \times \mathbf{p}_{\alpha}) + \mathbf{s}_{\alpha}].$$

Here, $\mathbf{r}_{\alpha}$, $\mathbf{p}_{\alpha}$ and $\mathbf{s}_{\alpha}$ denote the position, linear momentum and spin of body α . We can identify the total spin and orbital angular momenta of the system as the sum of the individual quantities for each of the particles. Hence our total spin $\mathbf{S}$ and orbital angular momenta $\mathbf{L}$ are

$$\mathbf{S} = \sum_{\alpha} \mathbf{s}_{\alpha},$$

$$\mathbf{L} = \sum_{\alpha} \mathbf{r}_{\alpha} \times \mathbf{p}_{\alpha}.$$

Intrinsic and extrinsic angular momenta are often identified with the spin and orbital angular momenta. It is more meaningful, however, to define these terms by the manner in which they change if we displace the origin of the axes of rotation. Hence, the external angular momentum $\mathbf{J}_{\text{ext}}$ will be the cross product of the position of the centre of mass with the total linear momentum, $\mathbf{P}$

$$\mathbf{J}_{\text{ext}} = \mathbf{R} \times \mathbf{P}$$

where

$$\mathbf{R} = \sum_{\alpha} \frac{m_{\alpha}}{M} \mathbf{r}_{\alpha}; \quad M = \sum_{\alpha} m_{\alpha}; \quad \mathbf{P} = \sum_{\alpha} \mathbf{p}_{\alpha}.$$

The intrinsic angular momentum comprises the spin angular momentum together with the orbital angular momentum *relative to* the centre of mass. If we make simple use of the fact that momentum depends on velocity, it follows that

$$\mathbf{J}_{\text{int}} = \sum_{\alpha} m_{\alpha} (\mathbf{r}_{\alpha} - \mathbf{R}) \times (\dot{\mathbf{r}}_{\alpha} - \dot{\mathbf{R}}) + \mathbf{S}.$$

Clearly, displacement of the axes of rotation will change the extrinsic angular momentum but leave the intrinsic angular momentum unchanged. If there is only a single body then the orbital and spin angular momenta are the same as the extrinsic and intrinsic angular momenta respectively. In general, however, they are different quantities.

Many of the properties of the mechanical system can be usefully applied to optical angular momentum. It is clear from the definition of the external angular momentum that its component parallel to the total momentum must be zero. This means that the angular momentum carried by a Laguerre–Gaussian beam about its axis, or parallel to it, is entirely intrinsic (2.7, **Paper 2.5**) as the total linear momentum points along the beam axis. It is interesting to ask how the orbital and spin angular momenta are distributed within the beam (2.8, **Paper 2.6**). If only part of the beam is considered then the balance between the orbital contributions to the intrinsic and extrinsic angular momenta will change. In the mechanical system this arises because, if only a subset of the bodies is considered, the centre of mass will in general be different. If we consider only a part of a Laguerre–Gaussian beam then an extrinsic angular momentum (2.9, **Paper 2.7**) is found. The spin angular momentum, as it is wholly intrinsic, remains

unchanged. The spin of a macroscopic body can, of course, be re-interpreted in terms of the orbital angular momentum of its constituent particles. It is interesting to note that a similar procedure is possible, even for a single electron, if its spin is associated with an orbital angular momentum for its wave field (2.10). However, for light fields, the association of optical spin angular momentum with polarisation is the most useful approach.

Our previous discussion has focused on optical angular momentum within the paraxial approximation and by analogy with quantum theory. It is highly desirable, however, to be able to ground such a fundamental concept within electromagnetic theory. We can write the total angular momentum very simply in terms of the momentum density ($\epsilon_0 \mathbf{E} \times \mathbf{B}$) as:

$$\mathbf{J} = \int \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) d^3x.$$

Much has been written about suitable spin and orbital parts of this angular momentum. Yilmaz identifies a spin or intrinsic angular momentum, but notes that it is not in general gauge invariant (2.11). Gauge invariant quantities can be derived in terms of the gauge invariant transverse part of the vector potential $\mathbf{A}^\perp$ (2.12, 2.13). No distinction is made between spin and orbital, or extrinsic and intrinsic, angular momenta but the total angular momenta is written as a the sum of “spin” and “orbital” parts:

$$\begin{aligned} \mathbf{S} &= \int \epsilon_0 \mathbf{E}^\perp \times \mathbf{A}^\perp d^3x, \\ \mathbf{L} &= \sum_l \int \epsilon_0 E_l^\perp (\mathbf{r} \times \nabla) A_l^\perp d^3x. \end{aligned}$$

The spin part gives the difference between the numbers of right and left circularly polarised photons (2.13), but it has been suggested that neither quantity is itself physically observable (2.12). A more dramatic problem arises from the fact that neither of these quantities is, by itself, an angular momentum (2.14, 2.15, **Paper 2.8**). These difficulties question the validity of the simple picture obtained within the paraxial approximation. A study of exact, or non-paraxial, beams with $\exp(-il\phi)$ dependence revealed that the ratio of the angular momentum per unit length to the energy per unit length is only approximately $(l + \sigma_z)/\omega$ (2.16, **Paper 2.9**). It is more correct, however, to analyse the properties of optical beams in terms of fluxes. The recent introduction of the optical angular momentum flux has resolved some of the outstanding problems with the electromagnetic description of optical spin and orbital angular momentum (2.17, **Paper 2.10**). In particular, the angular momentum flux for a beam about its axis can be separated into well-behaved spin and orbital components and the ratio of these to the energy flux are precisely σ_z/ω and l/ω respectively.

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Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes

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Laser light with a Laguerre-Gaussian amplitude distribution is found to have a well-defined orbital angular momentum. An astigmatic optical system may be used to transform a high-order Laguerre-Gaussian mode into a high-order Hermite-Gaussian mode reversibly. An experiment is proposed to measure the mechanical torque induced by the transfer of orbital angular momentum associated with such a transformation.

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I. INTRODUCTION

It is well known from Maxwell's theory that electromagnetic radiation carries both energy and momentum. The momentum may have both linear and angular contributions; angular momentum has a spin part associated with polarization [1] and an orbital part associated with spatial distribution [2]. Any interaction between radiation and matter is inevitably accompanied by an exchange of momentum. This often has mechanical consequences some of which are related to radiation pressure. Although an experimental demonstration of the mechanical torque created by the transfer of angular momentum of a circularly polarized light beam was performed over 50 years ago [1], the work associated with the mechanical influence of light beams on atoms and matter has been almost exclusively concerned with linear momentum [3–5].

Beth [1] made the first observation of the angular momentum of light following Poynting [6], who inferred from a mechanical analogy that circularly polarized light should exert a torque on a birefringent plate and that the ratio of angular to linear momentum is equal to $\lambda/2\pi$. In his experiment a half-wave plate was suspended by a fine quartz fiber [see Fig. 1(a)]. A beam of light, circularly po-

larized by a fixed quarter-wave plate, passed through the plate which transformed right-handed circularly polarized light into left-handed circularly polarized light and transferred $2\hbar$ of spin angular momentum for each photon to the birefringent plate. It was found that the measured torque agreed in sign and magnitude with that predicted by both wave and quantum theories of light. The ratio of the angular momentum of N photons in the beam, $J = \pm N\hbar$, to their energy, $W = N\hbar\omega$, is $\pm 1/\omega$ and Beth's measurement is sometimes referred to as the measurement of the spin angular momentum of the photon.

The purpose of this paper is to investigate whether a Gaussian mode may be said to possess orbital angular momentum and to propose a study of the mechanical consequences which might then arise. The amplitude of a Laguerre-Gaussian mode has an azimuthal angular dependence of $\exp(-il\phi)$, where l is the azimuthal mode index. Analogy between quantum mechanics and paraxial optics [7] suggests that such modes are the eigenmodes of the angular momentum operator L_z and carry an orbital angular momentum of $l\hbar$ per photon. The transverse amplitude distribution of laser light is usually described in terms of a product of Hermite polynomials $H_n(x)H_m(y)$ and associated with TEM_{nmq} modes. Laguerre polynomial distributions of amplitude, TEM_{plq} modes, are also possible but occur less often in actual lasers. This allows the analogy with quantum mechanics to be stressed even more strongly. It is well known [8] that the one-dimensional quantum-mechanical harmonic oscillator has a solution in the form of a Hermite polynomial and that in two dimensions the solutions may be written as Laguerre polynomials with energy $(n+m+1)\hbar\omega$, while the eigenvalue of the two-dimensional angular momentum operator is $(n-m)\hbar$. It seems likely, therefore, that TEM_{plq} modes possess well-defined orbital angular momenta.

II. ORBITAL ANGULAR MOMENTUM OF A LAGUERRE-GAUSSIAN MODE

The angular momentum density associated with the transverse electromagnetic field may be shown [2] to be

$$\mathbf{M} = \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) \quad (1)$$

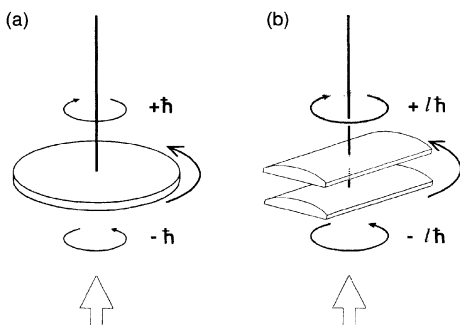


FIG. 1. (a) A suspended $\lambda/2$ birefringent plate undergoes torque in transforming right-handed into left-handed circularly polarized light. (b) Suspended cylindrical lenses undergo torque in transforming a Laguerre-Gaussian mode of orbital angular momentum $-l\hbar$ per photon, into one with $+l\hbar$ per photon.

while the total angular momentum of the field is

$$\mathbf{J} = \epsilon_0 \int \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) d\mathbf{r}. \quad (2)$$

In atomic physics it is normal to expect that

$$\mathbf{J} = \mathbf{L} + \mathbf{S}, \quad (3)$$

where the first term is identified with the orbital angular momentum $\mathbf{L}$, and the second with the spin $\mathbf{S}$. But there is doubt as to whether $\mathbf{L}$ and $\mathbf{S}$ are, in general, separately physically observable [9] for vector fields.

Clearly the linear momentum of a transverse plane wave, $\mathbf{E} \times \mathbf{B}$, is in the direction of propagation, z , and there cannot be a component of angular momentum $\mathbf{r} \times (\mathbf{E} \times \mathbf{B})$ in the same direction [10]. However, the fields of the laser modes TEM_{nmq} or TEM_{plq} , unlike those in a coaxial metal waveguide of infinite length, are not strictly transverse [11]. They have small components in the direction of propagation, z . A convenient representation of a linearly polarized laser mode is achieved [12] in the Lorentz gauge using the vector potential

$$\mathbf{A} = \mathbf{x} u(x, y, z) e^{-ikz} \quad (4)$$

where $\mathbf{x}$ is the unit vector in the x direction. The expression $u(x, y, z)$, or $u(r, \phi, z)$, is the complex scalar function describing the distribution of the field amplitude which satisfies the wave equation in the paraxial approximation. In this approximation the second derivatives of E and B fields, and the products of first derivatives, are ignored and $\partial u / \partial z$ taken to be small compared with ku . The cylindrically symmetric solutions $u_{pl}(r, \phi, z)$ which describe Laguerre-Gaussian beams are given by

$$\begin{aligned} u_{pl}(r, \phi, z) = & \frac{C}{(1 + z^2/z_R^2)^{1/2}} \left[\frac{r\sqrt{2}}{w(z)} \right]^l \left[L_p^l \left(\frac{2r^2}{w^2(z)} \right) \right] \\ & \times \exp \left[\frac{-r^2}{w^2(z)} \right] \exp \left[\frac{-ikr^2 z}{2(z^2 + z_R^2)} \right] \exp(-il\phi) \\ & \times \exp \left[i(2p + l + 1) \tan^{-1} \frac{z}{z_R} \right], \end{aligned} \quad (5)$$

where z_R is the Rayleigh range, $w(z)$ is the radius of the beam, L_p^l is the associated Laguerre polynomial, C is a constant, and the beam waist is at $z=0$. The Lorentz gauge has the advantage of being readily amenable in all coordinate systems and leads in this case to considerable symmetry in the x and y directions although the results are best expressed in cylindrical coordinates.

Within this description we have shown that the time average of the real part of $\epsilon_0 \mathbf{E} \times \mathbf{B}$, which is the linear momentum density, is given by

$$\begin{aligned} \frac{\epsilon_0}{2} (\mathbf{E}^* \times \mathbf{B} + \mathbf{E} \times \mathbf{B}^*) = & i\omega \frac{\epsilon_0}{2} (u^* \nabla u - u \nabla u^*) \\ & + \omega k \epsilon_0 |u|^2 \mathbf{z}, \end{aligned} \quad (6)$$

for a beam of unit amplitude, where $\mathbf{z}$ is the unit vector in the z direction. We may recognize that $u^* \nabla u$ closely echoes the quantum-mechanical expression for the expectation value of linear momentum of a wave function. To

achieve this appealing form we have retained the term $\partial u / \partial z$ in the expression for the magnetic field $\mathbf{B}$, which could have been ignored.

When applied to the Laguerre-Gaussian distribution given by Eq. (5) for linearly polarized light, the momentum density per unit power is found to be

$$\mathcal{P} = \frac{1}{c} \left[\frac{rz}{(z^2 + z_R^2)} |u|^2 \mathbf{r} + \frac{l}{kr} |u|^2 \boldsymbol{\phi} + |u|^2 \mathbf{z} \right], \quad (7)$$

where $\mathbf{r}$ and $\boldsymbol{\phi}$ are unit vectors and $|u|^2 \equiv |u(r, \phi, z)|^2$. Here the $\partial u / \partial z$ term has now been neglected. It may be seen that the Poynting vector, given by $c^2 \mathcal{P}$, spirals along the direction of propagation; see Fig. 2. The $\mathbf{r}$ component relates to the spread of the beam; the $\boldsymbol{\phi}$ component gives rise to orbital angular momentum in the z direction and the $\mathbf{z}$ component relates to the linear momentum P in the direction of propagation.

Calculation of the time averaged angular momentum density, $\epsilon_0 \mathbf{r} \times \langle \mathbf{E} \times \mathbf{B} \rangle$, per unit power yields

$$\mathbf{M} = -\frac{l}{\omega} \frac{z}{r} |u|^2 \mathbf{r} + \frac{r}{c} \left[\frac{z^2}{(z^2 + z_R^2)} - 1 \right] |u|^2 \boldsymbol{\phi} + \frac{l}{\omega} |u|^2 \mathbf{z}. \quad (8)$$

The radial and azimuthal components are symmetric about the axis, so that integration over the beam profile leaves only the z component. The ratio of the flux of angular momentum to that of energy is $L/cP = l/\omega$, while the ratio of angular momentum to linear momentum is now $L/P = l(\lambda/2\pi)$. Our conviction that the Laguerre-Gaussian mode possesses a well-defined orbital angular momentum has thus been justified.

At position (r, ϕ, z) the magnitude of angular momentum density per unit power is given by $M = l/\omega(1 + z^2/r^2)^{1/2} |u|^2$, oriented at an angle $\theta = \tan^{-1} z/r$ to the z axis. Locally we have $M_z/P_z = l(\lambda/2\pi)$ where M_z is the z component of angular momentum density and P_z that of the linear momentum density. There is, however, also a local radial component.

We have so far considered linearly polarized light; when the vector potential is generalized to arbitrary polarization we find

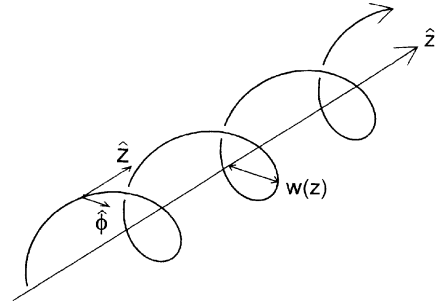


FIG. 2. The spiraling curve represents the Poynting vector of a linearly polarized Laguerre-Gaussian mode of radius $w(z)$.

$$\frac{\epsilon_0}{2}(\mathbf{E} \times \mathbf{B} + \mathbf{E} \times \mathbf{B}^*) = i\omega \frac{\epsilon_0}{2}(u^* \nabla u - u \nabla u^*) + \omega k \epsilon_0 |u|^2 \mathbf{z} \\ + \omega \sigma_z \frac{\epsilon_0}{2} \frac{\partial |u|^2}{\partial r} \Phi, \quad (9)$$

where the first two terms are polarization independent and relate to orbital angular momentum and the final term is a polarization, or spin, part. These lead to a $\mathbf{z}$ component of total angular momentum density per unit power

$$M_z = \frac{l}{\omega} |u|^2 + \frac{\sigma_z r}{2\omega} \frac{\partial |u|^2}{\partial r}. \quad (10)$$

The ratio of the angular momentum flux to energy flux now becomes $J/cP = (l + \sigma_z)/\omega$ where $\sigma_z = \mp 1$ for right-handed or left-handed circularly polarized light and $\sigma_z = 0$ for linearly polarized light.

We may note that in the paraxial approximation the spin-dependent part of angular momentum density depends upon the gradient of the intensity. Thus at a particular local point the $\mathbf{z}$ component of angular momentum flux divided by energy flux does not yield a simple value. However, when the total angular momentum flux is calculated, the integration across the beam profile leads to the simple result of the preceding paragraph.

III. THE DETECTION OF ORBITAL ANGULAR MOMENTUM

In the Beth experiment right-handed circularly polarized light, $-\hbar$, was converted to left-handed circularly polarized light, $+\hbar$, so that $2\hbar$ of spin angular momentum per photon was imparted to the birefringent plate.

In the same way the maximum transfer of orbital angular momentum would take place if a Laguerre-Gaussian beam possessing $l\hbar$ angular momentum per photon were converted into one with $-l\hbar$ per photon. The torque arising from the transfer of momentum must then be measured by an appropriate equivalent of the birefringent plate. It is, however, more convenient to discuss the conversion to a beam with zero orbital angular momentum. This is readily possible by transforming a Laguerre-Gaussian to a Hermite-Gaussian distribution, which can be done using a mode convertor with astigmatic optical components.

Abramochkin and Volostnikov [13] have considered the mathematical transformation of laser beams undergoing astigmatism and found an integral transformation of Hermite-Gaussian into Laguerre-Gaussian beams. They recognized that passing the beam through a cylindrical lens can perform the desired conversion. Tamm and Weiss [14] have similarly employed a "mode convertor" involving cylindrical lenses.

Some insight into the mechanism for transformation of modes of arbitrarily high order is readily possible. A Hermite-Gaussian laser beam with its axes along the axes of a cylindrical lens is not changed, apart from a difference in size along the two axes. Therefore the operation of a cylindrical lens on an arbitrary beam pattern is most easily discussed in terms of its Hermite-Gaussian components along the axes of the lens.

The decomposition of a Laguerre-Gaussian mode in terms of Hermite-Gaussian modes can be seen in Eq. (A3) in the Appendix of Abramochkin and Volostnikov's paper [13]. It connects a combination of products of Hermite polynomials to a Laguerre polynomial,

$$\sum_{k=0}^{n+m} (2i)^k P_k^{(n-k, m-k)}(0) H_{n+m-k}(x) H_k(y) = 2^{n+m} \times \begin{cases} (-1)^m m! (x+iy)^{n-m} L_m^{n-m}(x^2+y^2) & \text{for } n \geq m \\ (-1)^n n! (x-iy)^{m-n} L_n^{m-n}(x^2+y^2) & \text{for } m > n \end{cases}, \quad (11)$$

where

$$P_k^{(n-k, m-k)}(0) = \frac{(-1)^k}{2^k k!} \frac{d^k}{dt^k} [(1-t)^n (1+t)^m] \Big|_{t=0}.$$

Also in the appendix of Ref. [13] there is an unnumbered equation which connects the same Hermite polynomials to their 45° transformed value,

$$\sum_{k=0}^{n+m} (-2)^k P_k^{(n-k, m-k)}(0) H_{n+m-k}(x) H_k(y) = (\sqrt{2})^{n+m} H_n \left[\frac{x-y}{\sqrt{2}} \right] H_m \left[\frac{x+y}{\sqrt{2}} \right]. \quad (12)$$

It may be seen that the summation in Eq. (11) is the same as that in Eq. (12), save for a factor of $(-i)^k$ where k is the integer associated with the Hermite polynomial in the y direction, $H_k(y)$. But $(-i)^k$, of course, corresponds to an additional change of phase $\pi/2$ for each integer increase in the value of k .

A simple example perhaps makes the result of this clear. From Eq. (12) we may show that

$$(\sqrt{2})^5 H_3 \left[\frac{x-y}{\sqrt{2}} \right] H_2 \left[\frac{x+y}{\sqrt{2}} \right] = H_5(x) H_0(y) - H_4(x) H_1(y) - 2H_3(x) H_2(y) \\ + 2H_2(x) H_3(y) + H_1(x) H_4(y) - H_0(x) H_5(y) \quad (13)$$

while from Eq. (11)

$$\begin{aligned} 32re^{i\phi}L_2^1(r^2) &= H_5(x)H_0(y) + iH_4(x)H_1(y) \\ &\quad + 2H_3(x)H_2(y) + 2iH_2(x)H_3(y) \\ &\quad + H_1(x)H_4(y) + iH_0(x)H_5(y). \end{aligned} \quad (14)$$

We see that a Hermite-Gaussian mode with spatial dependence $H_n(x)H_m(y)$, may, provided the appropriate $\pi/2$ change of phase is achieved, become a single Laguerre-Gaussian mode well defined by L_p^l , or specifically $L_{m-n}^{n-m}(r^2)$ for $n \geq m$. We may identify $(n-m)$ as the orbital angular momentum of the photon in units of $\hbar$. Therefore we have converted a Hermite-Gaussian mode with zero orbital angular momentum into a Laguerre-Gaussian mode with $L=l\hbar$. If a change of phase other than $\pi/2$ is introduced, distributions between a Laguerre-Gaussian and a Hermite-Gaussian mode will result with ill-defined orbital angular momentum because such a mode is not an eigenstate of L_z .

As $H_k(y) = (-1)^k H_k(-y)$ we should note that if the left-hand side of Eq. (11) is multiplied inside the summation by $(-1)^k$, the right-hand side then relates to $(x-iy)$ for $n \geq m$ and $(x+iy)$ for $m \geq n$. Therefore the sign of the ϕ dependence of the Laguerre mode becomes changed. In other words, an additional π change of phase for all terms will transform a Laguerre-Gaussian mode with an angular momentum of $l\hbar$ to one with $-l\hbar$.

The equations allow us to recognize the nature of the decomposition of mode patterns into modes along the axes of a cylindrical lens. The presence of the focusing along only one of the axes and not the other, allows the components to propagate with different Gouy phase shifts [15]. In this way the necessary $\pi/2$ phase change to convert a Laguerre-Gaussian mode to a Hermite-Gaussian mode, or vice versa, may be achieved as Tamm and Weiss [14] realized for first-order modes.

Only for the TEM₁₀ mode is the intuition that it must be combined with an out of phase version of itself to produce a radially continuous field or intensity distribution, actually true. Using Eqs. (11) and (12) it is easy to show

$$H_1\left[\frac{x-y}{\sqrt{2}}\right]H_0\left[\frac{x+y}{\sqrt{2}}\right]e^{-r^2/w^2} \rightarrow rL_0^1(r^2)e^{-r^2/w^2}e^{i\phi}. \quad (15)$$

It follows from Eq. (14) that to obtain higher-order Laguerre modes it is necessary to combine a number of Hermite-Gaussian distributions: an example is shown in Fig. 3. We have thus established a general recipe for the transformation of Laguerre-Gaussian modes of well-defined angular orbital momentum into Hermite-Gaussian modes, or into a Laguerre-Gaussian mode of opposite angular symmetry. In the experiments currently in progress in our laboratory Hermite-Gaussian modes have been converted into pure, nondegenerate, Laguerre-Gaussian modes by a simple astigmatic optical arrangement which confirms the analysis outlined in the preceding section. The full analysis of the effect of astigmatism on the Gouy phase of laser modes and the detailed design

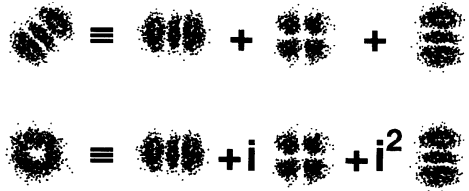


FIG. 3. The decomposition of a TEM₀₂ Hermite-Gaussian mode at 45° into a set of Hermite-Gaussian modes and the decomposition of a Laguerre-Gaussian mode into the same, re-phased, set.

of appropriate mode converters are planned to be published elsewhere [16].

IV. THE MEASUREMENT OF MECHANICAL TORQUE

Spin angular momentum in the form of circularly polarized light is both produced and detected by birefringence. In the Beth experiment [1] the torque per unit volume is given by $\mathbf{P} \times \mathbf{E}$. As $\mathbf{D} = \kappa \mathbf{E} = \epsilon_0 \mathbf{E} + \mathbf{P}$ this may be written as

$$\frac{d\mathbf{J}}{dt} = (\mathbf{D} \times \mathbf{E}). \quad (16)$$

The torque on the birefringent material arises because κ is a tensor; that is, $\mathbf{E}$ is not parallel to $\mathbf{P}$; in fact $\kappa_{xx} = n_x^2$ and $\kappa_{yy} = n_y^2$. The same “tensorial” attributes are required in the measurement of orbital angular momentum.

We have already indicated that a cylindrical lens can introduce phase differences between the Hermite-Gaussian components. Various combinations of cylindrical, or tilted spherical lenses, may be used to yield “retardation plates” of arbitrary thickness for orbital angular momentum. Just as appropriate phasing of the x and y components of the electric field in any light beam is capable of transforming linearly polarized light into circularly, or elliptically, polarized light so the rearrangement of phase introduced by an astigmatic optical element can destroy or create orbital angular momentum. Any measurement of mechanical torque must rely on such a change.

The measurement of a mechanical torque arising from orbital angular momentum should, thus, closely resemble that of spin. A suspended combination of two cylindrical lenses may be made to transform a Laguerre-Gaussian mode to the one with opposite orbital angular momentum, employing the anisotropy of the Gouy phase in the space in between the two lenses [see Fig. 1(b)]. The resultant torque which is predictable in terms of intensity and the value of l will then be compared with that measured from the rotation of the fiber suspension. An appropriate astigmatic birefringent element, or a combination of astigmatic and birefringent parts, would be capable of responding to both orbital and spin contributions simultaneously. This would allow the observation of total angular momentum J . However, a purely astigmatic or

birefringent detector will respond only to the uncoupled values of L and S , respectively. It appears that there is no sense in which the spin and orbital contributions are intrinsically coupled except through the mechanical torque applied to such as astigmatic and birefringent device. It may be that perfect decoupling of L and S is limited to those cases where the paraxial approximation is valid; the decoupling of polarization state and scalar field amplitude is a feature of this approximation.

V. CONCLUSIONS

We have demonstrated that a Laguerre-Gaussian laser mode has a well-defined orbital angular momentum equal to $l\hbar$ per photon, with l the azimuthal mode index. We have outlined how such orbital angular momentum may be removed from the mode and converted into a mechanical torque. We have shown that such a transformation may be achieved by the use of astigmatic optical elements, which may also be used to produce Laguerre-Gaussian modes from the more commonly occurring Hermite-Gaussian modes.

It would appear that all light beams which possess field gradients, and which are not therefore plane waves, will possess a measure of orbital angular momentum. Indeed a badly phased transformation between transverse laser amplitude distributions will in general lead to ill-defined orbital angular momentum. For this reason it is important that stable, nondegenerate, propagating Laguerre-Gaussian polynomial modes are created and entirely transformed. A meaningful measurement of orbital angular momentum will not otherwise result. Although Laguerre-Gaussian modes may be created within the laser [14,19] they are usually degenerate, either simultaneously possessing $\exp(\pm il\phi)$ components or randomly

fluctuating between them in time, and so have zero average orbital angular momentum. The reason why Hermite-Gaussian modes are the dominant ones in lasers is because of the presence of intracavity astigmatism [17].

We have been concerned in this paper with the orbital, or the total, angular momentum of the whole light beam. The ratio of the flux of total angular momentum to that of energy only gives $(l + \sigma_z)/\omega$ when we integrate over the whole beam profile. We have not investigated the consequences of the local ratio l/ω inside a linearly polarized Laguerre-Gaussian beam. It is in this regime that interactions with either atoms or small particles will take place; what the effect of orbital momentum of the light will be in this case remains to be investigated.

Other areas of study currently exist which invoke inhomogeneous, transverse, laser fields with field gradients. These include laser cooling, the manipulation of atoms and, expressly, mechanical effects in the interaction of neutral atoms and cylindrically symmetric modes: see [4,18,19]. It is interesting to speculate on whether such investigations might be changed by the existence of orbital angular momentum. It would also be interesting to know the extent to which the language of angular momentum might be a meaningful alternative to that already used in these areas.

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Eigenfunction description of laser beams and orbital angular momentum of light

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The propagation of light beams through astigmatic lens systems is accompanied by a transfer of orbital angular momentum. We develop a method to describe propagating light beams by operators of which the field is an eigenfunction. This method is applied to determine when an astigmatic lens system transforms gaussian beams into other gaussian beams and where in the system angular momentum is transferred. We show that the Gouy phase is equal to the dynamical phase of a quantummechanical harmonic oscillator with time-dependent energy.

1. Introduction

With the transverse intensity gradient of a beam of light corresponds a density of transverse momentum. This in turn may lead to a nonvanishing orbital angular momentum along the beam axis. Recently, it was demonstrated that astigmatic lens systems may change this angular momentum [1]. Since lenses normally do not change the polarization of the light, the spin is conserved. In an experiment [2], Hermite-gaussian beams which possess no orbital angular momentum were transformed into Laguerre-gaussian beams, which have a well-defined orbital angular momentum per photon. Since the total angular momentum of matter and light is conserved, this transformation is accompanied by a small torque on the lens system. The transformation of laser beams using astigmatic lenses has recently been investigated experimentally by Tamm and Weiss [3].

In this paper we present a novel theoretical description of light beams, and we apply it to this type of experiments. The method exploits some well-known similarities between quantum mechanics and paraxial wave optics, and makes use of a state-vector representation developed by Stoler [4]. Instead of giving analytical expressions for the electromagnetic field, we specify the field by (hermitian) operators of which the field is an eigenfunction, and by the cor-

responding eigenvalues. Two operators suffice to determine the spatial dependence of a monochromatic beam and one operator determines its polarization. The propagation of laser light through optical systems can then be described by the evolution of the 'eigenoperators' of the light field. In this way, we derive conditions under which gaussian beams are transformed into other gaussian beams. A case of special interest occurs when Hermite-gaussians are converted into Laguerre-gaussian modes. Analogously, the change of angular momentum of a field by lenses and lens systems is described by the evolution of the corresponding operator $\hat{L}_z$, in the same manner as in the Heisenberg picture in quantum mechanics. We also demonstrate that the Gouy phase is related to the dynamical phase of a time-dependent harmonic oscillator.

2. Linear and angular momentum of classical electromagnetic fields

The local densities of linear momentum and angular momentum of the source-free classical electromagnetic (em) field in vacuum are given by [5,6]

$$\begin{aligned} \mathbf{p} &= \epsilon_0 \mathbf{E} \times \mathbf{B}, \\ \mathbf{j} &= \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}), \end{aligned} \quad (1)$$

in terms of the electric field $\mathbf{E}$ and the magnetic field $\mathbf{B}$. These densities correspond to the Noether currents associated with the invariance of the free Maxwell equations under spatial translations and rotations, respectively [6,7]. The total linear and angular momentum of the em field, defined by

$$\mathbf{P} = \int d\mathbf{r} \mathbf{p}, \quad \mathbf{J} = \int d\mathbf{r} \mathbf{j}, \quad (2)$$

are therefore conserved quantities in vacuum.

We now consider monochromatic fields with frequency ω , and we use the complex notation

$$\begin{aligned} \mathbf{E} &= (\mathbf{E} e^{-i\omega t} + \mathbf{E}^* e^{i\omega t})/2, \\ \mathbf{B} &= (\mathbf{B} e^{-i\omega t} + \mathbf{B}^* e^{i\omega t})/2, \end{aligned} \quad (3)$$

where the asterisk denotes complex conjugation. Then we can eliminate the magnetic field from eqs. (2) by using the Maxwell equation

$$i\omega \mathbf{B} = \nabla \times \mathbf{E}. \quad (4)$$

For fields that vanish sufficiently fast for $|\mathbf{r}| \rightarrow \infty$, partial integration in eqs. (2) leads to the following expression for the total linear momentum

$$\mathbf{P} = \frac{\epsilon_0}{2i\omega} \int d\mathbf{r} \sum_{j=x,y,z} \mathbf{E}_j^* \nabla \mathbf{E}_j, \quad (5)$$

where we used that $\nabla \cdot \mathbf{E} = 0$. The total momentum is manifestly independent of time. The expression (5) takes a form similar to the quantum-mechanical expression for the expectation value of the linear momentum of a particle. This has been noted before in a reciprocal (Fourier) representation [6–8]. In a similar way the total angular momentum can be written as

$$\begin{aligned} \mathbf{J} &= \frac{\epsilon_0}{2i\omega} \int d\mathbf{r} \sum_{j=x,y,z} \mathbf{E}_j^* (\mathbf{r} \times \nabla) \mathbf{E}_j + \frac{\epsilon_0}{2i\omega} \int d\mathbf{r} \mathbf{E}^* \times \mathbf{E} \\ &\equiv \mathbf{L} + \mathbf{S}. \end{aligned} \quad (6)$$

The interpretation of the first and second term on the right hand side of eq. (6) as orbital angular momentum and spin is, although seemingly obvious, not without fundamental difficulties [6–8].

3. The paraxial approximation

Paraxial wave optics [9–11] describes the propagation of light beams whose transverse dimensions are much smaller than the typical longitudinal distance over which the field changes in magnitude. We start by recalling some results from ref. [9]. The transverse dimensions of the beam have the order of magnitude of the beam waist w_0 , which is assumed to be much smaller than the diffraction length $l = kw_0^2$, with $k = \omega/c$ the wave number. One writes for the electric-field component of a beam propagating in the z direction

$$\mathbf{E} = e^{ikz} \mathbf{F}. \quad (7)$$

The small parameter w_0/l is used as an expansion parameter, which means that derivatives of $\mathbf{F}$ with respect to z can be neglected compared to the transverse derivatives. The field $\mathbf{F}$ satisfies to lowest order in w_0/l the paraxial wave equation

$$2ik \frac{\partial}{\partial z} \mathbf{F} = - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \mathbf{F}, \quad (8)$$

which determines the propagation of the field in the z direction for a given field distribution in a plane $z = z_0$. Furthermore, the z component of $\mathbf{F}$ is smaller than the transverse components by a factor w_0/l .

Within the paraxial approximation we find it convenient to consider quantities of the em field that are defined as averages over planes with $z = \text{constant}$. Thus, the linear and angular momentum per unit of length in a plane $z = z_0$ are given by

$$\begin{aligned} \mathcal{P}(z_0) &= \iint dx dy \mathbf{p}(x, y, z_0), \\ \mathcal{J}(z_0) &= \iint dx dy \mathbf{j}(x, y, z_0). \end{aligned} \quad (9)$$

Since the integrations in these definitions do not extend over all space, the steps leading to the appealing forms (5) and (6) are not necessarily valid here. For instance, the quantities (9) are not independent of time. Therefore, we will always average such quantities over a period $2\pi/\omega$. Then, in the paraxial approximation the (time-averaged) transverse components of $\mathcal{P}$, and the z component of $\mathcal{J}$ can be rewritten to lowest order in w_0/l as

$$\begin{aligned}
\mathcal{P}_x &= \sum_{j=x,y} \frac{\epsilon_0}{2i\omega} \iint dx dy F_j^* \frac{\partial}{\partial x} F_j, \\
\mathcal{P}_y &= \sum_{j=x,y} \frac{\epsilon_0}{2i\omega} \iint dx dy F_j^* \frac{\partial}{\partial y} F_j, \\
\mathcal{J}_z &= \sum_{j=x,y} \frac{\epsilon_0}{2i\omega} \iint dx dy F_j^* \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) F_j \\
&\quad + \frac{\epsilon_0}{2i\omega} \iint dx dy (F_x^* F_y - F_y^* F_x) \\
&\equiv \mathcal{L}_z + \mathcal{S}_z.
\end{aligned} \tag{10}$$

Here we explicitly neglected derivatives of $\mathbf{F}$ with respect to z compared with transverse derivatives, and used that the z component of $\mathbf{F}$ is small [9]. The z component of the linear momentum per unit length is to lowest order in w_0/l given by

$$\mathcal{P}_z = \sum_{j=x,y} \frac{\epsilon_0}{2\omega} \iint dx dy F_j^* k F_j. \tag{11}$$

From the well-known expression for the local energy density of the em field [5,6],

$$u = \frac{\epsilon_0}{2} (\mathbf{E}^2 + c^2 \mathbf{B}^2), \tag{12}$$

we obtain the time-averaged field energy per unit of length $\mathcal{E}$,

$$\mathcal{E} = \mathcal{N} \hbar \omega = \sum_{j=x,y} \frac{\epsilon_0}{2} \iint dx dy F_j^* F_j, \tag{13}$$

where $\mathcal{N}$ is the number of photons per unit of length. Division of quantities like eqs. (10) and (11) by $\mathcal{N}$ yields the value of that quantity per photon. For instance, the momentum in the z direction is, obviously, equal to $\hbar k$ per photon.

4. Operator formalism and notation

4.1. Schrödinger picture

The propagation of light beams can also be described in an operator formalism which is very similar to the operator description of the hamiltonian evolution of quantum-mechanical states. We use here the formalism developed by Stoler [4]. Moreover, the analogy between the expressions (10)–(13) for

time-averaged quantities of the classical em field, and the quantum-mechanical expectation values of the same quantities for particles, allows us to extend that formalism (see also ref. [11]). We use the following notations and conventions:

- the field $\mathbf{F}(x, y, z)$ is represented by the ket vector $|\mathbf{F}(z)\rangle$;
- operators acting upon the field $\mathbf{F}$ are represented by corresponding operators, denoted by a caret, acting upon the ket $|\mathbf{F}\rangle$;
- the evolution of a field propagating from $z = z_0$ to $z = z_1$ through a given lossless optical system is determined by a unitary operator $\hat{h}$, according to

$$|\mathbf{F}(z_1)\rangle = \hat{h} |\mathbf{F}(z_0)\rangle; \tag{14}$$

the coordinate z plays the same role as time in ordinary quantum mechanics;

- the scalar product $\langle \mathbf{F}(z) | \mathbf{G}(z) \rangle$ is defined as

$$\begin{aligned}
\langle \mathbf{F}(z) | \mathbf{G}(z) \rangle &= \frac{\epsilon_0}{2\omega} \iint dx dy \mathbf{F}^*(x, y, z) \cdot \mathbf{G}(x, y, z) \\
&= \sum_{j=x,y} \frac{\epsilon_0}{2\omega} \iint dx dy F_j^*(x, y, z) G_j(x, y, z),
\end{aligned} \tag{15}$$

to lowest order in w_0/l .

When $\hat{Q}$ is an hermitian operator, the quantity $\mathcal{Q}(z)$ defined by

$$\mathcal{Q}(z) = \langle \mathbf{F}(z) | \hat{Q} | \mathbf{F}(z) \rangle \tag{16}$$

is real. For example, when we introduce the operators for transverse momentum

$$\hat{p}_x = \frac{1}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{1}{i} \frac{\partial}{\partial y}, \tag{17}$$

then it follows from eqs. (10) that

$$\mathcal{P}_x = \langle \mathbf{F} | \hat{p}_x | \mathbf{F} \rangle, \quad \mathcal{P}_y = \langle \mathbf{F} | \hat{p}_y | \mathbf{F} \rangle. \tag{18}$$

The momentum operators obey the usual commutation relations with the position operators $\hat{x}$ and $\hat{y}$,

$$[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = i. \tag{19}$$

As another example, we introduce the operator $\hat{\mathcal{L}}_z$ for the z component of the orbital angular momentum acting upon the external degrees of freedom of the field, by

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x. \quad (20)$$

The spin operator $\hat{S}_z$ on the other hand acts upon the vectorial indices of the field,

$$\hat{S}_z |F_j\rangle = \sum_{k=x,y} i\epsilon_{jkz} |F_k\rangle, \quad (21)$$

with ϵ_{jkl} the completely anti-symmetrical Levi-Civita tensor. A matrix representation of the spin operator is therefore

$$\hat{S}_z = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}. \quad (22)$$

The eigenvalues of $\hat{S}_z$ are, of course, equal to ± 1 . With these operators substituted for $\hat{Q}$ we can write eqs. (10) for $\mathcal{L}_z$ and $\mathcal{S}_z$ in the form (16).

The evolution of a light beam propagating in vacuum is found from the formal solution of eq. (8) [4]. It can be written in the form (14), with $\hat{h}$ the free propagation operator describing evolution,

$$\hat{h} = \exp\left(-\frac{i(z_1 - z_0)}{2k} \hat{p}^2\right), \quad (23)$$

where we defined $\hat{p}^2 = \hat{p}_x^2 + \hat{p}_y^2$.

4.2. Heisenberg picture

When a field evolves according to eq. (14), the evolution of a quantity $\mathcal{Q}(z)$, defined in eq. (16), is given by

$$\mathcal{Q}(z_1) = \langle \mathbf{F}(z_0) | \hat{h}^\dagger \hat{Q} \hat{h} | \mathbf{F}(z_0) \rangle, \quad (24)$$

with $\hat{h}^\dagger$ the Hermitian conjugate of $\hat{h}$. As in the Heisenberg picture in quantum mechanics we can interpret the operator $\hat{h}^\dagger \hat{Q} \hat{h}$ as giving the evolution of the quantity corresponding to the operator $\hat{Q}$, where now the field is described by the state vector $|\mathbf{F}(z_0)\rangle$ at a single value z_0 of z . This picture is very useful when examining certain properties of physical quantities of the form (16) that are independent of the fields. For instance, if the operator $\hat{Q}$ commutes with the operator $\hat{p}^2$, it also commutes with the free propagation operator (23). This implies that

$$\hat{h}^\dagger \hat{Q} \hat{h} = \hat{Q}, \quad (25)$$

which in turn implies that the quantity $\mathcal{Q}(z)$ is conserved under free propagation, i.e.

$$\frac{d\mathcal{Q}}{dz} = 0. \quad (26)$$

Thus the z components of both orbital angular momentum and spin are conserved during free propagation, since the operators $\hat{L}_z$ and $\hat{S}_z$ both commute with $\hat{p}^2$. This means that their value per unit length is uniform along z . Also the energy per unit length (13) is a conserved quantity, as it is represented by the identity operator. Note that ‘conserved’ here does not refer to conservation in time, but rather to conservation with the position z .

5. Ideal lenses

A lens is called ideal when it is sufficiently thin, so that propagation within the lens is negligible. An ideal lens modifies an incoming field distribution $\mathbf{F}(x, y, z)$ by adding a local phase $\psi(x, y)$ to the field, where ψ is a real function of (x, y) [12]. If the lens has a constant refractive index n , the phase change ψ is due to the local change of the optical path, which is proportional to the local thickness $b(x, y)$ of the lens,

$$\psi(x, y) = kb(x, y)(n - 1). \quad (27)$$

In the operator formalism of the preceding section, the ideal lens is represented by the operator

$$\hat{T} = \exp[i\psi(\hat{x}, \hat{y})]. \quad (28)$$

In particular, a cylindrical lens with its axis oriented at an angle γ with the positive x axis is described by the phase

$$\psi(x, y) = -\frac{k}{2f}(x \cos \gamma + y \sin \gamma)^2, \quad (29)$$

with f the focal length of the lens along its axis. The focal length in the orthogonal direction is infinite. Within the Heisenberg picture it is easy to prove that the z component of the spin, $\mathcal{S}_z$, cannot be changed by a lens. Since the operator $\hat{S}_z$ does not act upon the coordinates of the field,

$$\hat{T}^\dagger \hat{S}_z \hat{T} = \hat{S}_z, \quad (30)$$

for arbitrary $\hat{T}$ defined in eq. (28). The orbital part of the angular momentum does change in general, since

$$\begin{aligned}\hat{T}^\dagger \hat{L}_z \hat{T} &= \hat{L}_z + \hat{x} \frac{\partial \psi}{\partial \hat{y}} - \hat{y} \frac{\partial \psi}{\partial \hat{x}} \\ &\equiv \hat{L}_z + \delta \hat{L}_z.\end{aligned}\quad (31)$$

For a general field $|\mathbf{F}_{\text{in}}\rangle$ incident on the lens, the difference in the value of $\mathcal{L}_z$ in the output field and its value in the input field is

$$\delta \mathcal{L}_z = \langle \mathbf{F}_{\text{in}} | \delta \hat{L}_z | \mathbf{F}_{\text{in}} \rangle. \quad (32)$$

Therefore the angular momentum absorbed by the lens per unit time is $-c\delta \mathcal{L}_z$. This is the total torque on the lens. In terms of the output field $|\mathbf{F}_{\text{out}}\rangle$ the relation (32) becomes

$$\delta \mathcal{L}_z = -\langle \mathbf{F}_{\text{out}} | \delta \hat{L}_z | \mathbf{F}_{\text{out}} \rangle, \quad (33)$$

which follows from the relation

$$\hat{T} \hat{L}_z \hat{T}^\dagger = \hat{L}_z - \delta \hat{L}_z. \quad (34)$$

For a spherical lens the operator $\delta \hat{L}_z$ vanishes. Hence, such a lens can never absorb angular momentum in the z direction from an em field. For a cylindrical lens satisfying eq. (29) one finds

$$\delta \hat{L}_z(\gamma) = \frac{k}{2f} [2\hat{x}\hat{y} \cos 2\gamma - (\hat{x}^2 - \hat{y}^2) \sin 2\gamma]. \quad (35)$$

From eq. (35) it follows that

$$\delta \hat{L}_z(\gamma \pm \pi/2) = -\delta \hat{L}_z(\gamma). \quad (36)$$

Thus, when a cylindrical lens is rotated over $\pi/2$ about the z axis, the amount of orbital angular momentum absorbed by the lens changes sign. This implies, that for any incoming field there is always an orientation γ_0 of the lens for which $\mathcal{L}_z$ of that field is not changed by the lens. Then it follows from eq. (35) again that the maximum change in the value of $\mathcal{L}_z$, $\delta \mathcal{L}_{\text{max}}$, is attained for an orientation angle $\gamma = \gamma_0 \pm \pi/4$. In fact, the change in orbital angular momentum of an arbitrary field due to a cylindrical lens can always be written as

$$\delta \mathcal{L}_z = \pm \delta \mathcal{L}_{\text{max}} \sin(2\gamma - 2\gamma_0). \quad (37)$$

6. Applications

6.1. Circular fields

As is well known, the operator $\hat{L}_z$ takes the simple form

$$\hat{L}_z = -i \frac{\partial}{\partial \phi} \quad (38)$$

in cylindrical coordinates

$$x = r \cos \phi; \quad y = r \sin \phi; \quad z = z. \quad (39)$$

We define a circular field as an eigenfunction of $\hat{L}_z$. These fields have an azimuthal dependence given by $\exp(il\phi)$. Since ϕ is a periodic variable, l must be an integer. In a circular field with eigenvalue l the orbital angular momentum per photon amounts to $l\hbar$ [1], and the intensity is independent of ϕ . Moreover, since $\hat{L}_z$ commutes with the free propagation operator, circular fields remain circular during free propagation. We shall denote these fields by kets $|l\rangle$. Examples of circular fields are the Laguerre–gaussian modes [10].

We can now calculate the effect of an arbitrary ideal lens on the orbital angular momentum of a circular field. The change $\delta \mathcal{L}_z$ in orbital angular momentum of an arbitrary input field $|\mathbf{F}\rangle$, eq. (32), can be rewritten in cylindrical coordinates:

$$\begin{aligned}\delta \mathcal{L}_z &= \langle \mathbf{F} | \partial \psi / \partial \phi | \mathbf{F} \rangle \\ &= \frac{\epsilon_0}{2\omega} \int_0^\infty dr \int_0^{2\pi} r d\phi \mathbf{F}^* \cdot \mathbf{F} \frac{\partial \psi}{\partial \phi} \\ &= -\frac{\epsilon_0}{2\omega} \int_0^\infty dr \int_0^{2\pi} r d\phi \psi \frac{\partial}{\partial \phi} (\mathbf{F}^* \cdot \mathbf{F}),\end{aligned}\quad (40)$$

where we used partial integration to obtain the last equality. From this equality it is clear that no ideal lens can change the orbital angular momentum of a circular field. Hence

$$\delta \mathcal{L}_z = \langle l | \delta \hat{L}_z | l \rangle = 0. \quad (41)$$

With eq. (33), this also shows that the lens cannot have changed the orbital angular momentum of a field when the *outcoming* beam is circular.

6.2. A cylindrical lens system

In this subsection we consider a particular optical system, which has been used in the experiments mentioned in the Introduction [2]. The system consists of two identical cylindrical lenses, oriented along the x axis, with focal length f , and separated from each other by a distance $2d$ (fig. 1). In ref. [3] a slightly different configuration was used, in that the two lenses had different focal lengths. Our results can easily be generalized to include this case. However, an arbitrary conversion of a gaussian beam can already be realized with lenses with equal focal lengths (see sect. 7). The propagation operator $\hat{h}$ for this system is

$$\hat{h} = \exp\left(\frac{-ik\hat{x}^2}{2f}\right) \exp\left(\frac{-id\hat{p}^2}{k}\right) \exp\left(\frac{-ik\hat{x}^2}{2f}\right). \quad (42)$$

The action of the system can be specified by giving the evolution of the coordinate operators $\hat{x}$ and $\hat{y}$, and of the momentum operators $\hat{p}_x$ and $\hat{p}_y$ in the Heisenberg picture,

$$\hat{h}^\dagger \hat{x} \hat{h} = \hat{x} \left(1 - \frac{2d}{f}\right) + \frac{2d}{k} \hat{p}_x,$$

$$\hat{h}^\dagger \hat{y} \hat{h} = \hat{y} + \frac{2d}{k} \hat{p}_y,$$

$$\hat{h}^\dagger \hat{p}_x \hat{h} = \hat{p}_x \left(1 - \frac{2d}{f}\right) - \frac{2k}{f} \left(1 - \frac{d}{f}\right) \hat{x},$$

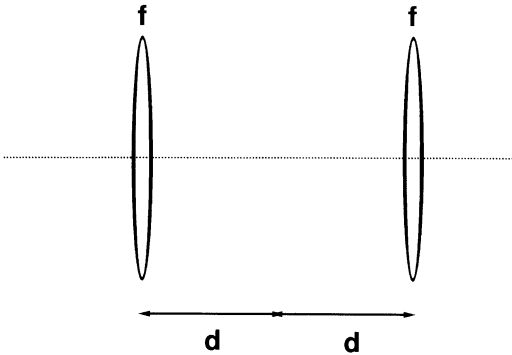


Fig. 1. Configuration of two parallel cylindrical lenses with focal length f at mutual distance $2d$.

$$\hat{h}^\dagger \hat{p}_y \hat{h} = \hat{p}_y, \quad (43)$$

obtained by means of the Baker–Campbell–Hausdorff formula [13]

$$\exp(B) A \exp(-B)$$

$$= A + [B, A] + \frac{1}{2!} [B, [B, A]] + \dots \quad (44)$$

From eqs. (43) one finds that the orbital angular momentum changes according to

$$\begin{aligned} \hat{h}^\dagger \hat{L}_z \hat{h} &= \hat{L}_z \left(1 - \frac{2d^2}{f^2}\right) + \frac{2k}{f} \left(1 - \frac{d}{f}\right) \hat{x} \hat{y} + \frac{4d^2}{kf} \hat{p}_x \hat{p}_y \\ &+ (\hat{x} \hat{p}_y + \hat{y} \hat{p}_x) \left(\frac{2d}{f} - \frac{2d^2}{f^2}\right). \end{aligned} \quad (45)$$

In particular, for incoming circular fields $|l\rangle$ only the first term on the right hand side of eq. (45) contributes, and one finds for the change in $\mathcal{L}_z$

$$\delta \mathcal{L}_z = -\frac{2d^2}{f^2} \langle l | \hat{L}_z | l \rangle, \quad (46)$$

so that the lens system absorbs an amount $2\hbar d^2/f^2$ of angular momentum per photon. However, the output field cannot be circular. In fact, when both in- and outcoming fields would be circular, then neither of the two lenses could have changed $\mathcal{L}_z$, as shown in section 6.1; furthermore, during the free propagation between the lenses $\mathcal{L}_z$ is conserved (section 4.2), so that the orbital angular momentum of in- and outcoming fields would have to be the same. This is in contradiction with eq. (46), since the multiplication factor is always negative.

From similar arguments is clear that the change in angular momentum of an incoming circular field takes place exclusively at the second lens.

Moreover, eq. (46) shows that we must have $d=f/\sqrt{2}$ in order that the lens system changes a circular field $|l\rangle$ with $l \neq 0$ into a field with zero angular momentum. Conversely, the same condition is required to produce a carries field $|l\rangle$ with $l \neq 0$ from an input field that carries no angular momentum (cf. ref. [2]).

6.3. A paradox

Geometrical optics suggests a simple way to convert a circular field $|l\rangle$ into the opposite state $|-l\rangle$.

This is achieved when the system inverts the x direction, without changing the y direction. For a well-collimated beam this simple reflection is produced by the configuration of fig. 2, with $d=f$. This configuration has been termed a π converter [2]. The picture of x inversion is confirmed by eqs. (43), provided that the last terms of the first two equations are negligible. This requires that

$$\delta \equiv d/kw^2 \ll 1, \quad (47)$$

where w is the transverse dimension of the incoming beam. Then the lens system transforms a field $|F\rangle$ according to

$$F(x, y) \rightarrow F(-x, y), \quad (48)$$

apart from a phase factor, and a small $\mathcal{O}(\delta)$ correction. When applied to a circular field $|l\rangle$, this transformation (48) yields another circular field $|-l\rangle$. This simple geometrical picture is confirmed by eq. (46), which shows that for $d=f$ the momentum transfer is exactly $-2\hbar l$ per photon. However, this transformation is in contradiction with the conclusion of section 6.2, namely that for a circular input field the output field cannot be circular. In fact, the conversion is only exact in the geometrical optics limit $k \rightarrow \infty$ (so that $\delta \rightarrow 0$). Hence, when the input field is exactly the circular state $|l\rangle$, then the output field must be written as $|-l\rangle + \delta|s\rangle$, where $|s\rangle$ stands for some superposition of circular fields. Then the input field has an angular momentum per photon equal to $\hbar l$, and for the output field this value is exactly $-\hbar l$ (when $d=f$). Furthermore, the transfer

of angular momentum must occur at the second lens, as we showed in section 6.2.

This raises another problem: on the one hand the second lens transfers $2\hbar$ per photon when δ is small but finite, while on the other hand there can be no transfer when the output field is circular (i.e. for $\delta=0$). This discontinuity can be explained from eq. (29). The phase that the lens adds to the field is of the order of $1/\delta$. Therefore the effect of the second lens on the orbital angular momentum can be arbitrarily large for $\delta \rightarrow 0$. Thus, for an *exact* circular output field there is no transfer of $\mathcal{L}_z$, but small $\mathcal{O}(\delta)$ corrections to a pure circular field give in general rise to a finite transfer of $\mathcal{L}_z$ of order unity in units of $\hbar$ per photon. More precisely, since the output field is given by $|-l\rangle + \delta|s\rangle$, one finds from eqs. (33) and (35) the finite change in $\mathcal{L}_z$ by the second cylindrical lens in the limit $\delta \rightarrow 0$,

$$\delta \mathcal{L}_z = -2 \operatorname{Re} \langle s | \hat{x} \hat{y} / w^2 | -l \rangle. \quad (49)$$

A similar argument is valid when the *input* field is not exactly circular.

We conclude that the total transfer of angular momentum is close to $-2\hbar l$ per photon when the input field is nearly circular, and when d is close to f . However, the location where the angular momentum is transferred is very sensitive to the input state: a small admixture of other states can give a drastic variation in the distribution of the transfer over the two lenses.

For noncircular input states, eq. (45) shows that the total amount of angular momentum transfer depends in a very sensitive way on d (or f). This is due to the second term in eq. (45), which is proportional to $1/\delta$. When the average of xy vanishes, which is the case for circular fields, this term does not contribute, and the sensitivity disappears.

7. Eigenoperators of fields

We earlier defined circular fields by requiring them to be eigenfunctions of the operator $\hat{L}_z$. In a similar way we can specify a field $|F(z)\rangle$, for given wave number k , by giving three commuting hermitian operators of which the field is an eigenmode, and the corresponding eigenvalues. One of the operators, $\hat{S}(z)$, refers to the vector character (polarization or spin) of the field. The other two operators $\hat{K}_i(z)$, for

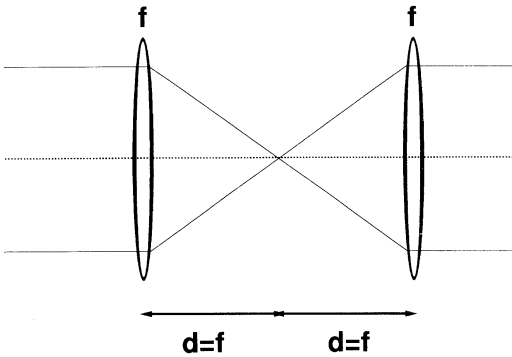


Fig. 2. In the geometrical optics limit the lens system of fig. 1. inverts the image in one direction when $d=f$.

$i=1, 2$, act upon the translational degrees of freedom of the field, and determine its (x, y) dependence. Thus, the field $|F(z)\rangle$ is uniquely determined, up to a normalization factor and an arbitrary phase, by

$$\begin{aligned}\hat{S}(z)|F(z)\rangle &= s|F(z)\rangle, \\ \hat{K}_i(z)|F(z)\rangle &= \mu_i|F(z)\rangle.\end{aligned}\quad (50)$$

The operators $\hat{S}$ and $\hat{K}_i$ may be called eigenoperators of the field. The eigenvalue of the polarization operator $\hat{S}$ equals $s = \pm 1$, and the eigenvalues μ_i are real, and independent of z (see below). When a field propagates through a given optical system, it evolves according to eq. (14), and it will in general no longer satisfy eqs. (50). Instead, the field $\hat{h}|F\rangle$ is an eigenfunction of the operators

$$\begin{aligned}\hat{S}' &= \hat{h}\hat{S}\hat{h}^\dagger, \\ \hat{K}_i' &= \hat{h}\hat{K}_i\hat{h}^\dagger,\end{aligned}\quad (51)$$

with the same eigenvalues as in eqs. (50). The evolution of an em field is thus also described (up to a phase factor) by the evolution (eq. (51)) of its eigenoperators. This description is sometimes more convenient than dealing with the explicit spatial dependence of the fields. Notice that the operators $\hat{K}_i(z)$ as a function of z do not obey the Heisenberg equation, but rather the Liouville–Von Neumann equation.

Since the optical systems we consider in this paper do not contain polarization-changing elements, polarization is conserved, and will henceforth be left out of consideration. From now on we only discuss scalar fields.

7.1. Hermite- and Laguerre-gaussian modes

We apply the eigenoperator description to Hermite–gaussian (HG) and Laguerre–gaussian (LG) modes, which are commonly used in laser physics [10]. We first give the eigenoperators of these fields in their focal plane, which is the plane where the transverse dimension of the beam has a minimum. The focal plane is chosen to be the plane $z=0$. In an arbitrary plane $z=z_0$ the fields can then be constructed formally by letting the fields propagate freely from $z=0$ to $z=z_0$. In the next section we will show that the modes defined in this way are indeed the HG and LG modes.

We define a HG field in its focal plane by the eigenvalue equations

$$\begin{aligned}\frac{1}{2}\left(\frac{k}{\beta}\hat{x}^2 + \frac{\beta}{k}\hat{p}_x^2\right)|H_{nm}^\beta(0)\rangle &= (n + \tfrac{1}{2})|H_{nm}^\beta(0)\rangle, \\ \frac{1}{2}\left(\frac{k}{\beta}\hat{y}^2 + \frac{\beta}{k}\hat{p}_y^2\right)|H_{nm}^\beta(0)\rangle &= (m + \tfrac{1}{2})|H_{nm}^\beta(0)\rangle.\end{aligned}\quad (52)$$

Thus a HG field $|H_{nm}^\beta(0)\rangle$ is an eigenfunction of a ‘Hamiltonian’ of two identical harmonic oscillators in the x and y direction, with ‘energy’ $n + \frac{1}{2}$ and $m + \frac{1}{2}$, respectively. The parameter β can be identified with the Rayleigh range or half the confocal parameter of the beam [10]. Analogously, LG fields with the same Rayleigh range β can be defined by

$$\begin{aligned}\hat{L}_z|L_{lN}^\beta(0)\rangle &= l|L_{lN}^\beta(0)\rangle, \\ \frac{1}{2}\left(\frac{k}{\beta}\hat{r}^2 + \frac{\beta}{k}\hat{p}^2\right)|L_{lN}^\beta(0)\rangle &= (N+1)|L_{lN}^\beta(0)\rangle,\end{aligned}\quad (53)$$

where $\hat{r}^2 = \hat{x}^2 + \hat{y}^2$. Thus a LG field $|L_{lN}^\beta(0)\rangle$ is an eigenfunction of $\hat{L}_z$, with eigenvalue l , and of a 2-D degenerate harmonic oscillator, with ‘energy’ $N+1$. The latter operator is just the sum of the two harmonic oscillators in eqs. (52). From quantum mechanics it is known that one obtains solutions only for $|l| \leq N$. The two eigenvalue equations (53) determine the ϕ and r dependence of LG fields. By applying the free propagation operator (23) to the fields at $z=0$ we obtain the HG and LG modes for arbitrary z ,

$$\begin{aligned}|H_{nm}^\beta(z)\rangle &= \exp\left(\frac{-iz}{2k}\hat{p}^2\right)|H_{nm}^\beta(0)\rangle, \\ |L_{lN}^\beta(z)\rangle &= \exp\left(\frac{-iz}{2k}\hat{p}^2\right)|L_{lN}^\beta(0)\rangle.\end{aligned}\quad (54)$$

For $z \neq 0$ these fields are no longer eigenfunctions of the same, or of any other, harmonic oscillator. Namely, we find that the eigenoperators of the HG mode evolve during free propagation as

$$\begin{aligned}\hat{K}_1(z) &= \frac{1}{2}\left[\frac{k}{\beta}\hat{x}^2 + \left(\frac{\beta}{k} + \frac{z^2}{k\beta}\right)\hat{p}_x^2 - \frac{z}{\beta}\hat{t}_x\right], \\ \hat{K}_2(z) &= \frac{1}{2}\left[\frac{k}{\beta}\hat{y}^2 + \left(\frac{\beta}{k} + \frac{z^2}{k\beta}\right)\hat{p}_y^2 - \frac{z}{\beta}\hat{t}_y\right],\end{aligned}\quad (55)$$

and the eigenvalues remain $n + \frac{1}{2}$ and $m + \frac{1}{2}$. Relation (55) follows from eqs. (51) with eq. (23) substituted for $\hat{h}$ while using the BCH formula (44). We defined here the operators $\hat{l}_x$ and $\hat{l}_y$ and their sum $\hat{l}$ by

$$\begin{aligned}\hat{l}_x &= \hat{x}\hat{p}_x + \hat{p}_x\hat{x}, \\ \hat{l}_y &= \hat{y}\hat{p}_y + \hat{p}_y\hat{y}, \\ \hat{l} &= \hat{l}_x + \hat{l}_y.\end{aligned}\quad (56)$$

The LG mode $|L_{lN}^\beta(z)\rangle$ is an eigenmode of $\hat{L}_z$ with eigenvalue l for all z , since $\hat{L}_z$ commutes with $\hat{p}^2$. Its second eigenoperator is given by $\hat{K}(z) = \hat{K}_z(z) + \hat{K}_2(z)$, and the eigenvalue remains $N+1$.

7.2. Gaussian modes

We have shown that HG and LG modes have one eigenoperator in common, of the general form

$$\hat{K}(\mathbf{a}) = a_1 \hat{r}^2 + a_2 \hat{p}^2 + a_3 \hat{l}, \quad (57)$$

which is invariant under rotations about the z axis. In their focal plane both fields are eigenfunctions of a 2D degenerate harmonic oscillator (DHO), since there a_3 vanishes. Fields with an eigenoperator of the form (57) with arbitrary values a_1 , a_2 and a_3 will be called gaussian fields. This is in accordance with textbook nomenclature [10], where gaussian fields are written as an eigenfunction of a DHO multiplied by a position-dependent phase factor (see next section). Such fields remain gaussian during their free propagation. Furthermore, it is easily proved that any gaussian field can be written as a linear superposition of HG modes $|H_{nm}^\beta(z)\rangle$ with $n+m=\text{constant}$, and with a properly selected value of β and of the position of the focal plane. The converse statement is obviously also true. Thus, any HG mode $|H_{nm}^\beta(z)\rangle$ that is rotated about the z axis can be written as such a superposition. The same is true for the LG mode $|L_{lN}^\beta(z)\rangle$, where $n+m$ must equal N .

7.3. Mode convertors

The eigenoperator (57) of a gaussian field will be transformed by a cylindrical lens into an operator that is no longer invariant under rotations about the z axis. Then the output field is not gaussian. However, in special cases, a lens system may convert a

gaussian field into another gaussian field. Then we call the lens system a mode convertor [1,2]. The condition for a mode convertor is, that the operator

$$\hat{K}' = \hat{h}\hat{K}(\mathbf{a})\hat{h}^\dagger \quad (58)$$

be of the form (57) again. We substitute for $\hat{h}$ the evolution operator (42) of the lens system from section 6.2. For a given incoming field, this yields the conditions under which the lens system is a mode convertor for this field. We obtain

$$\begin{aligned}d &= k \frac{a_3}{a_1}, \\ f &= k \frac{a_3 a_2}{a_2 a_1 - 2a_3^2},\end{aligned}\quad (59)$$

in terms of the coefficients a_i of the eigenoperator (57) of the incoming field. For incoming HG modes with eigenoperators (55) we can substitute $a_1 = k/\beta$ and $a_3 = -z/\beta$ in the conditions (59), which then become

$$\begin{aligned}z &= -d, \\ f &= d \frac{\beta^2 + d^2}{\beta^2 - d^2}.\end{aligned}\quad (60)$$

The first condition means that for a mode convertor the focal plane of the incoming HG mode must be halfway between the two lenses [2].

7.4. Production of circular fields

We now wish to investigate when it is possible to convert a HG mode $|H_{nm}^\beta(z)\rangle$ into a field that has the orbital angular momentum operator $\hat{L}_z$ as an eigenoperator, that is, into a circular field.

The two eigenoperators of the HG mode were given in eqs. (55). Eigenoperators of this form will keep the same form during their propagation through the cylindrical lens system of section 6.2. The reason is (i) that the evolution operator (42) of the lens system consists of exponentials of the operators occurring in (55) and (ii) that these operators form a closed (Lie) algebra. In fact, they generate the direct product $\text{SU}(1,1) \times \text{SU}(1,1)$ [13]. The operators (55) will therefore evolve, according to eqs. (51), within that algebra during the entire propagation. This algebra does not contain $\hat{L}_z$. Therefore, the lens system satisfying eq. (42) alone is not sufficient to

produce an eigenfunction of $\hat{L}_z$ from a HG mode $|H_{nm}^\beta(z)\rangle$. Both eigenoperators of this field are and remain of the form (55).

However, application of a rotation about the z axis to a HG mode before it enters the lens system will transform eigenoperators in such a way that they will belong to a larger algebra which does contain $\hat{L}_z$. The reason is, that rotations around the z axis are generated by $\hat{L}_z$: the rotation operator $\hat{R}_\gamma$ describing a rotation around the z axis over an angle γ is given by

$$\hat{R}_\gamma = \exp(i\gamma\hat{L}_z). \quad (61)$$

If the HG mode $|H_{nm}^\beta(z)\rangle$, rotated over an angle γ and having passed the lens system, is to be an eigenfunction of $\hat{L}_z$, we must require $\hat{L}_z$ to be a linear combination of the two eigenoperators of the output field

$$\hat{K}'_i = \hbar\hat{R}_\gamma\hat{K}_i\hat{R}_\gamma^{-1}\hat{h}^\dagger, \quad (62)$$

for $i=1, 2$, with $\hat{K}_1$ and $\hat{K}_2$ the eigenoperators of the incoming field, defined in eqs. (55). Conversely, the operator

$$\hat{L}'_z = \hat{R}_\gamma^{-1}\hat{h}^\dagger\hat{L}_z\hat{R}_\gamma \quad (63)$$

must be a linear combination of the operators $\hat{K}_1$ and $\hat{K}_2$. Using eq. (45) one finds

$$\begin{aligned} \hat{R}_\gamma^{-1}\hat{h}^\dagger\hat{L}_z\hat{R}_\gamma &= \hat{L}_z \left(1 - \frac{2d^2}{f^2}\right) \\ &+ \left(\frac{2d}{f} - \frac{2d^2}{f^2}\right) [(\hat{x}\hat{p}_y + \hat{y}\hat{p}_x) \cos 2\gamma \\ &+ \sin 2\gamma(\hat{y}\hat{p}_y - \hat{x}\hat{p}_x)] \\ &+ \frac{4d^2}{kf} [\hat{p}_x\hat{p}_y \cos 2\gamma + \frac{1}{2} \sin 2\gamma(p_y^2 - \hat{p}_x^2)] \\ &+ \frac{2k}{f} \left(1 - \frac{d}{f}\right) [\hat{x}\hat{y} \cos 2\gamma + \frac{1}{2} \sin 2\gamma(\hat{y}^2 - \hat{x}^2)]. \end{aligned} \quad (64)$$

Inspection of eq. (64) shows that this expression is indeed a linear combination of $\hat{K}_1$ and $\hat{K}_2$:

$$\hat{R}_\gamma^{-1}\hat{h}^\dagger\hat{L}_z\hat{R}_\gamma = \pm (\hat{K}_2 - \hat{K}_1), \quad (65)$$

if and only if

$$\begin{aligned} \gamma &= \pm \pi/4, \quad f = d\sqrt{2}, \\ z &= -d, \quad \beta = d(1 + \sqrt{2}). \end{aligned} \quad (66)$$

These restrictions are the same as those given for the ' $\pi/2$ convertor' [2]. Here we have proved that the rotation angle γ must equal $\pm\pi/4$, and that the focal plane of the HG mode must be half way between the two lenses. For the eigenvalue l of $\hat{L}_z$ one finds, from eq. (65),

$$l = \pm(m-n), \quad (67)$$

because the eigenvalues corresponding to $\hat{K}_2$ and $\hat{K}_1$ are $m + \frac{1}{2}$ and $n + \frac{1}{2}$, respectively. Since the relations (66) are consistent with eqs. (60), the lens system acts in this case also as a mode convertor. That is, the outcoming mode is not just an eigenfunction of $\hat{L}_z$, it is in fact the LG field $|L_{l,n+m}^\beta(d)\rangle$.

8. Properties of gaussian modes

We show here that the eigenoperators provide a simple way to explain that a freely propagating field, which is an eigenfunction of a 2D degenerate harmonic oscillator (DHO) in one plane, can be written as the product of a local phase factor and an eigenfunction of another DHO, in any other plane $z = \text{constant}$. Subsequently, we show that the Gouy phase of gaussian beams [10] can be identified with the dynamical phase of a quantum-mechanical DHO with time-dependent energy.

We consider a field $|F_N(0)\rangle$ that is an eigenfunction of the DHO

$$\hat{K}(0) = \frac{1}{2} \left[\frac{\hat{r}^2}{\alpha_0} + \alpha_0 \hat{p}^2 \right], \quad (68)$$

with eigenvalue $N+1$. When the field propagates through the vacuum, it becomes an eigenfunction of the operator

$$\hat{K}(z) \equiv \exp\left(\frac{-iz}{2k}\hat{p}^2\right) \frac{1}{2} \left[\frac{\hat{r}^2}{\alpha_0} + \alpha_0 \hat{p}^2 \right] \exp\left(\frac{iz}{2k}\hat{p}^2\right), \quad (69)$$

at position z . We now search for parameters R and α such that this eigenoperator $\hat{K}(z)$ is again a DHO up to a position-dependent phase factor. More precisely, we require

$$\hat{K}(z) = \exp\left(\frac{ik}{2R}\hat{r}^2\right) \frac{1}{2} \left[\frac{\hat{r}^2}{\alpha} + \alpha \hat{p}^2 \right] \exp\left(\frac{-ik}{2R}\hat{r}^2\right). \quad (70)$$

This relation implies that the field $|F(z)\rangle$ at position z is the product of the phase factor $\exp(ik\hat{r}^2/2R)$ and an eigenfunction of a DHO of the form (68) with α_0 replaced by $\alpha(z)$. Its eigenvalue is $N+1$. Using the BCH formula (44) we find the unique solution of eq. (70) to be

$$R(z) = \frac{z^2 + \alpha_0^2 k^2}{z},$$

$$\alpha(z) = \frac{z^2 + \alpha_0^2 k^2}{k^2 \alpha_0}. \quad (71)$$

For HG and LG modes we may substitute the value $\alpha_0 = \beta/k$, according to eqs. (52) and (53). Then we find

$$R(z) = \frac{z^2 + \beta^2}{z},$$

$$\alpha(z) = \frac{z^2 + \beta^2}{k\beta}. \quad (72)$$

These parameters are, indeed, equal to the local radius of curvature of these modes, and to the square of the local beam radius, respectively [10].

8.1. Gouy phase

As just shown we can rewrite the field $|F_N(z)\rangle$ as

$$|F_N(z)\rangle = \exp\left(\frac{ik\hat{r}^2}{2R(z)}\right) |H_N(\alpha, z)\rangle, \quad (73)$$

where $|H_N(\alpha, z)\rangle$ is an eigenfunction of a DHO with eigenvalue $N+1$. Using the fact that $|F_N(z)\rangle$ satisfies the paraxial wave equation, we find, after substituting eq. (73),

$$\begin{aligned} 2ik \frac{d}{dz} |H_N(\alpha, z)\rangle &= \left[k^2 \frac{1}{R^2} \frac{\partial R}{\partial z} \hat{r}^2 \right. \\ &\quad \left. + \exp\left(\frac{-ik}{2R} \hat{r}^2\right) \hat{p}^2 \exp\left(\frac{+ik}{2R} \hat{r}^2\right) \right] |H_N(\alpha, z)\rangle \\ &= \left[\frac{\hat{r}^2}{\alpha^2} + \hat{p}^2 + \frac{k}{R} \hat{r} \right] |H_N(\alpha, z)\rangle \\ &= \left[\frac{\hat{r}^2}{\alpha^2} + \hat{p}^2 + \frac{k}{2\alpha} \frac{\partial \alpha}{\partial z} \hat{r} \right] |H_N(\alpha, z)\rangle, \end{aligned} \quad (74)$$

with $\hat{r}$ defined in eq. (56). The field distribution corresponding to $|H_N(\alpha, z)\rangle$ can be written as a real

function of x and y , represented by the ket $|G_N(\alpha)\rangle$, multiplied by an overall z -dependent phase factor,

$$|H_N(\alpha, z)\rangle = \exp(-i\phi_N(z)) |G_N(\alpha)\rangle. \quad (75)$$

The last term on the right hand side of eq. (74) being proportional to $d\alpha/dz$ results from the z dependence of the real part $|G_N(\alpha)\rangle$ in eq. (75) through $\alpha(z)$. Indeed one can show that

$$\hat{r} |G_N(\alpha)\rangle = 4i\alpha \frac{\partial}{\partial \alpha} |G_N(\alpha)\rangle. \quad (76)$$

The remaining two terms on the right hand side of eq. (74) therefore determine the explicit z dependence of $|H_N(\alpha, z)\rangle$, and hence the phase $\phi_N(z)$. This remaining part has the form of a Hamiltonian for an harmonic oscillator with constant mass and position-dependent frequency. The ket $|H_N(\alpha, z)\rangle$ is an eigenfunction of this Hamiltonian with position-dependent eigenvalue ('energy') $2(N+1)/\alpha(z)$. Therefore, the phase $\phi_N(z)$ is given by

$$\phi_N(z) = \int_0^z ds \frac{2(N+1)}{2k\alpha(s)}. \quad (77)$$

For HG and LG modes the substitution of $\alpha_0 = \beta/k$ leads to the well-known expression for the Gouy phase [10]:

$$\phi(z) = \tan^{-1}(z/\beta), \quad (78)$$

and $\phi_N = (N+1)\phi$. Thus HG and LG modes evolve, apart from the phase factor in eq. (73) in the same way as the state of a quantum-mechanical harmonic oscillator with time-dependent energy in the adiabatic regime, when it starts in an eigenstate. Namely, in the adiabatic approximation one neglects the time derivative of the hamiltonian. Here, on the other hand, the presence of the third term in eq. (74) proportional to the derivative of $\alpha(z)$ ensures that the ket $|H_N(\alpha, z)\rangle$ remains an *exact* eigenfunction of the Hamiltonian.

9. Conclusions

We discussed the change in orbital angular momentum of light beams during their passage through systems of cylindrical lenses. We presented a formalism which makes use of some formal analogies

between quantum mechanics and paraxial wave optics. For instance, a light field can be represented by a state vector, with the coordinate z as a propagation variable, which plays the same role as time in quantum mechanics [4]. We showed that physical quantities such as the linear and angular momentum per unit beam length can be given in the form of expectation values of corresponding operators in the state associated with the field.

In the ‘Schrödinger picture’, the propagation of fields along the z axis is described by a unitary operator. In the ‘Heisenberg picture’ the evolution of physical quantities rather than of the fields is given. By using the Heisenberg picture, we proved that, e.g., the z component of orbital angular momentum is uniform along z for arbitrary freely propagating light beams. Furthermore, we calculated the amount of angular momentum absorbed by ideal lenses, and described effects of cylindrical lenses for arbitrary incident fields.

We then discussed the action of cylindrical lens systems on the linear and orbital angular momentum of light beams, and determined exactly where and how much angular momentum is transferred. We also found a peculiar paradox for a particular lens system in the geometrical optics limit. It turns out that this limit displays a singularity for lens systems.

We argued that a light field is determined at each position z by three eigenoperators which have the field as an eigenfunction, and by its eigenvalues. Propagation of a field can alternatively be described by giving the evolution of its eigenoperators. In particular, we applied this formalism to define Hermite–gaussian (HG) and Laguerre–gaussian (LG) modes [10], and the definition for more general gaussian modes arises naturally. Using this formalism we were able to find in a straightforward way when an astigmatic lens system converts one gaus-

sain mode into another. We also derived the necessary and sufficient conditions under which HG modes can be converted into LG modes and conversely. This possibility was derived in a completely different way in ref. [2].

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Paraxial wave optics and harmonic oscillators

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The operator algebra of the quantum harmonic oscillator is applied to the description of Gaussian modes of a laser beam. Higher-order modes of the Hermite-Gaussian or the Laguerre-Gaussian form are generated from the fundamental mode by ladder operators. This approach allows the description of both free propagation and refraction by ideal astigmatic lenses. The paraxial optics analog of a coherent state is shown to be a light beam with a displaced beam axis which is refracted by lenses according to geometric optics. The expectation value of the orbital angular momentum of a paraxial beam of light is found to be expressible in terms of a contribution analogous to the angular momentum of the oscillator plus contributions which arise from the ellipticity of the wave fronts and of the light spot. This clarifies the process by which a transfer of orbital angular momentum between a light beam and astigmatic lenses or diaphragms occurs.

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I. INTRODUCTION

It is well known that the analytical form of Gaussian modes of a laser beam resembles the wave functions of the stationary states of a two-dimensional quantum-mechanical harmonic oscillator [1,2]. This suggests that algebraic treatments of the harmonic oscillator may be fruitfully applied to wave optics. Such a treatment can give insight into the structure and the properties of paraxial modes, and the connection between modes of different order. Moreover, algebraic methods often greatly simplify explicit calculations of physical quantities. Recently it has been pointed out that a Laguerre-Gaussian beam carries an orbital angular momentum along its propagation direction [3]. This angular momentum arises from the transverse momentum density of the field. It might be expected that the orbital angular momentum of a beam of light be analogous to the angular momentum of the harmonic oscillator.

In this paper, operator algebra is applied to describe Gaussian modes of a laser beam in the presence of ideal, but possibly, astigmatic lenses. Raising and lowering operators are introduced, which generate all higher-order modes from the fundamental one. These operators depend in a simple way on the coordinate in the propagation direction, and they can be expressed as a unitary transformation of the real harmonic-oscillator ladder operators. The ladder operators are characterized by three z -dependent beam parameter for each transverse dimension. These parameters are the radius of curvature, the spot size, and the phase. The algebraic connection between modes with different mode indices is the same as that for the number states of the harmonic oscillator. The fundamental mode is the eigenvector of the lowering operator with eigenvalue zero. The connection between modes of different order, and between Laguerre-Gaussian and Hermite-Gaussian modes, follows directly. Eigenvectors with different eigenvalues, which are analogous to the coherent states, represent beams with a displaced

axis. The axis of the beam simply obeys the rules of geometric optics while leaving unaffected the three beam parameters, which reflect the wave-optical nature of the beam. This separation between geometric-optical and wave-optical aspects of the beam during its propagating through lens systems follows in a direct fashion from the algebraic treatment. Finally, we apply the operator algebra to the description of the orbital angular momentum of a Gaussian beam, the expression for which is found to separate into a term that is analogous to the orbital angular momentum of the isotropic two-dimensional harmonic oscillator, a contribution expressing the ellipticity of the spot size, and a term arising from the ellipticity of the wave fronts. Only this final term is modified when the beam passes an astigmatic lens. The torque exerted on the lens vanishes when the intensity distribution of the beam is symmetric with respect to one of the axes of the lens. This explains why a nonastigmatic Laguerre-Gaussian beam cannot transfer angular momentum to an astigmatic lens [4]. However, we demonstrate that such a beam transfers a significant amount of angular momentum to the lens if one applies angular aperturing of the beam just in front of the lens. Such a transfer depends only on the radial mode index.

II. OPERATOR ALGEBRA FOR FREE PARAXIAL MODE

As is well known, a complete set of solutions of the wave equation of a beam of light in the paraxial approximation consists of the products of a Gaussian with a Hermite polynomial H_n [1,2]. For a light beam propagating in the z direction, and a single transverse dimension x , these Hermite-Gaussian modes can be expressed as the normalized functions [2]

$$u_n(x, z) = \exp \left[\frac{ikx^2}{2s(z)} - i\chi(z)(n + \frac{1}{2}) \right] \times \frac{1}{\sqrt{\gamma(z)}} \psi_n \left[\frac{x}{\gamma(z)} \right]. \quad (2.1)$$

Here the functions ψ_n represent the real normalized eigenfunctions of the harmonic oscillator, defined by the Hamiltonian

$$H = \frac{1}{2} \left[-\frac{\partial^2}{\partial \xi^2} + \xi^2 \right]. \quad (2.2)$$

Their explicit expressions are

$$\psi_n(\xi) = [2^n n! \sqrt{\pi}]^{-1/2} \exp(-\xi^2/2) H_n(\xi). \quad (2.3)$$

The formal analogy between the Hermite-Gaussian modes and the harmonic-oscillator eigenstates offers the possibility of applying the operator algebra of the harmonic oscillator to paraxial beam optics.

The Hermite-Gaussian mode functions (2.1) depend on three z -dependent mode parameters γ , s , and χ . The expressions for the spot size γ and radius of curvature of the wave front s can be combined in the single complex equality

$$\frac{1}{\gamma^2} - \frac{ik}{s} = \frac{k}{b + iz}, \quad (2.4)$$

with k the wave number of the light. The parameter b is the Rayleigh range of the beam, which determines the size of the focal region. Finally, the phase factor χ is given by

$$\tan \chi = z/b. \quad (2.5)$$

In the present case of Gaussian beams, this factor describes the phase jump of π that occurs over the focal region of any spherical converging wave, which was first recognized by Gouy [5,2]. The basis set (2.1) depends on the value of the Rayleigh range b and on the location of the focal plane, which we have chosen at $z=0$. The spot size at focus is equal to $\sqrt{b/k}$, and it increases as $z/\sqrt{bk}$ for $z \gg b$. Hence a small spot size at focus implies a large divergence angle of the beam.

Equation (2.1) displays both the analogy and the difference between the propagation of a Hermite-Gaussian mode of light and the evolution of a harmonic oscillator. The diffraction of the beam is expressed by a variation of the spot size γ and the radius of curvature s with z . This is absent for the oscillator. As may be noted in (2.1), the variation of the Gouy phase χ during propagation, which multiplies the level energy $n + \frac{1}{2}$, is analogous to the phase Ωt of the oscillator with frequency Ω during its evolution. In this sense, the free propagation of a Gaussian beam from $-\infty$ to ∞ corresponds to half a cycle of the oscillator.

The Hermite-Gaussian modes (2.1) are solutions of the paraxial wave equation [1,2,6]

$$\frac{\partial^2}{\partial x^2} u(x, z) = -2ik \frac{\partial}{\partial z} u(x, z). \quad (2.6)$$

When the variations of u in the transverse direction x are small over a wavelength, a solution u of (2.6) determines, to a good approximation, the electric field $\mathbf{E}$ of a monochromatic light wave with frequency $\omega = ck$, of the form

The constant transverse vector $\mathbf{E}_0$ determines the polarization and the amplitude of the paraxial beam. Equation (2.6) has the same form as Schrödinger's equation for a free particle in one dimension, with the z coordinate replacing the time variable [7]. The formalism of quantum mechanics in terms of a Hilbert state space, and with linear operators representing observables, can be carried over directly to a classical light beam in the paraxial approximation, as has been done by Stoler [8]. In the same spirit, we introduce state vectors $|u(z)\rangle$, which have $u(x, z)$ for the mode function. Furthermore, we define the coordinate operator X , and the momentum operator P , according to the equalities

$$Xu(x, z) = xu(x, z), \quad Pu(x, z) = \frac{1}{i} \frac{\partial}{\partial x} u(x, z). \quad (2.8)$$

(Operators acting on states vectors are indicated by roman capitals throughout this paper.) Then the propagation equation (2.6) can be put in the form

$$\frac{d}{dz} |u(z)\rangle = -\frac{i}{2k} P^2 |u(z)\rangle. \quad (2.9)$$

Equation (2.9) has the formal solution

$$|u(z)\rangle = U(z) |u(0)\rangle, \quad (2.10)$$

with the propagation operator U defined by

$$U(z) = \exp \left[-\frac{i}{2k} P^2 z \right]. \quad (2.11)$$

To complete the analogy with quantum mechanics, we assume normalized state vectors obeying the identity

$$\langle u(z) | u(z) \rangle \equiv \int dx u^*(x, z) u(x, z) = 1. \quad (2.12)$$

When $|u\rangle$ is normalized for one position z , the normalization for all other z values is automatic. For an inner product defined as in (2.12), the operators X and P are Hermitian. The expectation value of operators is defined in the standard way as

$$\langle P(z) \rangle \equiv \langle u(z) | P | u(z) \rangle = \int dx u^*(x, z) P u(x, z). \quad (2.13)$$

It may be shown that with this normalization (2.12) $\hbar \langle P(z) \rangle$ is equal to the transverse momentum per photon in the beam [4].

A. Ladder operators

The expressions (2.1), which are solutions of the paraxial wave equation (2.6), can be obtained by operator algebra techniques. Our starting point is the introduction of z -dependent field operators, which allow a series of solutions to be created from one. For $z=0$, we define the operators

$$A(0) = \frac{1}{\sqrt{2bk}} [kX + ibP], \quad A^\dagger(0) = \frac{1}{\sqrt{2bk}} [kX - ibP], \quad (2.14)$$

real: they transform a real function into another real function. The z dependence of these operators is defined by the requirement that, for any solution $|u(z)\rangle$ of (2.9), $A^\dagger(z)|u(z)\rangle$ is also a solution. This implies that

$$A(z) = U(z) A(0) U^\dagger(z), \quad A^\dagger(z) = U(z) A^\dagger(0) U^\dagger(z). \quad (2.15)$$

For free propagation, explicit expressions for these z -dependent operators follow from the operator identity

$$\exp\left[-\frac{i}{2k}P^2z\right]X\exp\left[\frac{i}{2k}P^2z\right] = X - \frac{z}{k}P. \quad (2.16)$$

Combining (2.15) and (2.16) gives the simple z dependence

$$A(z) = \frac{1}{\sqrt{2bk}}[kX + i(b + iz)P], \quad (2.17)$$

$$A^\dagger(z) = \frac{1}{\sqrt{2bk}}[kX - i(b - iz)P]. \quad (2.18)$$

These operators obey the standard commutation rule

$$[A(z), A^\dagger(z)] = 1 \quad (2.19)$$

for boson raising and lowering operators and, as shown in every quantum mechanics textbook, this commutation rule is sufficient to prove that the number operator

$$N(z) = A^\dagger(z)A(z) \quad (2.20)$$

has the natural numbers as its eigenvalue spectrum. If we indicate the eigenstates for $z=0$ as $|u_n(0)\rangle$, and define their z dependence by

$$|u_n(z)\rangle = U(z)|u_n(0)\rangle, \quad (2.21)$$

then these states $|u_n(z)\rangle$ are solutions of (2.9), obeying the eigenvalue relation

$$N(z)|u_n(z)\rangle = n|u_n(z)\rangle, \quad (2.22)$$

for $n=0, 1, 2, \dots$. Moreover, the phases of these states can be chosen so that

$$A^\dagger(z)|u_n(z)\rangle = \sqrt{n+1}|u_{n+1}(z)\rangle, \quad (2.23)$$

$$A(z)|u_n(z)\rangle = \sqrt{n}|u_{n-1}(z)\rangle.$$

Hence, as for the harmonic oscillator, the higher-order modes can be obtained from the fundamental mode by applying the raising operator, according to

$$|u_n(z)\rangle = \frac{1}{\sqrt{n!}}[A^\dagger(z)]^n|u_0(z)\rangle. \quad (2.24)$$

We have thus proved the existence of a set of normalized solutions $|u_n(z)\rangle$ of (2.9), which are coupled by the operators $A^\dagger(z)$ and $A(z)$ according to the usual relations for the ladder operators of the harmonic oscillator.

These results allow the analytic form of the Hermite-Gaussian modes (2.1) to be explained in terms of the harmonic-oscillator eigenfunctions. In order to show this, we rewrite the operators (2.17) as a transformation of real operators, by using the transformation

$$\exp\left[\frac{ik}{2s}X^2\right]P\exp\left[-\frac{ik}{2s}X^2\right] = P - \frac{k}{s}X, \quad (2.25)$$

which is fully analogous to (2.16). This allows us to write (2.17) in the alternative form

$$A(z) = \exp\left[\frac{ik}{2s}X^2\right]B(z)\exp\left[-\frac{ik}{2s}X^2\right]\exp(i\chi), \quad (2.26)$$

with $B(z)$ the real operator

$$B(z) = \frac{1}{\sqrt{2}}\left[\frac{X}{\gamma} + i\gamma P\right]. \quad (2.27)$$

The conjugate expressions hold for $A^\dagger$ and $B^\dagger$. As is obvious from (2.1), when A operates on a mode $u_n(x, z)$, the real operator B acts on the harmonic-oscillator state ψ_n . The z -dependent quantities s , γ , and χ are given in (2.4)–(2.6). As the higher-order modes $|u_n(z)\rangle$ can be generated from the fundamental one by repeated action of $A^\dagger$, it is sufficient to find the analytical form of the mode function $u_0(x, z)$. This is not difficult if one realizes that the lowering operators $A(z)$ must give zero when operating on the fundamental mode. For the normalized mode function, this gives as a solution of (2.6)

$$u_0(x, z) = \left[\frac{bk}{\pi}\right]^{1/4} \frac{1}{\sqrt{b + iz}} \exp\left[-\frac{kx^2}{2(b + iz)}\right], \quad (2.28)$$

which may be rewritten in terms of the ground state of the harmonic oscillator as

$$u_0(x, z) = \exp\left[\frac{ik}{2s}x^2\right]\exp\left[-\frac{i\chi}{2}\right]\frac{1}{\sqrt{\gamma}}\psi_0\left[\frac{x}{\gamma}\right]. \quad (2.29)$$

The analytical form (2.1) of the higher-order modes follows directly after applying Eq. (2.23) and the recognition that the operator $B^\dagger$ is the raising operator for the harmonic-oscillator eigenfunctions with Gaussian width γ .

B. Coherent states and the Gouy phase

In view of the analogy of the complete set of Hermite-Gaussian modes with a harmonic oscillator, or equivalently with a single quantized field mode, it is natural to consider the analog of a coherent state [9]. Such a state arises from the ground state after the application of a displacement in phase space. The displacement operator

$$D(0) = \exp[iq_0X - ia_0P] \quad (2.30)$$

displaces position over a_0 , and momentum over q_0 , according to the relations [10]

$$D^\dagger(0)XD(0) = X + a_0, \quad D^\dagger(0)PD(0) = P + q_0. \quad (2.31)$$

We consider the mode which, for $z=0$, is equivalent to the displaced ground state

$$|u(0)\rangle = D(0)|u_0(0)\rangle. \quad (2.32)$$

This state is an eigenstate of the lowering operator $A(0)$ with complex eigenvalue

$$\alpha_0 = \frac{1}{\sqrt{2bk}} [ka_0 + ibq_0]. \quad (2.32)$$

The solution of (2.9), which is given by the coherent state (2.31) for $z=0$, can be expressed as

$$|u(z)\rangle = D(z)|u_0(z)\rangle, \quad (2.33)$$

with

$$D(z) = U(z)D(0)U^\dagger(z). \quad (2.34)$$

Obviously, $|u(z)\rangle$ is an eigenvector of $A(z)$ with eigenvalue α_0 . Application of (2.16) shows that $D(z)$ is again a displacement operator, and takes the form

$$D(z) = \exp[iq(z)X - ia(z)P], \quad (2.35)$$

with the z -dependent displacements of position and momentum

$$a(z) = a_0 + \frac{z}{k}q_0, \quad q(z) = q_0. \quad (2.36)$$

Hence the state $|u(z)\rangle$, which is the analog of a coherent state, is simply the fundamental mode $|u_0(z)\rangle$, with the transverse momentum displaced by the constant amount q_0 , and the position by $a(z)$, as given by (2.36). This equation simply describes a ray of light, tilted with respect to the z axis at an angle q_0/k .

In view of the analogy between the Gouy phase χ and the phase of the oscillator, we may expect that the variation of the displacements (2.36) with z corresponds to a variation in time of the displacement of a coherent state of the oscillator during its evolution. This may be illustrated by rewriting the displacement operator in the form

$$D(z) = \exp[\alpha_0 A^\dagger(z) - \alpha_0^* A(z)], \quad (2.37)$$

with α_0 given in (2.32). If we substitute the alternative form (2.25) of A and $A^\dagger$ into (2.37), we obtain an expression for D in terms of the real field operators

$$D(z) = \exp\left[\frac{ik}{2s}X^2\right] \exp[\alpha(z)B^\dagger(z) - \alpha^*(z)B(z)] \times \exp\left[-\frac{ik}{2s}X^2\right], \quad (2.38)$$

with

$$\alpha(z) = \alpha_0 e^{-i\chi}. \quad (2.39)$$

In Eq. (2.39), the z -dependent displacement is described by the dimensionless quantity α , which indicates the relative displacement in units of γ . In contrast, the displacements in position and momentum in Eq. (2.35) are indicated by a and q on an absolute scale. Hence the Gouy phase jump over π near a focus, which corresponds to an oscillation over half a cycle of the relative displacement, corresponds to the rectilinear motion of the light on an absolute scale.

III. IDEAL LENSES

For the description of experiment, it is necessary to extend the field-operator description by including optical elements, such as lenses and diaphragms. An ideal lens is sufficiently thin so that no propagation occurs in the lens. Its only effect is to add an x -dependent phase factor to the field. When a lens with focal length f with its center on the z axis is located at the position z_1 , the relation between the field incident on the lens and the outgoing field is given [8] by

$$u(x, z_+) = \exp\left[-\frac{ik}{2f}x^2\right] u(x, z_-), \quad (3.1)$$

with z_+ (z_-) a position immediately behind (before) the lens. This local phase jump can be described by writing for the propagation operator

$$U(z_+) = \exp\left[-\frac{ik}{2f}X^2\right] U(z_-). \quad (3.2)$$

When the lens at position z_1 is the only one in the interval $[0, z]$, the propagation operator is equal to

$$U(z) = \exp\left[-\frac{i}{2k}P^2(z - z_1)\right] \exp\left[-\frac{ik}{2f}X^2\right] \times \exp\left[-\frac{i}{2k}P^2z_1\right], \quad (3.3)$$

for $z \geq z_1$. In this way, we can compose the propagation operator for an optical axis with an arbitrary set of lenses, with free propagation in between lenses.

A. Ladder operators and fundamental mode

If we start with a definition of a lowering operator $A(0)$, as in Eq. (2.14), then the propagation operator U defines the field operators for all values of z , according to (2.15). As the commutation relation (2.18) is unaffected by the presence of lenses, we still have the fundamental mode as an eigenstate of $A(z)$ with eigenvalue zero, together with the higher-order modes, which are related by (2.22). The analytical expressions are modified to account for the presence of lenses, but the operator algebra remains the same.

It is important to recognize that the explicit analytical expressions for all Hermite-Gaussian modes are fully determined by the radius of curvature s , the spot size γ , and the phase χ . The z dependence of these three parameters, which fully describe the fundamental mode for a given configuration of lenses, defines the framework of the operator algebra. To understand this, it is sufficient to note that a field operator A , defined as an arbitrary linear combination of the operators X and P for a single z value, keeps a similar form for all positions. This is obvious from the transformation rules (2.16) and (2.24). Physically, this reflects the fact that a Gaussian beam remains Gaussian during free propagation and in passage through (ideal) lenses.

If an optical axis with an arbitrary number of lenses of arbitrary focal length is considered, we can define the

lowering operator A at one position as a linear combination of X and P , and define the z dependence of $A(z)$ in terms of the propagation operator, as in (2.15). Then A retains a similar form for all z , and we may write

$$A(z) = \frac{1}{\sqrt{2}} [\kappa(z)X + i\beta(z)P], \quad (3.4)$$

with κ and β complex-valued functions of z . In order that A and $A^\dagger$ obey the commutation rule (2.18), κ and β must be related by the equality

$$\text{Re}\kappa\beta^* = 1. \quad (3.5)$$

Upon inversion, Eq. (3.4) takes the form

$$X = \frac{1}{\sqrt{2}} [\beta^* A + \beta A^\dagger], \quad P = \frac{1}{i\sqrt{2}} [\kappa^* A - \kappa A^\dagger]. \quad (3.6)$$

In an interval without lenses, where free propagation occurs, the z dependence of κ and β may be determined using the transformation (2.16). We find that κ is constant in such an interval, whereas β varies linearly with z . The derivatives are

$$\frac{d\beta}{dz} = \frac{i\kappa}{k}, \quad \frac{d\kappa}{dz} = 0. \quad (3.7)$$

On the other hand, application of (2.24) shows that a lens with focal length f does not change β , but it does cause a jump in κ . The result is

$$\beta(z_+) = \beta(z_-), \quad \kappa(z_+) = \kappa(z_-) + \frac{i\kappa\beta}{f}. \quad (3.8)$$

Hence, β is a continuous function that varies linearly between lenses with a slope proportional to κ . Conversely, κ makes jumps at the lens positions with a jump size that depends on the local value of β . In the absence of any lens, we again find the ladder operators to be given by (2.17).

The z -dependent lowering operator A has an eigenvector $|u_0(z)\rangle$ with eigenvalue zero, which is a solution of (2.7) with a propagation operator that is modified by the lenses. This light beam is the fundamental Gaussian mode for this arbitrary lens configuration, and its normalized wave function is equal to

$$u_0(x, z) = [\beta\sqrt{\pi}]^{-1/2} \exp\left[-\frac{\kappa x^2}{2\beta}\right]. \quad (3.9)$$

It is important to note that the intensity distribution has width $|\beta|$, whereas the momentum distribution, which is determined by the Fourier transform of (3.9), has width $|\kappa|$. When κ and β vary according to the rules (3.6) and (3.7), this expression obeys the evolution equation (2.6) between two lens positions and makes the phase jump (3.1) at the lens positions. This result generalizes that for which no lenses are present given in (2.27).

B. Higher-order modes

In order to obtain the analytic form for the higher-order modes, it is convenient to express the lowering operator (3.4) and the fundamental mode function (3.9) in terms of the radius of curvature s , the spot size γ , and the

phase χ , as in (2.25) and (2.28). The quantities β and κ , with the normalization (3.5), uniquely determine the values of the fundamental mode parameters s , γ , and χ , by the relations

$$s = -k \left[\text{Im} \frac{\kappa}{\beta} \right]^{-1}, \quad \gamma = |\beta|, \quad \chi = \arg \beta. \quad (3.10)$$

These results generalize (2.4) and (2.5). In the intervals between lenses, the variation of s , γ , and χ is determined by the linear variation of β with z , as expressed in (3.7). In the region of negative values of s , the beam converges and γ decreases, while the beam diverges where s is positive. A focus occurs where κ has the same argument as β , so that $s=0$. The phase angle χ can only increase, and it varies most rapidly at a focus. Since a lens does not change β , both γ and χ remain unmodified by the lens. According to (3.8), κ/β changes by ik/f , and the only effect of the lens is a change of the radius of curvature,

$$\frac{1}{s(z_+)} = \frac{1}{s(z_-)} - \frac{1}{f}, \quad (3.11)$$

in accordance with (3.1). This is just the lens formula of geometric optics. Since the expressions (2.25)–(2.27) remain valid, we can apply the raising operator to the fundamental mode repeatedly to obtain the higher-order modes. We conclude that they are given by the same expression (2.1) as in the case of free propagation, but with the behavior of the fundamental mode parameters as determined in this section. In the presence of lenses, the spot size γ can pass through several foci. Correspondingly, the Gouy phase χ can execute a series of π jumps. This is equivalent to the harmonic oscillator, which is the analog of the light beam in the presence of lenses, performing a number of cycles. The algebraic connection between the modes of various orders is completely unaffected by these changes in the three fundamental mode parameters.

C. Coherent states and ray optics

The discussion of Sec. II B on displaced fundamental modes in analogy to coherent states remains largely valid in the presence of lenses. The displacement of the spot at $z=0$, as described by $D(0)$ defined in (2.29), leads to a displacement for arbitrary z , where the operator $D(z)$ is still defined in (2.34). The only difference is that now the propagation operator $U(z)$ is changed. Likewise, the expression (2.37) for $D(z)$ in terms of the z -dependent field operators A and $A^\dagger$ remains valid, and, moreover, these operators still obey Eqs. (2.25) and (2.26), and (2.38) and (2.39) still hold. Hence the expression for the displacement on the relative scale in units of γ remains the same. The effect of the lenses is completely hidden in the modified behavior of the fundamental mode parameters s , γ , and χ .

The refraction by the lenses becomes directly obvious when we express the displacement on the absolute scale, by rewriting $D(z)$ in the form (2.35) in terms of a momentum displacement $q(z)$ and a position displacement $a(z)$. In an interval between two lenses, where free propagation

occurs, the z dependence of q and a is determined by the transformation (2.16). One finds that a varies linearly with z , whereas q is constant. The derivatives are

$$\frac{da}{dz} = \frac{q}{k}, \quad \frac{dq}{dz} = 0. \quad (3.12)$$

Applying (3.11) shows that the effect of a lens with focal length f is described by the transformation

$$a(z_+) = a(z_-), \quad q(z_+) = q(z_-) - \frac{ka}{f}. \quad (3.13)$$

One should recall that q/k has the significance of the angle of a light ray with the z axis, whereas a is the distance of the ray from the axis at the position z . Hence Eq. (3.12) describes the rectilinear motion of a ray of light in free space, while (3.13) describes the refraction of the ray by a lens with focal length f .

Hence, the behavior of the axis of a beam in a displaced fundamental mode (the analog of a coherent state) simply obeys the laws of geometric optics. This behavior is completely independent of the variation of the fundamental beam parameters s , γ , and χ , which are a reflection of the wave-optical nature of the beam. The variation of these beam parameters is governed by the transformation (3.7) and (3.8) for β and κ , which strongly resemble the geometric-optical rules (3.12) and (3.13). This is no accident, as both pairs of rules follow from the transformations (2.16) and (2.24). In summary, a Gaussian beam is completely specified by its first and second moments. The first moments give the average position and momentum: namely a and q . They determine the geometric-optical properties of the beam. The second moments, which are specified by the complex quantities β and κ , determine the position and momentum widths of the fundamental mode, and describe the wave aspect of the beam.

IV. ORBITAL ANGULAR MOMENTUM

In this section, we apply the operator formalism to describe the orbital angular momentum of a paraxial beam of light. This requires the explicit consideration of the two transverse dimensions. Then we have two momentum operators P_x and P_y for the x and the y direction, and two position operators X and Y . We introduce the operator

$$L = XP_y - YP_x, \quad (4.1)$$

in analogy to the quantum operator of orbital angular momentum. In cylindrical coordinates r and ϕ , with

$$x = r \cos \phi, \quad y = r \sin \phi, \quad (4.2)$$

L takes the well-known form

$$L = \frac{1}{i} \frac{\partial}{\partial \phi}. \quad (4.3)$$

The total angular momentum of a radiation field is given by Maxwell's theory in the form [11]

$$\mathbf{J} = \epsilon_0 \int d\mathbf{r} \{ \mathbf{r} \times [\mathbf{E}(\mathbf{r}) \times \mathbf{B}(\mathbf{r})] \}. \quad (4.4)$$

This angular momentum can be separated into a term resulting from the derivatives of the amplitudes, and a term that derives from the polarization of the field. With some caution, this may be viewed as a separation into orbital angular momentum and spin of the field [12]. For a monochromatic field in the paraxial approximation (2.7), the z component J_z is separated as [4]

$$J_z = \frac{\epsilon_0}{2\omega} \int dx dy dz \left[\mathbf{E}_0^* \cdot \mathbf{E}_0 u^* Lu + \frac{1}{i} u^* u (E_{0x}^* E_{0y} - E_{0y}^* E_{0x}) \right]. \quad (4.5)$$

This implies that for an arbitrary normalized mode $|u(z)\rangle$, the expectation value $\hbar \langle L \rangle$ is equal to that of the orbital angular momentum in the z direction per photon [4].

For each transverse dimension, we have independent values for the fundamental mode parameters s , γ , and χ , which we distinguish with suffices x and y . The corresponding lowering operators A_x and A_y are determined by these parameters, as in (2.25), in terms of the real operators B_x and B_y . In the presence of cylindrical lenses, the operators for the two dimensions can be treated independently only when the lens axes coincide with the x and the y axis, which is what will be assumed from now on. Hence the spot size and the wave front will, in general, be elliptical and with coinciding axes. We exclude for the moment the general astigmatism that arises by using cylindrical lenses oriented at oblique angles to each other [13]. Then the Hermite-Gaussian modes $|u_{nn},(z)\rangle$ have mode functions $u_{nn},(\mathbf{r}) = u_n(x,z)u_n(y,z)$, which are products of one-dimensional modes. All higher-order modes can then be generated by repeated action of ladder operators on the fundamental mode $|u_{00}(z)\rangle$.

Laguerre-Gaussian modes are the laser mode analog of the angular-momentum eigenstates of the isotropic two-dimensional harmonic oscillator. It has been shown that a higher-order Hermite-Gaussian beam can be converted into a Laguerre-Gaussian beam by using astigmatic lenses [14–16]. We wish to demonstrate that this and similar conversions are conveniently understood by operator algebra. First, we give the connection between these modes in an operator form.

A. Hermite-Gaussian, Laguerre-Gaussian, and elliptical Gaussian modes

Generalizing the algebraic treatment of the two-dimensional harmonic oscillator [17], we introduce mixed ladder operators by the definition

$$A_{\pm} = \frac{1}{\sqrt{2}} [A_x \mp i A_y e^{i\theta}], \quad (4.6)$$

in terms of a phase θ that is independent of z . These operators obey the commutation rules

$$[A_{\pm}(z), A_{\pm}^{\dagger}(z)] = 1, \quad [A_{\pm}(z), A_{\mp}^{\dagger}(z)] = 0. \quad (4.7)$$

The operators for the sum and the difference of the number operators are

$$N(z) = N_+(z) + N_-(z), \quad M(z) = N_+(z) - N_-(z), \quad (4.8)$$

with

$$N_{\pm}(z) = A_{\pm}^{\dagger}(z) A_{\pm}(z). \quad (4.9)$$

In terms of the field operators A_x and A_y , the operators N and M are

$$N = A_x^{\dagger} A_x + A_y^{\dagger} A_y, \quad M = i A_y^{\dagger} A_x e^{-i\theta} - i A_x^{\dagger} A_y e^{i\theta}. \quad (4.10)$$

Because N and M commute, they have a common basis of eigenstates $|u_{nm}(z)\rangle$, with eigenvalues n and m . As A_+ or A_- decrease the eigenvalue of N_+ and N_- by one unit, the lowering operators $A_{\pm}$ both decrease the total excitation number n , whereas A_+ decreases m and A_- increases m by one unit. This is apparent from the commutation rules

$$[N(z), A_{\pm}(z)] = -A_{\pm}(z), \quad [M(z), A_{\pm}(z)] = \mp A_{\pm}(z). \quad (4.11)$$

With the proper phase convention of the eigenfunctions, the explicit action of the ladder operators may be given by

$$\begin{aligned} A_{\pm}(z) |u_{nm}(z)\rangle &= \left[\frac{n \pm m}{2} \right]^{1/2} |u_{n-1, m \mp 1}(z)\rangle, \\ A_{\pm}^{\dagger}(z) |u_{nm}(z)\rangle &= \left[\frac{n \pm m + 2}{2} \right]^{1/2} |u_{n+1, m \pm 1}(z)\rangle. \end{aligned} \quad (4.12)$$

The total mode number n can take all natural values $0, 1, 2, \dots$, and for each value of n , m can take the $n+1$ values $-n, -n+2, \dots, n-2, n$. The mode $|u_{nm}\rangle$ is created from the fundamental mode by applying the raising operators $A_+^{\dagger}$ and $A_-^{\dagger}$, and one finds

$$|u_{nm}(z)\rangle = \left[\frac{1}{p!q!} \right]^{1/2} [A_-^{\dagger}(z)]^q [A_+^{\dagger}(z)]^p |u_{00}(z)\rangle, \quad (4.13)$$

where we introduced the integers p and q , defined by

$$p = \frac{n+m}{2}, \quad q = \frac{n-m}{2} \quad (4.14)$$

in terms of n and m . This result allows the expansion of the eigenfunctions $u_{nm}(\mathbf{r})$ of N and M in terms of the Hermite-Gaussian modes. If we substitute the definitions (4.6) into (4.13), the brackets can be worked out in terms of the coefficients g_s , which are defined by the expansion

$$(1-t)^q (1+t)^p = \sum_{s=0}^n g_s t^s. \quad (4.15)$$

As the fundamental mode u_{00} is the product of two one-dimensional fundamental modes, we may directly apply (2.23) for each transverse dimension, with the result

$$\begin{aligned} u_{nm}(\mathbf{r}) &= \sum_{s=0}^{p+q} g_s i^s e^{-is\theta} \left[\frac{(p+q-s)! s!}{2^{p+q} p! q!} \right]^{1/2} \\ &\quad \times u_{p+q-s}(x, z) u_s(y, z). \end{aligned} \quad (4.16)$$

The equalities (4.10)–(4.13) and (4.16) hold for any beam specified by its fundamental mode parameters. In the special case of a nonastigmatic region of the beam, where $\gamma_x = \gamma_y = \gamma$ and $s_x = s_y = s$, the phases χ_x and χ_y can only differ by a constant. Then the expression for $A_{\pm}$ in terms of the real ladder operators B_x and B_y takes the form

$$A_{\pm} = e^{i\chi_x} \exp \left[\frac{ik}{2s} (X^2 + Y^2) \right] B_{\pm} \exp \left[-\frac{ik}{2s} (X^2 + Y^2) \right]. \quad (4.17)$$

Here we denote

$$B_{\pm} = \frac{1}{\sqrt{2}} (B_x \mp i e^{i\zeta} B_y), \quad (4.18)$$

where B_x and B_y are determined by the spot sizes $\gamma_x = \gamma_y = \gamma$, in analogy to (2.26), and where

$$\zeta = \chi_y - \chi_x + \theta. \quad (4.19)$$

As the fundamental mode function takes the form

$$\begin{aligned} u_{00}(\mathbf{r}) &= \exp \left[\frac{ikr^2}{2s} \right] \exp \left[-\frac{i}{2} (\chi_x + \chi_y) \right] \\ &\quad \times \frac{1}{\gamma} \psi_0 \left[\frac{x}{\gamma} \right] \psi_0 \left[\frac{y}{\gamma} \right], \end{aligned} \quad (4.20)$$

operating on u_{00} with $A_{\pm}$ is equivalent to operating on the harmonic-oscillator ground state with $B_{\pm}$. This demonstrates that it is the value of ζ that determines the nature of the mode $|u_{nm}\rangle$ in a nonastigmatic region.

When $\zeta = \pi/2$, the operators $B_{\pm}$ are equal to

$$B_{\pm} = \frac{1}{\sqrt{2}} [B_x \pm B_y]. \quad (4.21)$$

Substituting (4.17) and (4.20) in (4.13) then gives for the higher-order mode functions

$$\begin{aligned} u_{nm}(\mathbf{r}) &= \exp \left[\frac{ik}{2s} r^2 - \frac{i}{2} (\chi_x + \chi_y) - i\chi_x(p+q) \right] \\ &\quad \times \frac{1}{\gamma} \psi_p \left[\frac{x+y}{\gamma\sqrt{2}} \right] \psi_q \left[\frac{x-y}{\gamma\sqrt{2}} \right], \end{aligned} \quad (4.22)$$

with p and q defined by (4.14). The functions (4.22) are normalized Hermite-Gaussian modes with their symmetry axis rotated over 45° . If we combine (4.22) with (4.16), we obtain the transformation of these rotated Hermite-Gaussian modes in the nonrotated Hermite-Gaussian modes. In the special case that $\chi_x = \chi_y$, so that $\theta = \zeta = \pi/2$, this is equivalent to an analytical expansion of products of Hermite polynomials in $(x \pm y)/\sqrt{2}$, as given by Abramochkin and Volostnikov [14].

When $\zeta = 0$, we find that

$$B_{\pm} = \frac{1}{\sqrt{2}}[B_x \mp iB_y], \quad (4.23)$$

which are circular ladder operators. It is easy to verify that, in this case, the operator M is equal to the angular momentum L , and the quantum number $m = l$ is the corresponding eigenvalue. The beam carries an orbital angular momentum $\hbar l$ per photon [3]. For the mode functions, we obtain, after using (4.15),

$$u_{nl}(\mathbf{r}) = \exp\left[\frac{ik}{2s}r^2 - i\chi(n+1)\right] \frac{1}{\gamma} \psi_{nl}\left[\frac{x}{\gamma}, \frac{y}{\gamma}\right], \quad (4.24)$$

with ψ_{nl} the wave functions of the isotropic harmonic oscillator that are eigenfunctions of both energy and angular momentum, with quantum numbers n and l . The azimuthal dependence of ψ_{nl} is given by $\exp(il\phi)$, and its radial part has the form [17,2]

$$\exp(-r^2/2\gamma^2) r^{|l|} L_p^{|l|}(r^2/\gamma^2),$$

with $p = (n - |l|)/2$, and $L_p^{|l|}$ the generalized Laguerre polynomial. As both the Laguerre-Gaussian and the Hermite-Gaussian modes form a complete set, a basis transformation must exist. The analytical derivation of this transformation is painstaking and not very transparent [15]. The transformation, obtained by operator algebra, is given by (4.16) in the special case that $\theta = 0$. A similar result has recently been found by group-theoretic methods by Danakas and Aravind [18].

It is natural to compare the Laguerre-Gaussian modes to circular polarization and the Hermite-Gaussian modes to linear polarization. We have seen that they arise as special cases of the modes $|u_{nm}\rangle$ defined by (4.13) with (4.17) and (4.18), for $\zeta = 0$ and $\zeta = \pi/2$, respectively. For intermediate values of ζ , the modes $|u_{nm}\rangle$ have an elliptical nature. This is obvious when we rewrite (4.18) in the form

$$B_{\pm} = \left[\frac{1}{\sqrt{2}}(B_x \pm iB_y) \cos\left[\frac{\zeta}{2} - \frac{\pi}{4}\right] - \frac{i}{\sqrt{2}}(B_x \mp iB_y) \sin\left[\frac{\zeta}{2} - \frac{\pi}{4}\right] \right] \times \exp\left[i\left[\frac{\zeta}{2} - \frac{\pi}{4}\right]\right]. \quad (4.25)$$

The corresponding creation operators $A_{\pm}^{\dagger}$ acting on the fundamental mode generate elliptical patterns of the transverse momentum density, with the axes oriented in the xy plane along the lines $x = \pm y$. This ellipticity should not be confused with the elliptic intensity distribution in the case of an astigmatic beam.

B. Mode conversion

In general, a mode converter is defined as a configuration of lenses that transforms a nonastigmatic input beam into a nonastigmatic output beam with an astigmatic region between them. Let us consider a configuration of astigmatic lenses with their elliptical axes oriented along the x and the y axis. The lenses are

located between z_1 and z_2 . The incoming beam for $z \leq z_1$ is supposed to be nonastigmatic, so that $\gamma_x = \gamma_y = \gamma$ and $s_x = s_y = s$. The lens configuration determines the values of the fundamental mode parameters s_x , γ_x , χ_x and s_y , γ_y , and χ_y for all values of z . The output beam for $z \geq z_2$ is also nonastigmatic, provided that the last lens is located at a position where $\gamma_x = \gamma_y$, and that it has different focal lengths f_x and f_y along the two axes that make up for a possible difference in s_x and s_y , according to

$$\frac{1}{s_x(z_-)} - \frac{1}{f_x} = \frac{1}{s_y(z_-)} - \frac{1}{f_y}. \quad (4.26)$$

As the characteristics of the mode $|u_{nm}\rangle$ in a nonastigmatic region are determined by the value of the parameter ζ , the properties of a converter are fully specified by the difference between the ζ values in the output and the input beam, which is equal to

$$\Delta\zeta = (\zeta)_{\text{out}} - (\zeta)_{\text{in}} = (\chi_x - \chi_y)_{\text{out}} - (\chi_x - \chi_y)_{\text{in}}. \quad (4.27)$$

Consider a Hermite-Gaussian input beam with its axes oriented at 45° with respect to the axes of the lenses, so that

$$u_{\text{in}}(z) = u_p\left[\frac{x+y}{\sqrt{2}}, z\right] u_q\left[\frac{x-y}{\sqrt{2}}, z\right], \quad (4.28)$$

with u_p and u_q the one-dimensional Hermite-Gaussian mode functions (2.1). According to (4.22), this input mode is identical to the mode $|u_{nm}\rangle$ with $\zeta = \pi/2$, and with $n = p + q$, and $m = p - q$. In order that the output beam be in the Laguerre-Gaussian mode (4.24), we must have $(\zeta)_{\text{out}} = 0$, or $\Delta\zeta = -\pi/2$. The angular-momentum quantum number is $l = p - q$. Such a $\pi/2$ converter has recently been realized with a system consisting of two identical cylindrical lenses [16]. Then the Rayleigh range is the same at input and output. As the Gouy phases χ_x and χ_y cannot increase by more than π in a region of free propagation, the value of $\Delta\zeta$ for a system of only two lenses must obey the inequality $-\pi < \Delta\zeta < \pi$. For a system consisting of more lenses, $\Delta\zeta$ can be outside this range and the Rayleigh range can be different at input and output. Inspection of (4.17) and (4.18) shows that for $\zeta = \pi$, the operator M is equal to $-L$. Hence a lens system with $\Delta\zeta = \pi$ inverts the azimuthal mode index l of a Laguerre-Gaussian beam. It cannot, however, be realized with less than three lenses [4].

Our treatment demonstrates that the conversion properties of an arbitrary astigmatic lens system with a single axis with a nonastigmatic input and output is determined by the single parameter $\Delta\zeta$, which is the difference of the change in Gouy phase for the two axes of the lens system. As indicated in (4.18), this phase determines the change in the ladder operator $B_{\pm}$ from the input to the output region. An elliptical Gaussian beam specified by a value of ζ is realized by a converter with an appropriately selected value of $\Delta\zeta$ when the input beam is Hermite-Gaussian or Laguerre-Gaussian. The action of the converter results from the modification of the eigenoperator M , while the eigenvalues n and m remain unchanged. The propagation of the beam through the lens system is

fully analogous to the evolution of a two-dimensional harmonic oscillator, with a modification of the oscillation frequencies Ω_x and Ω_y . The change in the phase χ over a propagation distance plays the role of the oscillator phase $\int dt \Omega(t)$ over a time interval. In each case, it is the total phase difference between the two directions that determines the state change for an arbitrary initial state. When the mode function is simply the product of one-dimensional functions for the x and the y direction, this phase difference does not affect the nature of the mode. It only plays a role for an entangled state, which is a linear combination of different products of one-dimensional eigenstates for the two axes. This explains why an input Hermite-Gaussian mode must have axes that do not coincide with the symmetry axes of the lenses, in order for mode conversion to take place.

C. Angular-momentum transfer

It is informative to derive expressions for the transfer of orbital angular momentum when a cylindrical lens is traversed by a possibly astigmatic Gaussian beam that is an eigenfunction of N . Again, clearly, it is necessary to consider both transverse directions. The transverse field amplitude for a given position z of the beam is generally described by the function $u(x, y)$, that can be separated as

$$u(\mathbf{r}) = \exp \left[\frac{ik}{2} \left(\frac{x^2}{s_x} + \frac{y^2}{s_y} \right) \right] \frac{1}{\sqrt{\gamma_x \gamma_y}} \Psi(\xi, \eta) \quad (4.29)$$

We introduced the scaled transverse coordinates and momenta

$$\xi = x/\gamma_x, \quad \eta = y/\gamma_y, \quad P_\xi = \frac{1}{i} \frac{\partial}{\partial \xi}, \quad P_\eta = \frac{1}{i} \frac{\partial}{\partial \eta}, \quad (4.30)$$

so that Ψ is the normalized wave function of an energy eigenstate of a two-dimensional isotropic harmonic oscillator. As we are interested only in the distribution at a given position z , we suppressed the z dependence in (4.29). The Gouy phases χ_x and χ_y are absorbed in Ψ . When γ_x and γ_y are different, the Gaussian part of u has unequal widths along the two axes, whereas the function Ψ is unsqueezed. If we evaluate the orbital angular momentum per photon $\hbar \langle L \rangle = \hbar \langle u | L | u \rangle$ for the beam (4.29) while using (2.24), we can express $\langle L \rangle$ as an expectation value over the wave function Ψ in terms of the scaled coordinates, with the result

$$\begin{aligned} \langle L \rangle &= k \gamma_x \gamma_y \left[\frac{1}{s_y} - \frac{1}{s_x} \right] \langle \Psi | \xi \eta | \Psi \rangle \\ &+ \frac{1}{2} \left[\frac{\gamma_x}{\gamma_y} - \frac{\gamma_y}{\gamma_x} \right] \langle \Psi | (\xi P_\eta + \eta P_\xi) | \Psi \rangle \\ &\times \frac{1}{2} \left[\frac{\gamma_x}{\gamma_y} + \frac{\gamma_y}{\gamma_x} \right] \langle \Psi | (\xi P_\eta - \eta P_\xi) | \Psi \rangle. \end{aligned} \quad (4.31)$$

The first term on the right-hand side of (4.31) results from the difference in the two radii of curvature. This reflects the ellipticity of the wave front. The second term is due to the difference in spot size and results from the

elliptical shape of the light spot. The last term contains the angular momentum in scaled coordinates. It is the only surviving term in a nonastigmatic beam. Note that for a freely propagating astigmatic beam, all three terms in (4.31) vary with z , whereas their sum remains constant.

If the beam described by u passes a cylindrical lens, with focal length f in the x direction, the field leaving the lens is given by $u(\mathbf{r}) \exp(-ikx^2/2f)$. Hence the last two terms in (4.31) are unaffected and only the first one changes, due to the change in s_x . The net increase of angular momentum per photon passing the lens is

$$\hbar \langle \delta L \rangle = \frac{\hbar k}{f} \gamma_x \gamma_y \langle \Psi | \xi \eta | \Psi \rangle = \frac{\hbar k}{f} \langle u | xy | u \rangle. \quad (4.32)$$

This term determines the torque that the field exerts on the lens. It vanishes when the intensity distribution $|u|^2$ has the x or the y axis as symmetry axis. In particular, the torque on the lens vanishes when the input beam is an astigmatic Hermite-Gaussian mode $u_n(x, z)u_m(y, z)$. This is understandable, as the lens orientation is at an extremum of energy. Furthermore, the angular-momentum transfer (4.32) vanishes for a Laguerre-Gaussian input mode, and for symmetry reasons [4] it will also vanish when the beam leaving the lens is Laguerre-Gaussian. Notice that this angular-momentum transfer is proportional to the wave number k . Therefore, when this matrix element is nonvanishing, it can be as large as many units $\hbar$, even for modes of moderate (nonzero) order.

D. Angular aperturing and angular momentum

In Sec. IV C, it was argued that a Laguerre-Gaussian beam cannot transfer angular momentum to an astigmatic lens. However, when angular aperturing is applied just in front of the lens, an appreciable transfer of angular momentum can occur. Suppose that the Laguerre-Gaussian mode (4.24) passes a diaphragm with a transmittance that depends only on the azimuthal angle ϕ , and not on the radial coordinate r . Then the mode function leaving the diaphragm can be given in polar coordinates as

$$\begin{aligned} u(\mathbf{r}) &= \exp \left[\frac{ik}{2s} r^2 - i\chi(n+1) \right] \frac{1}{\gamma \sqrt{2\pi}} R_{nl}(r/\gamma) \\ &\times e^{il\phi} A(\phi), \end{aligned} \quad (4.33)$$

with R_{nl} the normalized radial-wave function of the angular-momentum eigenstates ψ_{nl} of the two-dimensional harmonic oscillator. For an ideal diaphragm, the transmittance function $A(\phi)$ attains only the value zero or 1. When a cylindrical lens with focal length f in the x direction is placed immediately after the diaphragm, so that no propagation occurs between them, the transfer of angular momentum to the lens is found after substitution of (4.33) in (4.32). The result is the product of a radial and an angular average. By using the well-known fact that the average potential and kinetic energy are equal in an energy eigenstate of the harmonic oscillator, we obtain for the transfer of angular momentum per incident photon the expression

$$\hbar \langle \delta L \rangle = \frac{\hbar k}{2f} \gamma^2 (n+1) \frac{1}{2\pi} \int_0^{2\pi} d\phi |A(\phi)|^2 \sin(2\phi). \quad (4.34)$$

This result is independent of the azimuthal mode index l , and it is proportional to a Fourier coefficient of the ϕ -dependent transmission function $|A(\phi)|^2$. The angular momentum transferred to the lens can be much larger than the angular momentum per photon $\hbar l$ in the incident beam. The transfer is maximal when $|A(\phi)|^2$ is 1 for $0 < \phi < \pi/2$, and $\pi < \phi < 3\pi/2$, and zero elsewhere.

V. CONCLUSIONS

The propagation of Gaussian laser beams in free space, and their refraction by ideal astigmatic lenses, are described by harmonic-oscillator operator algebra. A mode is characterized by z -dependent eigenoperators and constant eigenvalues. This explains why the Hermite-Gaussian form of a beam is conserved during its free propagation and at refraction by lens systems. Moreover, it shows that the displacement of the axis of a slightly misaligned beam follows the rules of geometric optics, which does not affect the beam diffraction. Such a displaced beam is the analog of a coherent state of the harmonic oscillator. The relationship between Laguerre-Gaussian and Hermite-Gaussian modes takes a simple

algebraic form. Conversion from one mode into another by astigmatic lenses is directly characterized by a single phase difference. The orbital angular momentum of a light beam in its propagation direction is separated into a contribution that resembles the angular momentum of a harmonic oscillator and terms that originate from the astigmatism of the beam. These latter terms can be considerably larger than the first. The torque of a beam exerted on an astigmatic lens is expressed in terms of a single expectation value. When a Laguerre-Gaussian beam, with angular momentum $\hbar l$ per photon, traverses an astigmatic lens after passing an angular aperture, it may transfer more angular momentum to the lens than $\hbar l$ per photon. Apart from the physical interest of these results, they demonstrate the advantage of the algebraic method in terms of ladder operators for paraxial optics.

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Matrix formulation for the propagation of light beams with orbital and spin angular momenta

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Jones matrices describe the polarization, or spin angular momentum, of a light beam as it passes through an optical system. We devise an equivalent of the Jones matrix formulation for light possessing orbital angular momentum. The matrices are then developed to account for light that has both spin and orbital angular momentum. [S1063-651X(99)07412-7]

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The behavior of polarized light as it passes through birefringent optical components is well described by an approach devised by Jones [1]. The beam is represented in terms of a column vector, the Jones vector, the elements of which are the complex amplitudes of the orthogonally polarized components. Optical components such as linear and circular polarizers, phase retarders, etc., are represented by 2×2 matrices called Jones matrices: matrix multiplication accounts for the passage of the light through a sequence of optical elements. An alternative formulation derived by Mueller [2], based on the Stokes parameters, is not able to deal fully with the superposition of coherent beams, but has the advantage of being applicable to partially as well as fully polarized light. A full account of both the Jones and Mueller matrices and their application, has been given by Gerrard and Burch [3].

Polarized light is associated with spin angular momentum. It is now well established [4–10] that light beams may possess well defined orbital angular momentum. Beams with an azimuthal phase dependence of $\exp(il\phi)$, such as Laguerre Gaussian beams, have an orbital angular momentum of $l\hbar$ per photon [4]. Such modes may be readily created in the laboratory by passing the Hermite Gaussian modes usually emitted by lasers, through a mode converter [5] which consists of two canonically disposed cylindrical lenses oriented at 45° to the axes of the mode. The indices (n, m) characterizing the Hermite Gaussian modes give the indices (l, p) of the Laguerre Gaussian modes, where $l = |n - m|$ and $p = \min(m, n)$ [4].

Hermite Gaussian and Laguerre Gaussian modes both form complete, orthogonal, basis sets from which any arbitrary field distribution may be described. The order of a mode is defined [5] by $N = n + m = 2p + |l|$ and does not change as the mode is converted or rotated. There is an exact analogy between a waveplate for polarized light and a mode converter; see Fig. 1. Just as a phase shift is introduced between orthogonal polarization components by birefringent waveplates, so mode converters based on cylindrical lenses introduce a phase shift between orthogonal modes of the same order. This equivalence allows polarization and mode structure to be treated in similar ways and leads to a formulation, analogous to the Jones matrix approach, for modes of order N .

The Poincaré sphere is an equivalent representation to the Jones matrix formulation for polarized light. Any state of polarization may be represented by a point on the sphere; waveplates and other polarizing elements then move the polarization state to another position on the surface of the sphere. It has been shown [11], that there exists an orbital angular momentum equivalent to the Poincaré sphere for modes of order $N = 1$. A Poincaré sphere approach to arbi-

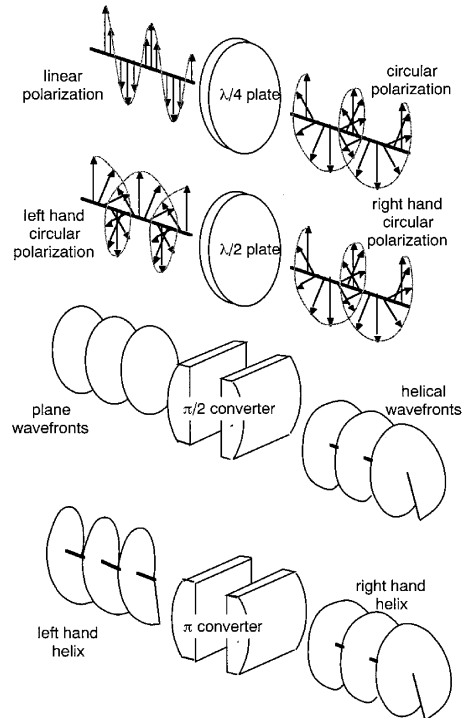


FIG. 1. Quarter and half wave plates and $\pi/2$ or π mode converters play equivalent roles for spin and orbital angular momentum.

TABLE I. $[(N+1) \times (N+1)]$ matrices for modes of order N .

$\pi/2$ converter									
$[C(\pi/2)] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & -i & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & -1 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & i & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & -i & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$									
π converter									
$[C(\pi)] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & -1 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 1 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & -1 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & -1 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$									
Mode filter (e.g. $HG_{N-2,2}$)									
$[F(N-2,2)] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 1 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$									
Mode rotation matrix for $N=1$									
$[\text{rot}(\phi)] = \begin{bmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{bmatrix}$									
Mode rotation matrix for $N=2$									
$[\text{rot}(\phi)] = \begin{bmatrix} \cos^2 \phi & \frac{\sin 2\phi}{\sqrt{2}} & \sin^2 \phi \\ \frac{-\sin 2\phi}{\sqrt{2}} & \cos 2\phi & \frac{\sin 2\phi}{\sqrt{2}} \\ \sin^2 \phi & \frac{-\sin 2\phi}{\sqrt{2}} & \cos^2 \phi \end{bmatrix}$									

trary order of N is, however, not easy to devise as higher order geometries are required. This is not a difficulty for a matrix formulation.

In this paper we show that just as there is a two element Jones column vector and a set of 2×2 matrices for polarized light, so there is an $(N+1)$ -element column vector and a set

of $[(N+1) \times (N+1)]$ matrices to describe the passage of modes of order N , which may possess orbital angular momentum, through a series of optical components such as mode converters and beam rotators. For these matrices to apply, N must be conserved and the modes must have the same Rayleigh range and beam waist. Most imaging systems

TABLE 1 (continued).

Mode rotation matrix for $N=3$				
$[\text{rot}(\phi)] =$	$\cos^3 \phi$	$\frac{\sqrt{3}\cos^2 \phi \sin \phi}{4}$	$\frac{\sqrt{3}\cos \phi \sin^2 \phi}{4}$	$\sin^3 \phi$
	$-\sqrt{3}\cos^2 \phi \sin \phi$	$\frac{\cos \phi + 3 \cos 3\phi}{4}$	$\frac{-\sin \phi + 3 \sin 3\phi}{4}$	$\sqrt{3}\cos \phi \sin^2 \phi$
	$\sqrt{3}\cos \phi \sin^2 \phi$	$\frac{\sin \phi - 3 \sin 3\phi}{4}$	$\frac{\cos \phi + 3 \cos 3\phi}{4}$	$\sqrt{3}\cos^2 \phi \sin \phi$
	$-\sin^3 \phi$	$\sqrt{3}\cos \phi \sin^2 \phi$	$-\sqrt{3}\cos^2 \phi \sin \phi$	$\cos^3 \phi$

satisfy this requirement, but those with optical components which may result in energy exchange between modes such as optical fibres, holograms apertures and nonlinear crystals [9], will not.

Any mode of order N can be expanded as the sum of $(N+1)$ Hermite Gaussian, or Laguerre Gaussian, modes of the same order [4,5]. It follows that such a mode may be represented by a column vector with $(N+1)$ elements

$$\begin{bmatrix} a_{N,0} \\ a_{N-1,0} \\ \dots \\ \dots \\ a_{1,N-1} \\ a_{0,N} \end{bmatrix},$$

where $a_{n,m}$ is the complex amplitude coefficient of the (n, m) Hermite Gaussian mode. For example, a Hermite Gaussian $(2, 0)$ mode oriented at 45° to the laboratory frame is represented by a three element column vector containing the complex amplitude weightings of the $(2,0)$, $(1,1)$, $(0,2)$ Hermite Gaussian modes, respectively. This rotated Hermite Gaussian mode is transformed to a Laguerre Gaussian, by passing the beam through a $\pi/2$ converter, which introduces phase shifts between the constituent modes of $(m-n)\pi/4$.

The matrices were developed by consideration of the role of each optical component. The way in which a $\pi/2$ converter transforms a (n,m) Hermite Gaussian mode to a (p,l) Laguerre Gaussian mode is well understood [5]. This converts a mode with no orbital angular momentum into one possessing it. The relative phases of the initial and final component modes are known and the nature of the required matrix follows easily enough. Likewise the matrix for a π converter, which converts a mode with orbital angular momentum l to one with $-l$, follows readily. It is rather more difficult to determine the form of the matrix for the rotation of a mode. Each element of the matrix can be deduced by considering the form of the decomposition of each of the individual modes as it is rotated through an angle ϕ . We may note that components of the higher orders of the rotation matrix follow from the transformation of angular momentum eigenfunctions under finite rotation, given by expressions (4.1.15) or (4.1.23), in the book by Edmonds [12] on angular momentum.

The optical component analogous to a polarizer is a mode filter which selects one constituent mode from those in the column vector. Such filters have been developed and employed to preferentially transmit specific modes [13]. The matrices for $\pi/2$ converters, π converters, mode filters and mode rotations are given in Table I.

To show how the matrices may be used, consider again the $(2,0)$ Hermite Gaussian mode. The column vector representing such an input mode is that for mode order $N=n+m=2$. The vector, therefore, has three components. The first term in the column vector indicates that the input mode is indeed the $(2,0)$ mode and the subsequent zeros show that there is no admixture of $(1,1)$ or $(0,2)$ modes. When the mode is oriented at 45° and passed through a $\pi/2$ converter, the column vector describing the output beam is given by

$$[C(\pi/2)] \times [\text{rot}(45^\circ)] \times [HG_{2,0}]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -i & 0 \\ 0 & 0 & -1 \end{bmatrix} \times \begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{i}{\sqrt{2}} \\ -\frac{1}{2} \end{bmatrix}.$$

From Beijersbergen *et al.* [5], we recognize the Hermite Gaussian coefficients of this output beam as a Laguerre Gaussian mode with indices $l=2$ and $p=0$.

The coefficients of the column vectors in the Jones matrix formulation correspond, respectively, to vertically and horizontally polarized light. It would be possible to formulate an equivalent representation in which the coefficients were for right and left handed circular polarizations. In the same way, our beam description in terms of Hermite Gaussian modes could be replaced with a description in terms of Laguerre Gaussian modes. Conceptually there is no difference between the two approaches, except that in our formulation the matrices for optical components are simple and the rotation matrix relatively complicated while for a column vector based on Laguerre Gaussian modes, the inverse would be the case.

TABLE II. $[2(N+1) \times 2(N+1)]$ matrices for polarized modes of order N .

$\pi/2$ converter	
$[C(\pi/2)] =$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$
π converter	
$[C(\pi)] =$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$
$\lambda/4$ waveplate	
$[W(\lambda/4)] =$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$

Light with a particular mode structure, and so a defined orbital angular momentum may also be polarized and have a defined spin angular momentum. In many instances the two attributes of the beam will remain totally uncoupled and the resultant beam may be determined by two independent cal-

culations: one with the Jones matrices and the other with the matrices developed for orbital angular momentum. However, this is not always the case; for example in the calculation of the eigenmodes of beams within laser resonators “out-of-plane” ring geometries and birefringent optical components

TABLE II (continued).

$\lambda/2$ waveplate													
$[W(\lambda/2)] =$	1	0	0	0	0	0	0	0	0	0	...	...	
	0	-1	0	0	0	0	0	0	0	0	...	...	
	0	0	1	0	0	0	0	0	0	0	...	...	
	0	0	0	-1	0	0	0	0	0	0	...	...	
	0	0	0	0	1	0	0	0	0	0	...	...	
	0	0	0	0	0	-1	0	0	0	0	...	...	
	0	0	0	0	0	0	1	0	0	0	...	...	
	0	0	0	0	0	0	0	-1	0	0	...	...	
	0	0	0	0	0	0	0	0	1	0	...	...	
	0	0	0	0	0	0	0	0	0	-1	...	...	
	...	...	...	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	...	...	...	
Mode filter (e.g., $HG_{N-2,2}$)													
$[F(N-2,2)] =$	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	1	0	0	0	0	0	...	...	
	0	0	0	0	0	1	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	...	...	...	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	...	...	...	
Polarizer; for vertical polarization													
$[P(\uparrow)] =$	0	0	0	0	0	0	0	0	0	0	...	...	
	0	1	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	1	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	1	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	1	0	0	...	...	
	0	0	0	0	0	0	0	0	0	0	...	...	
	0	0	0	0	0	0	0	0	0	1	...	...	
	...	...	...	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	...	...	...	

give rise to rotations of the mode profile and state of polarization, respectively [14]. The corresponding eigenvalues relate to the round trip loss and it is unclear how they could be determined from independent calculation of the orbital and spin eigenmodes. Another example, which we analyze be-

low, is the recently studied rotational frequency shift [10] where the orbital and spin angular momentum interact in such a way that the total angular momentum is the important parameter, not the individual contributions.

More generally, the study of the orbital angular momen-

TABLE II (continued).

Mode rotation matrix for $N=1$

$$[\text{rot}_{\text{mode}}(\phi)] = \begin{bmatrix} \cos(\phi) & 0 & \sin(\phi) & 0 \\ 0 & \cos(\phi) & 0 & \sin(\phi) \\ -\sin(\phi) & 0 & \cos(\phi) & 0 \\ 0 & -\sin(\phi) & 0 & \cos(\phi) \end{bmatrix}$$

Mode rotation matrix for $N=2$

$$[\text{rot}_{\text{mode}}(\phi)] = \begin{bmatrix} \cos^2 \phi & 0 & \frac{\sin 2\phi}{\sqrt{2}} & 0 & \sin^2 \phi & 0 \\ 0 & \cos^2 \phi & 0 & \frac{\sin 2\phi}{\sqrt{2}} & 0 & \sin^2 \phi \\ \frac{-\sin 2\phi}{\sqrt{2}} & 0 & \cos 2\phi & 0 & \frac{\sin 2\phi}{\sqrt{2}} & 0 \\ 0 & \frac{-\sin 2\phi}{\sqrt{2}} & 0 & \cos 2\phi & 0 & \frac{\sin 2\phi}{\sqrt{2}} \\ \sin^2 \phi & 0 & \frac{-\sin 2\phi}{\sqrt{2}} & 0 & \cos^2 \phi & 0 \\ 0 & \sin^2 \phi & 0 & \frac{-\sin 2\phi}{\sqrt{2}} & 0 & \cos^2 \phi \end{bmatrix}$$

Polarization rotation matrix for $N=1$

$$[\text{rot}_{\text{mode}}(\phi)] = \begin{bmatrix} \cos(\phi) & \sin(\phi) & 0 & 0 \\ -\sin(\phi) & \cos(\phi) & 0 & 0 \\ 0 & 0 & \cos(\phi) & \sin(\phi) \\ 0 & 0 & -\sin(\phi) & \cos(\phi) \end{bmatrix}$$

Polarization rotation matrix for $N=2$

$$[\text{rot}_{\text{mode}}(\phi)] = \begin{bmatrix} \cos \phi & \sin \phi & 0 & 0 & 0 & 0 \\ -\sin \phi & \cos \phi & 0 & 0 & 0 & 0 \\ 0 & 0 & \cos \phi & \sin \phi & 0 & 0 \\ 0 & 0 & -\sin \phi & \cos \phi & 0 & 0 \\ 0 & 0 & 0 & 0 & \cos \phi & \sin \phi \\ 0 & 0 & 0 & 0 & -\sin \phi & \cos \phi \end{bmatrix}$$

tum of light beams is a rapidly expanding field and it is not yet clear where it may lead. It is possible that there will be other applications where the interaction of polarization and orbital angular momentum is best examined within a single formalism. It is therefore of considerable interest to extend the matrices describing the mode composition to include simultaneously the polarization behavior previously described by Jones matrices. Such matrices would then be able to model the propagation of both spin and orbital angular momentum, through an optical system comprising both polarizing and mode transforming components.

Any combination of polarized modes of order N can be expressed as a $2(N+1)$ element column vector containing the complex weightings of orthogonally polarized Hermite Gaussian modes of the same order, as with

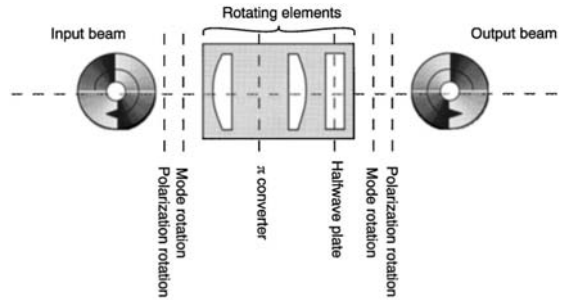


FIG. 2. Experimental configuration for the observation of the rotational frequency shift. (Arrows and shading on the mode distributions indicate the polarization and the sense of the azimuthal phase, respectively.)

$$\begin{bmatrix} a_{N,0}(\leftrightarrow) \\ a_{N,0}(\uparrow) \\ a_{N-1,0}(\leftrightarrow) \\ a_{N-1,0}(\uparrow) \\ \dots \\ \dots \\ \dots \\ \dots \\ a_{1,N-1}(\leftrightarrow) \\ a_{1,N-1}(\uparrow) \\ a_{0,N}(\leftrightarrow) \\ a_{0,N}(\uparrow) \end{bmatrix},$$

where the arrows in brackets denote the orientation of the polarization. The intensity is simply given by the square of the modulus of the vector representing the beam.

The $[2(N+1) \times 2(N+1)]$ matrices for the mode converters are based on those obtained previously, but each term must be repeated along the diagonal so that both polarizations are rephased. The $[2(N+1) \times 2(N+1)]$ matrices for waveplates are based on the corresponding Jones matrices, but the whole of each matrix must be repeated along the diagonal to act on each mode independently.

The rotation matrices are more complicated. The mode may be rotated with an image rotator and the polarization independently rotated by the use of a waveplate. The mode rotation matrix includes the same terms as our N dependent matrix given in Table I, but they must be diagonally repeated to rotate modes of both polarizations and interspaced with zeros to prevent their mixing. Similarly, the matrix for the rotation of the polarization is identical to that used within the Jones matrix formulation, but is diagonally repeated to rotate the polarization state of each of the individual Hermite Gaussian modes. The matrices for mode converters, waveplates, mode filters, polarizers and rotations are given in

Table II.

To confirm the effectiveness of the matrix formulation, we modelled our rotational frequency shift experiment [10] for a circularly polarized, $\sigma = \pm 1$, Laguerre Gaussian mode. This enables us to find the frequency shift which results from the rotation of a beam containing both spin and orbital angular momentum. In the experiment, beam rotation was introduced by simultaneously rotating a π converter and half-wave plate; see Fig. 2. The matrix description of the output beam becomes

$$\begin{aligned} & [\text{rot}_{\text{pot}}(\phi)] \times [\text{rot}_{\text{mode}}(\phi)] \times [W(\lambda/2)] \times [C(\pi)] \\ & \times [\text{rot}_{\text{mode}}(-\phi)] \times [\text{rot}_{\text{pot}}(-\phi)] \times [LG_{l,p}(\sigma = \pm 1)]. \end{aligned}$$

A change in the handedness of the circularly polarized light, such that the spin angular momentum component is either additive or subtractive from the orbital angular momentum, gives a total angular momentum of the beam of $(l \pm 1)\hbar$ per photon. The frequency shift may be correctly deduced by comparison with its nonrotated equivalent to be $(l \pm 1)\Omega$, where Ω is the angular frequency of the beam. This offers powerful evidence of the utility of the matrices.

We have shown that there is a set of matrices equivalent to the Jones matrices which can express the behavior of light possessing orbital angular momentum and have developed them to account simultaneously for polarization. These matrices can be used in any optical system which conserves mode order. The equivalent of the Mueller 4×4 matrices, which depend upon the four Stokes parameters, for orbital angular momentum is currently the basis of further investigation.

It is a pleasure to acknowledge the contribution of Professor S. M. Barnett who drew to our attention to the rotation matrix in Ref. [12].

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Paraxial beams of spinning light

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Abstract

A simple derivation is given for the known result that the component of total angular momentum along the propagation direction of a general paraxial beam can be separated into orbital and spin parts. Comments are made about orbital angular momentum and wave dislocations, and about how the orbital and spin angular momenta can be changed by propagation in a refracting medium.

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1. Angular momentum decomposition

It is now well established by theory and experiment that beams of light can carry both orbital and spin angular momentum [1-4], at least in the paraxial approximation [5]. In spite of this widespread understanding, it seems worthwhile to present the argument in its most general form, and that is my main purpose here - without claiming originality for the central ideas. The aims are to represent the light in terms of general transverse fields (i.e. not necessarily as a superposition of Gauss-Laguerre beams), to allow the polarization to vary with position, to introduce paraxiality in the most direct way, and to write the result in a way that resembles quantum mechanics, so that the separation into the two kinds of angular momentum is compelling. In addition, I will make some remarks about the significance of the two components, and how they can be changed by the medium through which the light is propagating.

Let monochromatic light with frequency ω (and wavenumber $k=\omega/c$) travel in vacuum in the paraxial direction z (unit vector $\mathbf{e}_z$). A

precise specification of z will be given later. Let positions in the transverse plane be denoted by $\mathbf{r}$, let the transverse part of a general vector be denoted by $\mathbf{V}$, and let the full three-dimensional counterparts of these quantities be denoted by the suffix "tot". Thus

$$\mathbf{r} = (x, y), \quad \mathbf{V} = (A_x, A_y), \quad \mathbf{r}_{\text{tot}} = (x, y, z), \quad \mathbf{V}_{\text{tot}} = (A_x, A_y, A_z) \quad (1)$$

We will calculate the z component of the total angular momentum per photon in unit length of a transverse slice of the beam about the axis $\mathbf{r}=0$, and denote this quantity by J_z . We use standard notation for the complex electromagnetic fields, and the well known results [6] that the angular momentum density is

$$\text{Re } \mathbf{r} \times (\mathbf{D}_{\text{tot}}^* \times \mathbf{B}_{\text{tot}}) \quad (2)$$

and the energy density is

$$\text{Re } \frac{1}{2} (\mathbf{E}_{\text{tot}}^* \cdot \mathbf{D}_{\text{tot}} + \mathbf{B}_{\text{tot}}^* \cdot \mathbf{H}_{\text{tot}}) \quad (3)$$

Then

$$J_z = \hbar \omega \frac{\text{Re} \iint dx dy \mathbf{r} \times (\mathbf{D}_{\text{tot}}^* \times \mathbf{B}_{\text{tot}}) \cdot \mathbf{e}_z}{\text{Re} \frac{1}{2} \iint dx dy (\mathbf{E}_{\text{tot}}^* \cdot \mathbf{D}_{\text{tot}} + \mathbf{B}_{\text{tot}}^* \cdot \mathbf{H}_{\text{tot}})} \quad (4)$$

Maxwell's equations and the constitutive relations can be used to eliminate all fields except $\mathbf{E}_{\text{tot}}$, with the result

$$J_z = \hbar \frac{\text{Re}(-i) \iint dx dy \mathbf{r} \times (\mathbf{E}_{\text{tot}}^* \times (\nabla \times \mathbf{E}_{\text{tot}})) \cdot \mathbf{e}_z}{\text{Re} \frac{1}{2} \iint dx dy \left(\mathbf{E}_{\text{tot}}^* \cdot \mathbf{E}_{\text{tot}} + \frac{1}{k^2} (\nabla \times \mathbf{E}_{\text{tot}}^*) \cdot (\nabla \times \mathbf{E}_{\text{tot}}) \right)} \quad (5)$$

In this exact formula, all components of $\mathbf{E}_{\text{tot}}$ can vary with x, y and z .

Now we invoke the paraxial approximation and then $\nabla \cdot \mathbf{E}_{\text{tot}} = 0$ to eliminate all z derivatives and z components to order $1/k$:

$$\partial_z \mathbf{E}_{\text{tot}} \approx ik \mathbf{E}_{\text{tot}}, \quad E_z \approx \frac{i}{k} \nabla \cdot \mathbf{E} \quad (6)$$

Thus, in the denominator in (5),

$$\mathbf{E}_{\text{tot}}^* \cdot \mathbf{E}_{\text{tot}} \approx \frac{1}{k^2} (\nabla \times \mathbf{E}_{\text{tot}}^*) \cdot (\nabla \times \mathbf{E}_{\text{tot}}) \approx \mathbf{E}^* \cdot \mathbf{E} \quad (7)$$

The numerator can be similarly simplified using (6) and also integration by parts with respect to x and y , leading after a little algebra to

$$J_z \approx \frac{\text{Re}(-i\hbar) \iint dx dy \left\{ \mathbf{E}^* \cdot [\mathbf{e}_z \cdot (\mathbf{r} \times (-i\nabla))] \mathbf{E} + \mathbf{e}_z \cdot \mathbf{E}^* \times \mathbf{E} \right\}}{\text{Re} \iint dx dy \mathbf{E}^* \cdot \mathbf{E}} \quad (8)$$

It is well known [5] that a similar expression, with all vectors replaced by their three-dimensional counterparts, can be obtained exactly - that is, without the paraxial approximation - if the quantity being calculated is the total angular momentum in the whole three-dimensional beam; but this depends on integration with respect to z , an option unavailable in our calculation for a transverse slice.

In the final step, we introduce a basis of circular polarizations, replacing $\mathbf{E}$ by the column vector

$$|\psi\rangle \equiv \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} E_x - iE_y \\ E_x + iE_y \end{pmatrix} \quad (9)$$

Then we define scalar products as

$$\langle \phi | \psi \rangle \equiv \iint dx dy \begin{pmatrix} \phi_+^*(\mathbf{r}) & \phi_-^*(\mathbf{r}) \end{pmatrix} \begin{pmatrix} \psi_+(\mathbf{r}) \\ \psi_-(\mathbf{r}) \end{pmatrix} \quad (10)$$

and recognize

$$\mathbf{p} = -i\hbar \nabla \quad (11)$$

as the momentum operator, and

$$l_z = \mathbf{r} \times \mathbf{p} \cdot \mathbf{e}_z, \quad s_z = \hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (12)$$

as the operators for the z components of the orbital angular momentum, and the spin angular momentum of a spin-1 particle. Thus (8) becomes the well known expression

$$J_z \approx \frac{\langle \psi | l_z | \psi \rangle + \langle \psi | s_z | \psi \rangle}{\langle \psi | \psi \rangle} \equiv L_z + S_z \quad (13)$$

This is valid in the paraxial approximation, that is when quantities of order $1/k$ are neglected. Henceforth we assume without loss of generality that the beam is normalized with $\langle\psi|\psi\rangle=1$.

2. Remarks on spin and orbital parts of the angular momentum

It might be thought that the difference between the spin and orbital contributions S_z and L_z in (13) is that S_z is independent of the axis $\mathbf{r}=0$ about which J_z is calculated, whereas L_z depends on this axis. But this can be wrong, as the following argument shows. Under a shift of origin, $\mathbf{r}\rightarrow\mathbf{r}_0+\mathbf{r}$, the orbital angular momentum shifts too:

$$L_z \rightarrow L_z + \mathbf{e}_z \cdot \mathbf{r}_0 \times \mathbf{P}, \quad \text{where } \mathbf{P} \equiv \langle\psi|\mathbf{p}|\psi\rangle \quad (14)$$

Here $\mathbf{P}$ is the transverse current in the beam. If $\mathbf{P}$ vanishes, the shift $\mathbf{r}\rightarrow\mathbf{r}_0+\mathbf{r}$ leaves L_z unaltered. But for a paraxial beam $\mathbf{P}$ will always be small compared with the current along the paraxial direction z . Therefore it is possible and indeed reasonable to stipulate z as the direction for which $\mathbf{P}$ is *exactly* zero. With paraxiality thus defined, *both* components of J_z are invariant under a shift of axis, so this invariance cannot be used to define S .

The true distinction between orbital and spin angular momentum is of course that L_z depends on the spatial structure of the beam irrespective of polarization, whereas S_z depends on the state of polarization of the beam. There are subtleties concealed here too. To appreciate these, note that, from (9), (10), (12) and (13),

$$S_z = \hbar \langle\psi \left| \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right| \psi \rangle = \hbar \iint dx dy \left(|\psi_+(\mathbf{r})|^2 - |\psi_-(\mathbf{r})|^2 \right) \quad (15)$$

so that $S_z=+\hbar$ for right-circularly polarized light ($\psi_-=0$), and $S_z=-\hbar$ for left-circularly polarized light ($\psi_+=0$), as expected for spin-1 particles. For linearly polarized light ($|\psi_+|=|\psi_-|$), $S_z=0$. A complete description of the polarization state involves the other two Pauli matrices, and suggests defining

$$S_x = \hbar \langle\psi \left| \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right| \psi \rangle, \quad S_y = \hbar \langle\psi \left| \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right| \psi \rangle \quad (16)$$

But S_x , S_y and S_z must not be regarded as the three components of a spin angular momentum vector of the light, although they give a complete description of the state of polarization of the beam. One reason is that they have the wrong commutation relations, e.g. $[S_x, S_y] = 2i\hbar S_z$ (rather than $i\hbar S_z$ as would be appropriate to a spin-1 particle); another is that $S_x^2 + S_y^2 + S_z^2 = 3\hbar^2$ (rather than $2\hbar^2$ as would be appropriate to a spin-1 particle).

One of several correct representations of spin-1 matrices is that implied by the three-dimensional version of the last term in (8), namely

$$\text{Re}\left(-i\hbar \mathbf{E}_{\text{tot}}^* \times \mathbf{E}_{\text{tot}}\right) = \mathbf{E}_{\text{tot}}^* \cdot \mathbf{S}_{\text{tot}} \cdot \mathbf{E}_{\text{tot}} \quad (17)$$

where

$$\mathbf{S}_{\text{tot}} = -i\hbar \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\} \quad (18)$$

Strangely, however, these spin-1 matrices, unlike those in (15) and (16), do not give a complete description of the state of polarization of the beam: for any transverse wave, whatever its polarization, the transverse components of the expectation value of $\mathbf{S}_{\text{tot}}$ vanish, leaving only the single (real) z component nonvanishing, and unable to discriminate, for example, between the different possible states of linear polarization. There is another well known 'paradox' associated with the full three-dimensional angular momentum of a transverse light wave (e.g. a plane wave): the exact equation (2) implies that all components of total angular momentum vanish for such a wave, whereas (13) predicts a nonvanishing value for J_z for a circularly polarized transverse wave. The resolution lies in the fact that (13) applies to a beam whose lateral extent is effectively finite, which can therefore not be exactly transverse; in the derivation of (13), the longitudinal component (approximated by (6)) contributed through the integration by parts.

Turning now to the orbital angular momentum L_z , we note the common opinion [7] that this quantity is associated with optical dislocation lines (phase singularities) [8-11]. In general, however, there is no such association, as the following argument shows.

Since L_z is associated with the spatial variation of the beam, it suffices to take for $|\psi\rangle$ a scalar wavefunction $\psi(\mathbf{r}, z)$, satisfying the paraxial wave equation. We let ψ be a 'spiral beam' [12, 13] with waist size w ; In the focal plane $z=0$ this has the form

$$\psi(\mathbf{r}, 0) = f(\xi) \exp\{-r^2 / 2w^2\} \quad \left(\xi \equiv x + iy, \quad r \equiv \sqrt{(x^2 + y^2)} \right) \quad (19)$$

Away from the focal plane, ψ preserves this form while expanding and rotating. A short calculation gives

$$L_z = \hbar \frac{\text{Re} \iint dx dy \exp(-r^2 / w^2) f(\xi)^* \xi \partial_{\xi} f(\xi)}{\text{Re} \iint dx dy \exp(-r^2 / w^2) |f(\xi)|^2} \quad (20)$$

If $f(\xi)$ is an M th order polynomial, that is

$$f(\xi) = \sum_{m=0}^M f_m \xi^m \quad (21)$$

then

$$L_z = \hbar \frac{\sum_{m=1}^M m m! |f_m|^2 w^{2m}}{\sum_{m=0}^M m! |f_m|^2 w^{2m}} \equiv \langle m \rangle \hbar \quad (22)$$

Now, dislocations are the zeros of ψ , that is the zeros of $f(\xi)$, and by the fundamental theorem of algebra there are exactly M of them for the spiral beam with f given by (21). Moreover, they all have the same sign, so the total dislocation strength [11] is also M . In the special case of a pure Gaussian beam, where only the term $m=M$ in the series (21) is nonzero, $\langle m \rangle = M$ and there is a sense in which it is correct to ascribe the (quantized) orbital angular momentum to the M th order (screw) dislocation. But in all other cases, $\langle m \rangle \neq M$ and the association between L_z and dislocations is false. The easy way to see that this must be so is to note that since dislocations are zeros they have zero weight in the integral (20) for L_z .

To avoid confusion, I should remark that there is an association between a dislocation line and the torque exerted on a small particle [2-4],

arising from the vortex structure close to the line. This very interesting effect concerns the angular momentum density in the vicinity of the particle, rather than the integrated quantity L_z considered here.

3. Beam propagation in a refracting medium

If the beam propagates paraxially in a homogenous and isotropic medium, no torque can be exerted on it, and L_z and S_z are separately conserved. However, an inhomogenous medium can exert a torque that changes L_z , and an anisotropic medium can exert a torque that changes S_z . To quantify these effects, let the medium have dielectric tensor

$$\mathbf{e}(\mathbf{r}) = \varepsilon_0(1 + 2\mathbf{n}(\mathbf{r})) \quad (23)$$

Here $\mathbf{n}$ is a transverse refractive index tensor, hermitean because the medium is assumed transparent, whose components

$$\mathbf{n}(\mathbf{r}) = \begin{pmatrix} n_{xx}(\mathbf{r}) & n_{xy}(\mathbf{r}) \\ n_{xy}(\mathbf{r}) & n_{zz}(\mathbf{r}) \end{pmatrix} \quad (24)$$

are all much less than unity since the propagation is paraxial. In this case contributions from $\mathbf{n}$ to J_z are of higher than paraxial order, and the argument leading from (2) and (3) to the final formula (13) is unaffected by the anisotropy and variations of the medium.

However, propagation of the angular momentum is affected, in a manner determined by the paraxial wave equation for $\mathbf{E}$. From Maxwell's equations, this is

$$ik\partial_z\mathbf{E} = -\frac{1}{2}(\partial_x^2 + \partial_z^2)\mathbf{E} - k^2\mathbf{n}(\mathbf{r}) \cdot \mathbf{E} \quad (25)$$

Transforming to the basis (9) of circular polarizations gives the 'Schrödinger lookalike' equation

$$ik\partial_z|\psi\rangle = \mathbf{H}|\psi\rangle \quad (26)$$

Here the 'hamiltonian' $\mathbf{H}$ (now setting $\hbar=1$ for simplicity of notation) is

$$\mathbf{H} = \frac{1}{2}\mathbf{p} \cdot \mathbf{p} - k^2\{W_0(\mathbf{r}) + \mathbf{W}(\mathbf{r}) \cdot \boldsymbol{\sigma}\} \quad (27)$$

in which $\boldsymbol{\sigma}$ is the (three-dimensional) vector of Pauli matrices and $W_0(\mathbf{r})$ and the three-dimensional vector $\mathbf{W}(\mathbf{r})$ are (invoking the hermiticity of $\mathbf{n}$)

$$\begin{aligned} W_0 &= n_{xx} + n_{yy}, \\ \mathbf{W} &= \{W_x, W_y, W_z\} = \{n_{xx} - n_{yy}, 2\operatorname{Re}n_{xy}, -2\operatorname{Im}n_{xy}\} \end{aligned} \quad (28)$$

Note that W_0 depends on the isotropic part of the refractive index of the medium, W_x and W_y to its birefringence, and W_z to its optical activity (chirality).

The rate of change in S_z is easily shown to be

$$\begin{aligned} \partial_z S_z &= 2k \langle \psi | \mathbf{W} \times \boldsymbol{\sigma} \cdot \mathbf{e}_z | \psi \rangle \\ &= 2k \langle \psi | \begin{pmatrix} 0 & -i(n_{xx} - n_{yy}) - 2\operatorname{Re}n_{xy} \\ i(n_{xx} - n_{yy}) - 2\operatorname{Re}n_{xy} & 0 \end{pmatrix} | \psi \rangle \end{aligned} \quad (29)$$

Thus the spin angular momentum can be changed by the birefringence of the medium but not by its optical activity.

Similarly, the rate of change of L_z is determined by the commutator in

$$\partial_z L_z = -ik \langle \psi | [W_0(\mathbf{r}) + \mathbf{W}(\mathbf{r}) \cdot \boldsymbol{\sigma}, l_z] | \psi \rangle \quad (30)$$

Since, from the definition (12),

$$l_z = -i\partial_\phi \quad (31)$$

where ϕ is the polar angle of $\mathbf{r}$, we recover the obvious result that the orbital angular momentum can be changed only by propagation in a medium that is not azimuthally symmetric, regardless of whether the asymmetry is in the isotropic part W_0 of the medium or its anisotropic (birefringent or chiral) part $\mathbf{W}$.

During any such changes of its angular momentum, the beam will, of course, exert equal and opposite reactive torques on the medium

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The Poynting vector in Laguerre–Gaussian beams and the interpretation of their angular momentum density

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Abstract

We use the local linear momentum density in Laguerre–Gaussian beams to investigate classically the trajectory of the Poynting vector for light of any polarisation. Some of the difficulties of interpretation of the local angular momentum density are discussed. © 2000 Published by Elsevier Science B.V.

Keywords: Poynting vector; Angular momentum density; Polarised light

1. Introduction

The rate of electromagnetic energy flow per unit area, the Poynting vector, is defined and discussed in all books on electromagnetism (see, e.g., Ref. [1]). The vector and its associated linear momentum density is, however, invariably examined for plane waves or at least waves with a plane wavefront. Recent interest in orbital angular momentum of light beams [2] has led to a considerable literature relating to Laguerre–Gaussian beams which have helical wavefronts. The angular momentum density, $\mathbf{j}$, is related [1] to the linear momentum density, $\mathbf{p}$, through

$$\mathbf{j} = \mathbf{r} \times \mathbf{p}, \quad (1)$$

where

$$\mathbf{p} = \epsilon_0 \mathbf{E} \times \mathbf{B}, \quad (2)$$

and the total angular momentum is naturally derived from the linear momentum [3]. The emphasis in the literature is, for obvious reasons, on orbital angular momentum, but as light of arbitrary polarisation is considered, it also accounts for spin angular momentum. In this work, we use the components of the linear momentum density to determine the behaviour of the Poynting vector for Laguerre–Gaussian modes and then examine the meaning of the angular momentum density.

2. Linear momentum density

It is easy to show within the paraxial approximation [2,3] that for any mode with an amplitude given by

$$u(r, \phi, z) = u(r, z) \exp(i l \phi), \quad (3)$$

the r , ϕ and z components of linear momentum density, $\epsilon_0 \mathbf{E} \times \mathbf{B}$, for a circularly polarised beam propagating in the z -direction are given by

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$$\begin{aligned}
p_r &= \varepsilon_0 \frac{\omega k r z}{(z_R^2 + z^2)} |u|^2, \\
p_\phi &= \varepsilon_0 \left[\frac{\omega l}{r} |u|^2 - \frac{1}{2} \omega \sigma \frac{\partial |u|^2}{\partial r} \right], \\
p_z &= \varepsilon_0 \omega k |u|^2.
\end{aligned} \tag{4}$$

The component p_r , relates to the spread of the beam as it propagates. The first term of the azimuthal component p_ϕ depends on l , where $l\hbar$ has been identified as the orbital angular momentum per photon [3]. Its second term relates to the spin contribution, where $\sigma = \pm 1$ for left-handed and right-handed circularly polarised light, respectively, and $-1 < \sigma < 1$ describes elliptically polarised light. It is this contribution which leads to the spin angular momentum of the beam. The final component, p_z , is the linear momentum in the direction of propagation.

The relative size of the components determines the trajectory of the Poynting vector (Fig. 1). We have investigated elsewhere this trajectory with respect to the orbital contribution alone [4]. This

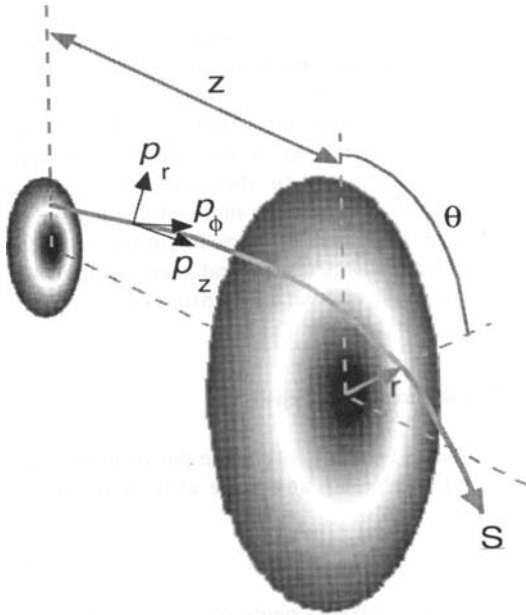


Fig. 1. The trajectory of the Poynting vector and the components of linear momentum density.

was for linearly polarised light where $\sigma = 0$; most of the interesting terms investigated here were therefore zero and played no role. When both orbital and spin momenta are included, the rate of change of azimuthal angle of the trajectory of the Poynting vector with z is given by

$$\frac{\partial \theta}{\partial z} = \frac{1}{r} \frac{p_\phi}{p_z} = \frac{l}{kr^2} - \frac{\sigma}{2kr} \frac{1}{|u|^2} \frac{\partial |u|^2}{\partial r}. \tag{5}$$

A Laguerre–Gaussian mode has an amplitude of

$$\begin{aligned}
u_{pl}^{\text{LG}} &= \frac{C_{pl}^{\text{LG}}}{w(z)} \left(\frac{r\sqrt{2}}{w(z)} \right)^{|l|} \exp \left[-\frac{r^2}{w^2(z)} \right] L_p^{|l|} \\
&\times \exp \left(-\frac{ikr^2 z}{2(z^2 + z_R^2)} \right) \exp(-il\phi) \\
&\times \exp \left(i(2p + |l| + 1) \tan^{-1} \frac{z}{z_R} \right),
\end{aligned} \tag{6}$$

where C_{pl}^{LG} is the normalisation constant; $L_p^{|l|}(2r^2/w^2(z))$ is a generalised Laguerre polynomial; the radius of the beam at position z is $w(z)$ where

$$w(z)^2 = \frac{2}{k} \frac{z_R^2 + z^2}{z_R}, \quad \text{and} \quad (2p + |l| + 1) \tan^{-1} \frac{z}{z_R}$$

is the Gouy phase, and z_R , the Rayleigh range. Here, p and l are mode indices, where l is the number of intertwined helices, and p , the number of radial nodes. It follows that for such a mode, Eq. (5) becomes

$$\begin{aligned}
\frac{\partial \theta}{\partial z} &= \frac{l}{kr^2} - \frac{\sigma |l|}{kr^2} + \frac{2\sigma}{kw^2(z)} \\
&+ \frac{4\sigma}{kw^2} \frac{L_{p-1}^{|l|+1}(2r^2/w^2(z))}{L_p^{|l|}(2r^2/w^2(z))},
\end{aligned} \tag{7}$$

where k may be expressed in terms of $w(z)$, as above.

For modes with $p = 0$, the final term in Eq. (7) is always zero. When $l = 0$ and $p = 0$ and the Laguerre–Gaussian is identical to the Hermite Gaussian (0,0)-mode, the orbital angular momentum is zero and the rotation of the Poynting vector for a polarised beam arises from the effect of spin alone. The first two terms are also zero and $\theta = \sigma \arctan z/z_R$ and $\theta \rightarrow \pi/2$ as $z \rightarrow \infty$.

For a single ringed Laguerre–Gaussian mode when $p = 0$ but $l \neq 0$,

$$\frac{\partial \theta}{\partial z} = \frac{l}{kr^2} \left(1 - \frac{\sigma |l|}{l} \right) + \frac{2\sigma}{kw^2}. \quad (8)$$

At peak intensity when $r^2 = w^2(z)|l|/2$, we find for all values of l and σ that $\theta = \pm \arctan z/z_R$. The trajectory of the Poynting vector is in general a spiral with a pitch that depends on the radius. Surprisingly, at peak intensity, the pitch is such that it becomes a straight line skewed with respect to the beam axis. Such a trajectory is consistent with a ray optical model where orbital angular momentum may be represented by skew rays in the optical beam.

When $\sigma = \pm 1$, that is for circularly polarised light, we see that if the handedness of spin and orbit are the same, then $\theta = \arctan z/z_R$ for all l and for all positions across the beam amplitude distribution, not just at the peak intensity.

In general, for a mode of arbitrary index p and any value of l and σ ,

$$\theta = \left\{ \frac{l - \sigma \frac{|l|}{l} w^2(0)}{2 r^2(0)} + \sigma + 2\sigma \frac{L_{p-1}^{|l+1|}(2r^2/w^2(z))}{L_p^{|l|}(2r^2/w^2(z))} \right\} \times \arctan \frac{z}{z_R}. \quad (9)$$

All previous values may be found from this, as well as the solely spin-dependent contribution which arises for higher-order modes. As any arbitrary light beam can be expressed in terms of a set of Laguerre–Gaussian modes, this result is of general applicability.

3. Angular momentum density

We now consider the angular momentum density, which is closely related to the azimuthal component of the Poynting vector. It follows from the linear momentum densities already given in Eq. (4), that the angular momentum density in the direction of propagation z derived from $\mathbf{j} = \mathbf{r} \times \mathbf{p}$ is

$$j_z = \varepsilon_0 \left\{ \omega l |u|^2 - \frac{1}{2} \omega \sigma r \frac{\partial |u|^2}{\partial r} \right\}. \quad (10)$$

When this quantity is integrated over the beam and divided by the energy, we find [3]

$$\begin{aligned} \frac{J_z}{W} &= \frac{\int j_z r dr d\phi}{\int w_z r dr d\phi} \\ &= \frac{l \int |u|^2 r dr d\phi}{\omega \int |u|^2 r dr d\phi} - \frac{\sigma}{2\omega} \frac{\int r^2 \frac{\partial |u|^2}{\partial r} d\phi}{\int |u|^2 r dr d\phi}. \end{aligned} \quad (11)$$

When the integration is over the whole cross-section of the beam, the total angular momentum J_z divided by total energy W reduces simply to

$$\frac{J_z}{W} = \frac{l}{\omega} + \frac{\sigma}{\omega}. \quad (12)$$

When the integration extends over only part of the beam, the first term is always constant, l/ω , while the second term depends on the intensity gradient.

If the energy is radiated in photons of energy $\hbar\omega$, then the angular momentum is transferred in units of $\hbar$. As Biedenharn and Louck [5], “It is in this way that Maxwell’s equations, which do not contain Planck’s constant, are nonetheless compatible with the quantum mechanics of photons. (The agreement fails for order $\hbar^2$.)” It is this simple relationship which has led to the quantity $\hbar h$ being designated as the orbital angular momentum [3]. There are, however, problems. A classical treatment is not straightforwardly in accord with the quantum theory. For example, the square of the angular momentum divided by the square of the energy for a classical electromagnetic multipole (J, J_z) turns out to be J_z^2/ω^2 and not, as might have been expected, $J(J+1)/\omega^2$. In fact, Morette-de Witt and Jensen [6] showed using quantised fields that the correct value is $\{N^2 J_z^2 + N[J(J+1) - J_z^2]\}/N^2 \omega^2$, where N is the number of quanta in the mode (J, J_z). For large N , the classical value J_z^2/ω^2 occurs, but at the single photon level, we get $J(J+1)/\omega^2$.

A more serious problem is one which has been tackled ever since Beth’s famous experimental determination of $\hbar$ [7]. A circularly polarised plane wave with electric and magnetic fields only in the x - and y -directions has a linear momentum density only in the z -direction. When this is crossed with r to give the angular momentum density, there is no contribution in the z -direction. Thus, such a beam has no angular momentum to transfer to a

waveplate, yet, Beth was able to make such a transfer – a paradox.

The explanation has a considerable literature (see list of references spanning 1936–54 given in the book by Jauch and Rohrlich [8]). Probably Simmons and Guttman [9] give the most readily accessible discussion of the problem in their unusual, but fascinating, book. They do not solve the wave equation directly but their rejection of second-order derivatives when examining Maxwell's equations, makes their argument equivalent to the paraxial approximation. The explanation of the paradox is that the finite extent of the waveplate, or the finite extent of the field whichever is smaller, creates an aperturing that prevents the field being a plane wave as its extent is no longer infinite. It then follows that in the direction of propagation, there are electric and magnetic field components, which give azimuthal and radial components to the momentum density. When these are crossed with r , there is an angular momentum density in the z -direction. The essential part of the Simmons and Guttman's argument is that the aperturing at the edge of the waveplate creates a large intensity gradient, and in this region, the angular momentum components are, therefore, very strong. The term involving the intensity gradient shown in Eq. (10) is to be found in the book in the integrands of the two unnumbered equations prior to Eq. (9.22). Remarkably, when averaged with the zero angular momentum over the area of the waveplate, the result is that the spin angular momentum divided by energy is σ/ω per photon, as expected.

The problem of Gaussian beams is, however, not quite the same. They already have a natural radial intensity gradient because they go to zero intensity at infinity. Such fields, irrespective of the presence of any waveplate, have field components in the direction of propagation. It is because these fields exist that there is a permanent spin angular momentum density, and the fundamental paradox associated with the idealised plane wave is absent. For Laguerre–Gaussian beams, the fields are more complex because of the helical wavefronts and their associated azimuthal phase term. Indeed, it is these more complex axial fields which specifically create the orbital angular momentum with which L–G modes are now associated. The key question

is whether the edge effect of a waveplate, or some other aperture or obstacle, still plays as significant a role as that played in the case of a plane wave? The question is important because, as the spin angular momentum density depends on intensity gradient and because these fields do not have uniform intensity, a different amount of angular momentum might be expected to be transferred at different positions in the wavefront.

For the commonly occurring $p = 0$ modes, the expression for total angular momentum divided by total energy Eq. (11) is readily shown to be

$$\frac{J_z}{W} = \frac{l}{\omega} - \frac{\sigma}{\omega} \times \frac{\int_0^b \frac{1}{2} r^2 \left\{ \frac{2|l|}{r} - \frac{4r}{w^2(z)} \right\} r^{2|l|} \exp \left[-\frac{2r^2}{w^2(z)} \right] dr}{\int_0^b r r^{2|l|} \exp \left[-\frac{2r^2}{w^2(z)} \right] dr}. \quad (13)$$

The local value of the orbital angular momentum given by the first term is of little interest to this discussion, as it is a constant. We see that the second term although associated with spin also depends on the magnitude of l . This does not imply a coupling between spin and orbital angular momentum, it is merely a reflection of the fact that the spin contribution depends on the field intensity gradient and this depends on $|l|$. The integral is easily evaluated to show that when the beam is apertured at any radius b to give a rapid decline in intensity to zero, we find the result given in expression (12) in just the same way as for a plane wave. If the expression equivalent to Eq. (13) is derived for modes with a value of $p > 0$, then it may be readily shown that the parts of the wavefront that lie between any of the regions of zero intensity also integrate neatly to the same result; there is, of course, no plane wave equivalent.

What, however, remains of interest is the possibility that there is some regime where there might be a difference in the detailed behaviour of the spin angular momentum of Laguerre–Gaussian beams compared with plane waves. The intensity-gradient-dependent local value of spin is given by the second term of Eq. (10). If we write the local spin angular momentum per energy, we find

$$\frac{j_{z,\text{spin}}}{w} = -\frac{\sigma r}{2\omega} \frac{(\partial|u|^2/\partial r)}{|u|^2} \quad (14)$$

which for a $p = 0$ mode, gives a local spin angular momentum per photon of

$$j_{z,\text{spin}} = -\sigma\hbar \left(|l| - \frac{2r^2}{w^2(z)} \right). \quad (15)$$

Even though the beam has been macroscopically polarised with a unique value of σ , the z -component of spin per photon in a $(p = 0, l)$ -mode clearly depends in this picture on the position across the beam; in some places, it is negative and in others it is positive, and its absolute value is unlimited. It is easy to see that there is a spread of values of $j_{z,\text{spin}}$ and that it is not limited to a maximum value of unity. The spin angular momentum goes from $-|l|\sigma$ near to the beam axis, to zero at the peak intensity and thereafter becomes positive and arbitrarily high. Nevertheless, on average, the spin per photon remains $\sigma\hbar$.

Is it really possible that matter can respond to the local value of angular momentum, which may not only be different in magnitude but have a different sign from that expected by the state of polarisation of the beam? Clearly any attempt to separate the different regions by means of an aperture is doomed to failure, because the rapid intensity variation of the edge effect will again produce the expected, “correct” result. It would seem that an experiment involving atoms might be more promising. If there were still an equivalent to the edge effect, the dipole approximation would seem to have no meaning, and yet, it is the basic assumption of virtually all theoretical approaches to atom–field interactions.

Another concern is that it appears that a net spin component per photon greater than unity is possible; it is hard to see what this really means. Nevertheless, in the Simmons and Guttman exposition for a plane wave at the edge of the waveplate, the same thing is true, otherwise the net

angular momentum per unit energy cannot average to unity, as in some areas the density is zero. But, as they write, “a classical quantity associated with the electromagnetic field does not necessarily indicate the value of that quantity which will be measured.” There is considerable scope for the classical argument to fail; it could well be that there are significantly different observable effects at the single photon level. Nevertheless, the classical approach although simple, has been remarkably successful so far [2] in accounting for so many phenomena. It would be very exciting to find a regime where the experimental results manifest the sign and the varying value of the local angular momentum density in the way the above discussion suggests.

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Intrinsic and Extrinsic Nature of the Orbital Angular Momentum of a Light Beam

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We explain that, unlike the spin angular momentum of a light beam which is always intrinsic, the orbital angular momentum may be either extrinsic or intrinsic. Numerical calculations of both spin and orbital angular momentum are confirmed by means of experiments with particles trapped off axis in optical tweezers, where the size of the particle means it interacts with only a fraction of the beam profile. Orbital angular momentum is intrinsic only when the interaction with matter is about an axis where there is no net transverse momentum.

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Introduction.—Some 65 years ago Beth [1] demonstrated that circularly polarized light could exert a torque upon a birefringent wave plate suspended in the beam by the transfer of angular momentum. The angular momentum associated with circular polarization arises from the spin of individual photons and is termed spin angular momentum.

More recently, Allen *et al.* [2] showed that for beams with helical phase fronts, characterized by an $\exp(il\phi)$ azimuthal phase dependence, the orbital angular momentum in the propagation direction has the discrete value of $l\hbar$ per photon. Such beams have a phase dislocation on the beam axis that in related literature is sometimes referred to as an optical vortex [3]. In general, any beam with inclined phase fronts carries orbital angular momentum about the beam axis which, when integrated over the beam, can be an integer or noninteger [4,5] multiple of $\hbar$.

In this paper, we experimentally examine the motion of particles trapped off axis in an optical tweezers and are able to associate specific aspects of the motion with the distinct contributions of spin and orbital angular momentum of the light beam. The interpretation of the experiments, when combined with a numerical calculation of the spin and orbital contributions derived from established theory, allows a distinction to be made between the intrinsic and extrinsic aspects of the angular momentum of light.

Angular momentum of a light beam.—The cycle-averaged linear momentum density, $\mathbf{p}$, and the angular momentum density, $\mathbf{j}$, of a light beam may be calculated from the electric, $\mathbf{E}$, and magnetic, $\mathbf{B}$, fields [6]:

$$\mathbf{p} = \varepsilon_0 \langle \mathbf{E} \times \mathbf{B} \rangle, \quad (1)$$

$$\mathbf{j} = \varepsilon_0 (\mathbf{r} \times \langle \mathbf{E} \times \mathbf{B} \rangle) = \mathbf{r} \times \mathbf{p}. \quad (2)$$

Equation (2) encompasses both the spin and orbital angular momentum density of a light beam.

Within the paraxial approximation, the local value of the linear momentum density of a light beam is given by [2]

$$\begin{aligned} \mathbf{p} = & i\omega \frac{\varepsilon_0}{2} (u^* \nabla u - u \nabla u^*) + \omega k \varepsilon_0 |u|^2 \mathbf{z} \\ & + \omega \sigma \frac{\varepsilon_0}{2} \frac{\partial |u|^2}{\partial r} \Phi, \end{aligned} \quad (3)$$

where $u = u(r, \phi, z)$ is the complex scalar function describing the distribution of the field amplitude. Here σ describes the degree of polarization of the light; $\sigma = \pm 1$ for right- and left-hand circularly polarized light, respectively, and $\sigma = 0$ for linearly polarized.

The cross product of this momentum density with the radius vector $\mathbf{r} = (r, 0, z)$ yields an angular momentum density. The angular momentum density in the z direction depends upon the Φ component of $\mathbf{p}$, such that

$$j_z = r p_\phi. \quad (4)$$

The final term in Eq. (3) depends upon the polarization but is independent of the azimuthal phase and, consequently, this term may be linked directly to the spin angular momentum. The first term in Eq. (3) depends upon the phase gradient and not the polarization, and so gives rise to the orbital angular momentum.

For many mode functions, u , such as for circularly polarized Laguerre-Gaussian modes, Eqs. (3) and (4) can be evaluated analytically such that the local angular momentum density in the direction of propagation is given by [2]

$$j_z = \varepsilon_0 \left\{ \omega l |u|^2 - \frac{1}{2} \omega \sigma r \frac{\partial |u|^2}{\partial r} \right\}. \quad (5)$$

The angular momentum integrated over the beam is readily shown to be equivalent to $\sigma\hbar$ per photon for the spin and $l\hbar$ per photon for the orbital angular momentum [2], that is

$$J_z = (l + \sigma)\hbar. \quad (6)$$

A theoretical discussion of the behavior of local momentum densities has been published elsewhere [7], and it should be noted that the local spin and orbital angular momentum do not have the same functional form.

As is well known, spin angular momentum does not depend upon the choice of axis and so is said to be *intrinsic*. The angular momentum which arises for any light beam from the product of the z component of linear momentum about a radius vector, may be said to be an *extrinsic* because its value depends upon the choice of calculation axis.

Berry showed [8] that the orbital angular momentum of a light beam does not depend upon the lateral position of the axis and can therefore also be said to be intrinsic, provided the direction of the axis is chosen so that the transverse momentum is zero. When integrated over the whole beam the angular momentum in the z direction is

$$\mathbf{J}_z = \varepsilon_0 \iint dx dy \mathbf{r} \times \langle \mathbf{E} \times \mathbf{B} \rangle. \quad (7)$$

If the axis is laterally displaced by $\mathbf{r}_0 \equiv (r_{0x}, r_{0y})$ it is easy to show that the change in the z component of angular momentum is given by

$$\begin{aligned} \Delta J_z &= (r_{0x} \times P_y) + (r_{0y} \times P_x) \\ &= r_{0x} \varepsilon_0 \iint dx dy \langle \mathbf{E} \times \mathbf{B} \rangle_y \\ &\quad + r_{0y} \varepsilon_0 \iint dx dy \langle \mathbf{E} \times \mathbf{B} \rangle_x. \end{aligned} \quad (8)$$

The angular momentum is intrinsic only if ΔJ_z equals zero for all values of r_{0x} and r_{0y} . This condition is satisfied only if z is stipulated as the direction for which the transverse momenta $\varepsilon_0 \iint dx dy \langle \mathbf{E} \times \mathbf{B} \rangle_x$ and $\varepsilon_0 \iint dx dy \langle \mathbf{E} \times \mathbf{B} \rangle_y$ are exactly zero.

For Laguerre-Gaussian light beams truncated by apertures, Eqs. (3) and (8) can only be evaluated numerically. Nevertheless, for all apertures, of whatever size or position, the spin angular momentum remains $\sigma\hbar$ irrespective of the choice of calculation axis and so is, as expected, intrinsic; see Fig. 1. Any beam with a helical phase front apertured symmetrically about the beam axis has zero transverse momentum and, consequently, an orbital angular momentum of $l\hbar$ per photon, independent of the axis of calculation. The orbital angular momentum of the light beam may therefore be described as intrinsic. However, when the beam is passed through an off-axis aperture, its transverse momentum is nonzero and the orbital angular momentum depends upon the choice of calculation axis and so must be described as extrinsic; see Fig. 1. An interesting result occurs when the orbital angular momentum of the apertured beam is calculated about the original beam axis. Even though the transverse momentum is nonzero, the orbital angular momentum remains $l\hbar$ per photon because r_{0x} and r_{0y} are both zero. However, it does not follow that the angular momentum of the apertured beam is intrinsic as the result does depend upon the choice of calculation axis. When any beam is apertured off axis, it is simpler and more accurate to

understand its interaction with particles by considering the components of $\mathbf{p}$ in the x - y plane. For beams with helical phase fronts, these transverse components are in the Φ direction with respect to the beam axis. It is this distinction between spin and orbital angular momentum which gives rise to differences in behavior for the interaction of light with matter.

The transfer of spin and orbital angular momentum to small particles.—The interaction of small particle with the angular momentum of a light beam has been investigated by a number of groups with the use of optical tweezers. Usually implemented by use of a high numerical-aperture microscope, optical tweezers rely on the gradient force to confine a dielectric particle near the point of highest light intensity [9]. For particles trapped on the beam axis, both the spin and orbital angular momentum have been shown to cause rotation of birefringent [10] and absorptive [11] particles, respectively. For absorbing particles, both spin and orbital angular momenta are transferred with the same efficiency so that the applied torque is proportional to the total angular momentum [12], that is $(\sigma + l)\hbar$ per photon.

In this present work we also use optical tweezers, but in this instance the particles are trapped away from the beam axis. This allows us to demonstrate the difference between particle interactions with spin and orbital angular momentum. The experimental configuration is shown in Fig. 2. Our optical tweezers are based on a 1.3 numerical aperture, $\times 100$ objective lens, configured with the trapping beam directed upwards, which allows easier access to the sample plane. This beam is generated from the 100 mW output of a commercial Nd:YLF laser transformed, using a computer generated hologram, to give a Laguerre-Gaussian mode of approximately 30 mW. The beam is circularly polarized, $\sigma = \pm 1$ with a high azimuthal mode index, $l = \pm 8$. The sign of the spin or the orbital angular momentum may be reversed by the insertion of a half-wave plate or a Dove prism, respectively [13]. The radius of maximum intensity, $r_{\max}$ of a Laguerre-Gaussian mode is given by [14],

$$r_{\max} = \sqrt{\frac{z_R l}{k}}, \quad (9)$$

where z_R is the Rayleigh range of the beam. Even under the tight focusing associated with optical tweezers, the peak intensity ring of a Laguerre-Gaussian mode of high index l may be made several μm in diameter and, consequently, be much larger than the particles it is attempting to trap.

It is not surprising, for such conditions, that we observe the particles to be confined by the gradient force at the radius of maximum light intensity and not on the beam axis. When a birefringent particle such as a calcite fragment is trapped, and circularly polarized light is converted to linear, we observe that the particle spins about its own axis. The sense of rotation is governed by the handedness of the circular polarization.

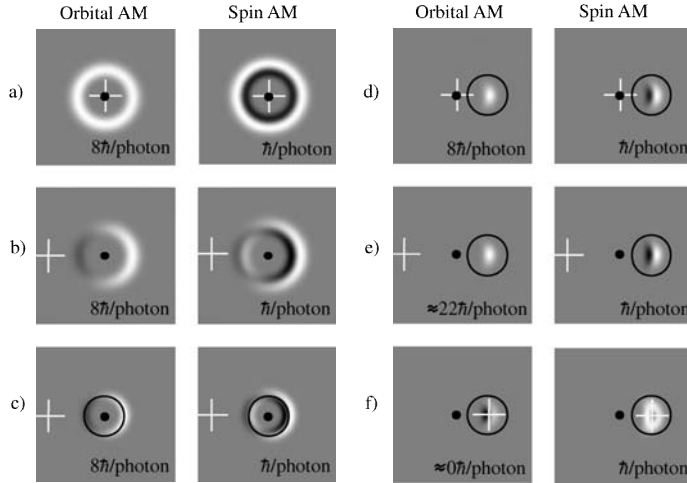


FIG. 1. Numerically calculated local spin and orbital angular momentum densities in the direction of propagation for a $l = 8$ and $\sigma = 1$ Laguerre-Gaussian mode. A positive contribution is shown in white, gray represents zero, and black a negative contribution; the black spot marks the axis of the original beam, the white cross marks the axis about which the angular momenta are calculated and, where appropriate, the black circle marks the position of a soft edged aperture. Note that the spin angular momentum is equivalent to $\sigma \hbar$ per photon irrespective of the choice of aperture or calculation axis, whereas the orbital angular momentum is only $l \hbar$ per photon if the aperture or calculation axes coincide with the axis of the original beam.

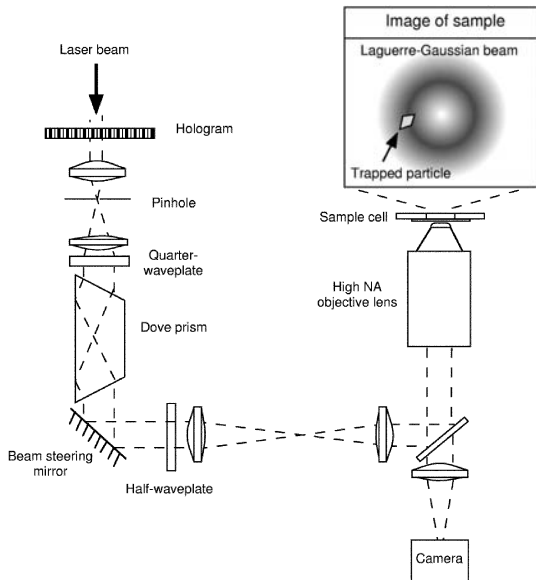


FIG. 2. The optical tweezers use a high-order Laguerre-Gaussian beam to trap the particle in the region of maximum light intensity, away from the beam axis. The circularly polarized Laguerre-Gaussian beam is generated using a quarter-wave plate and a hologram. The sense of the orbital and spin angular momentum can be reversed by the inclusion of a half-wave plate or a Dove prism, respectively.

For small particles the force arising from the light scattering, the momentum recoil force, becomes important. For a tightly focused Laguerre-Gaussian mode, the dominant component of the scattering force lies in the direction of beam propagation. The gradient force again constrains the particle to the annulus of maximum beam intensity. However, as the intensity distribution is cylindrically symmetric, the particle is not constrained azimuthally. Because the particle is trapped off the beam axis, the inclination of the helical phase fronts and the corresponding momentum result in a tangential force on the particle. We observe that a small particle, while still contained within the annular ring of light, orbits the beam axis in a direction determined by the handedness of the helical phase fronts; see Fig. 3. We conclude that the larger calcite and small particles are interacting with intrinsic spin and extrinsic orbital angular momentum, respectively. In principle, it should be possible to observe both the orbital and spin angular momenta acting simultaneously upon the same small birefringent particle. However, our observations have been inconclusive as birefringent particles small enough for the scattering force to induce a rotation about the beam axis are typically too small to see whether they are spinning about their own axis.

This orbital and spin behavior is entirely consistent with the formulation summarized in Eqs. (7) and (8). If one considers the cross section of an off-axis trapped particle to play the rôle of an aperture, then we see that the intrinsic spin of the angular momentum creates a torque about the

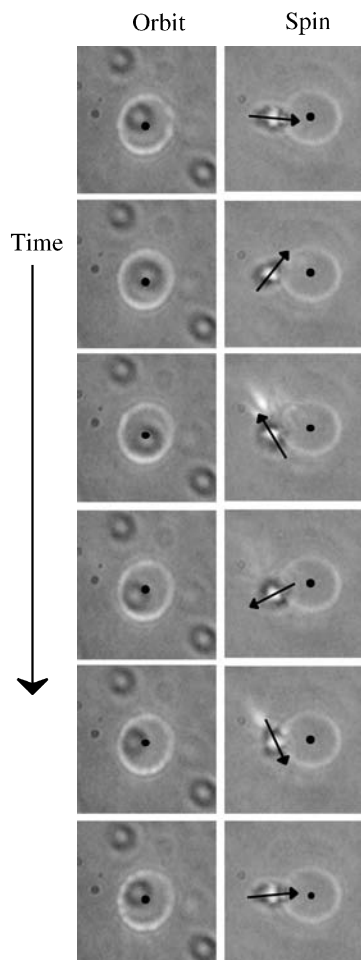


FIG. 3. Successive video frames showing particles trapped near the focus of an $l = 8$ and $\sigma = 1$ Laguerre-Gaussian mode. The left column shows particles of $\approx 1 \mu\text{m}$ diam. These particles are sufficiently small to be subject to a well-defined scattering force, allowing them to interact with the orbital angular momentum of the beam. They are set in motion, orbiting the beam axis at a frequency of ≈ 1 Hz. The right column shows a calcite fragment with a length of $\approx 3 \mu\text{m}$ and a width of about $\approx 1.5 \mu\text{m}$, which is large enough not to interact detectably with the beam's orbital angular momentum. However, due to its birefringence it interacts with the spin angular momentum of the beam and is set spinning about its own axis at ≈ 0.3 Hz.

particle's own axis, causing it to spin. A calculation of the particle's angular momentum about an arbitrary axis shows a clear distinction between the intrinsic angular momentum associated with its spinning motion and the extrinsic angular momentum associated with its orbital motion. In this situation, orbital angular momentum is better described as the result of a linear momentum component directed at a tangent to the radius vector.

Unlike the spin angular momentum of a light beam which is always intrinsic, the z component of the orbital angular momentum can be described as intrinsic only if the z direction can be stipulated such that the transverse momentum integrated over the whole beam is zero. If an interaction is with only a fraction of the beam cross section, then the orbital angular as measured about the original axis is extrinsic.

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Spin and Orbital Angular Momentum of Photons.

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Abstract. – We consider the separation of the total angular momentum of the electromagnetic field into a «spin» and an «orbital» part. Though this separation is normally considered to be unphysical and not observable, we argue that both members in the separation are separately measurable quantities. However, the commutation relations for the associated quantum operators reveal that neither of them is an angular-momentum operator.

The angular momentum $\mathbf{J}$ of the classical electromagnetic field is defined by [1]

$$\mathbf{J} = \varepsilon_0 \int d\mathbf{r} \mathbf{r} \times [\mathbf{E} \times \mathbf{B}]. \quad (1)$$

It is a conserved quantity for the free field as a result of the invariance of the free Maxwell equations under arbitrary rotations. Two contributions to (1) may be distinguished, originating from the longitudinal and the transverse part of the electric field $\mathbf{E}$ [2], respectively. The former gives the angular momentum of the instantaneous Coulomb field of the charges, the latter is the contribution from the radiation field, and is denoted by $\mathbf{J}_{\text{rad}}$. It is well known that this $\mathbf{J}_{\text{rad}}$ can be separated into two parts, called the «orbital» and the «spin» part [2,3]. One finds

$$\mathbf{J}_{\text{rad}} = \mathbf{L}_{\text{rad}} + \mathbf{S}_{\text{rad}}, \quad (2)$$

with

$$\mathbf{S}_{\text{rad}} = \varepsilon_0 \int d\mathbf{r} \mathbf{E}_{\perp} \times \mathbf{A}_{\perp}, \quad \mathbf{L}_{\text{rad}} = \varepsilon_0 \sum_l \int d\mathbf{r} \mathbf{E}_l^{\perp} (\mathbf{r} \times \nabla) A_l^{\perp}, \quad (3)$$

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where the symbol $\perp$ denotes the transverse part. Since these quantities are defined in terms of the transverse part of the vector potential $\mathbf{A}$, they are gauge invariant. The spin part $\mathbf{S}_{\text{rad}}$ is also called the intrinsic part, since it is independent of the definition of the origin of the coordinate system. In this sense this separation is similar to the separation for the total angular momentum $\mathbf{J}$ of a system of matter particles in external and internal parts.

It may be shown that for a free field $\mathbf{L}_{\text{rad}}$ and $\mathbf{S}_{\text{rad}}$ are conserved. Only in interaction with matter do they change in time [4]. They can be measured by detecting the angular-momentum effect of the radiation field interacting with matter. Indeed, already in 1936 Beth had observed one component of the spin angular momentum of light [5], by measuring the torque on a birefringent plate exerted by a circularly polarized beam of light. (See also the discussion in [6].) Moreover, it has recently been shown that also the orbital angular momentum of a paraxial laser beam can be measured [7-9]. In both cases the measurement concerns the component along the propagation direction. Both experiments can be understood within a classical framework.

However, it has generally been assumed that the separation (2) is unphysical [2, 3, 10]. In the first place, the spin part should correspond to the total angular momentum of a particle in its rest frame, but this frame does not exist for a photon. This argument does not, of course, prevent one from assigning a definite value to the spin of a photon, but it does preclude the definition of a spin vector. In fact, for a massless spin- s particle the argument implies that one can only define the component of the spin along the propagation direction, which assumes the values $\pm s$. Secondly, the action of the quantum-mechanical operators for spin and orbital angular momentum on a physical state of the radiation field is believed to yield unphysical states (we will elaborate this second argument below), so that $\mathbf{S}_{\text{rad}}$ and $\mathbf{L}_{\text{rad}}$ would not be observables in the quantum-mechanical sense. We show here that the latter objection is wrong, while the former does not apply to (2). The question remains whether in a quantum-mechanical theory orbital angular momentum and spin are still observables of the radiation field, and what physical meaning they have. In this letter we investigate the quantum-mechanical operators $\mathbf{S}_{\text{rad}}$ and $\mathbf{L}_{\text{rad}}$.

The transverse electromagnetic field can be quantized by first expanding the fields in any complete set of transverse vector functions. We assume that we have defined a complete orthonormal set of transverse mode functions $\mathbf{F}_\lambda$ with λ denoting the quantum numbers of the field, which we take discrete for simplicity. These functions are chosen as solutions of

$$\nabla^2 \mathbf{F}_\lambda = -k^2 \mathbf{F}_\lambda; \quad \nabla \cdot \mathbf{F}_\lambda = 0, \quad (4)$$

so that $\hbar\omega_\lambda \equiv \hbar kc$ is the energy of a photon in a mode λ . For our purpose the remaining (three) quantum numbers may be left unspecified. Orthonormality implies that

$$\langle \mathbf{F}_\lambda | \mathbf{F}_{\lambda'} \rangle \equiv \int d\mathbf{r} \mathbf{F}_\lambda^* \cdot \mathbf{F}_{\lambda'} = \delta_{\lambda\lambda'}. \quad (5)$$

One can expand the transverse part of the vector potential $\mathbf{A}_\perp$ in this complete set. After quantization $\mathbf{A}_\perp$ and the electric and magnetic fields $\mathbf{E}_\perp$ and $\mathbf{B}$ become operators,

$$\begin{cases} \mathbf{A}_\perp = \sum_\lambda \mathcal{A}_\lambda [a_\lambda \mathbf{F}_\lambda + a_\lambda^\dagger \mathbf{F}_\lambda^*], \\ \mathbf{E}_\perp = \sum_\lambda i\mathcal{E}_\lambda [a_\lambda \mathbf{F}_\lambda - a_\lambda^\dagger \mathbf{F}_\lambda^*], \\ \mathbf{B} = \sum_\lambda \mathcal{B}_\lambda [a_\lambda \nabla \times \mathbf{F}_\lambda + a_\lambda^\dagger \nabla \times \mathbf{F}_\lambda^*], \end{cases} \quad (6)$$

where a_λ and $a_\lambda^\dagger$ are the annihilation and creation operators for photons in the mode λ , which

satisfy the usual commutation rules

$$[a_\lambda, a_{\lambda'}^\dagger] = \delta_{\lambda\lambda'}. \quad (7)$$

The normalization factors in (6) are $\mathcal{G}_\lambda = \sqrt{\hbar\omega_\lambda/2\varepsilon_0}$ and $\mathcal{A}_\lambda = \sqrt{\hbar/2\varepsilon_0\omega_\lambda}$. The field operators (6) act, just as a_λ and $a_\lambda^\dagger$, in Fock space [2,3]. This space is spanned by the Fock state vectors, which describe the quantum state of the radiation field.

The operator for the total angular momentum of the transverse electromagnetic field has the same form (1) as for the classical field. There is no need to symmetrize this expression, since $\mathbf{E}_\perp$ and $\mathbf{B}$, when evaluated at the same position and time, commute. In the same way as for the classical field this operator can be decomposed into two terms, corresponding to «spin» and «orbital» angular momentum. By substituting the expansions (6), one finds

$$\begin{cases} S_{\text{rad}} = \varepsilon_0 \int d\mathbf{r} \mathbf{E}_\perp \times \mathbf{A}_\perp = \frac{1}{2} \sum_{\lambda\lambda'} [a_\lambda^\dagger a_{\lambda'} + a_{\lambda'} a_\lambda^\dagger] \langle \mathbf{F}_\lambda | \hat{\mathbf{S}} | \mathbf{F}_{\lambda'} \rangle, \\ L_{\text{rad}} = \varepsilon_0 \sum_l \int d\mathbf{r} E_l^\perp (\mathbf{r} \times \nabla) A_l^\perp = \frac{1}{2} \sum_{\lambda\lambda'} [a_\lambda^\dagger a_{\lambda'} + a_{\lambda'} a_\lambda^\dagger] \langle \mathbf{F}_\lambda | \hat{\mathbf{L}} | \mathbf{F}_{\lambda'} \rangle. \end{cases} \quad (8)$$

Similar expressions have been derived before in reciprocal (Fourier) space [2,3]. The operators $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$ are defined as

$$\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla), \quad (\hat{\mathbf{S}}_k)_{ij} = -i\hbar\varepsilon_{ijk}, \quad (9)$$

for $i, j, k = x, y, z$, with ε_{ijk} the Levi-Civita pseudotensor. Notice that the operator $\hat{\mathbf{L}}$ acts on the position dependence of $\mathbf{F}_\lambda$, whereas $\hat{\mathbf{S}}$ operates as a vector of 3×3 matrices on the Cartesian components of $\mathbf{F}_\lambda$. These operators, which we distinguish from field operators by a caret, have the same form as the quantum-mechanical operators for orbital angular momentum and spin of a spin-1 particle, respectively. Expressions (8) for S_{rad} and L_{rad} thus closely resemble the expectation values for the quantum-mechanical spin and orbital angular momentum in a given quantum state of a spin-1 particle. Here the state of the radiation field is represented by the transverse mode function $\mathbf{F}_\lambda$ of the electromagnetic field. Now the operators $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$ do not preserve this transversality. Hence, it appears that $\hat{\mathbf{S}}|\mathbf{F}_\lambda\rangle$ and $\hat{\mathbf{L}}|\mathbf{F}_\lambda\rangle$ no longer represent physical states of the radiation field. The conclusion would be that S_{rad} and L_{rad} cannot be observables [2,3]. This is the second objection against the separation (2), as mentioned above. However, the operators S_{rad} and $\hat{\mathbf{S}}$ act in different spaces. The former is an operator that acts in Fock space, whereas the latter operator acts on the classical field modes, and not on quantum states of the radiation field. Since S_{rad} is a gauge-invariant Hermitian operator acting on the quantum states of the radiation field, S_{rad} is an observable. The same conclusion holds for the orbital angular momentum L_{rad} . Hence we conclude that the second argument that S_{rad} and L_{rad} are not observable is invalid.

The operators $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$ satisfy the standard commutation rules for angular-momentum operators

$$\begin{cases} [\hat{\mathbf{S}}_i, \hat{\mathbf{S}}_j] = \sum_k i\hbar\varepsilon_{ijk} \hat{\mathbf{S}}_k, \\ [\hat{\mathbf{L}}_i, \hat{\mathbf{L}}_j] = \sum_k i\hbar\varepsilon_{ijk} \hat{\mathbf{L}}_k. \end{cases} \quad (10)$$

Since the total angular-momentum operator $\mathbf{J}_{\text{rad}}$ generates rotations in space, as was

explicitly verified in [11], also $\mathbf{J}_{\text{rad}}$ obeys this same rule

$$[J_i, J_j] = \sum_k i\hbar \varepsilon_{ijk} J_k. \quad (11)$$

Relations (8) may suggest that the operators $\mathbf{S}_{\text{rad}}$ and $\mathbf{L}_{\text{rad}}$ obey the same commutation rules (10). This turn out not to be true. This is most easily shown when one chooses circularly polarized plane waves as the field modes, so that

$$\mathbf{F}_\lambda = \varepsilon_{\mathbf{k},s} \exp[i\mathbf{k} \cdot \mathbf{r}] / \sqrt{N}, \quad (12)$$

with $\mathbf{k}$ the wave vector and with $\varepsilon_{\mathbf{k},s}$ for $s = \pm 1$ the (orthogonal) right and left circular-polarization vectors. The number N is a normalization factor. In this representation the spin operator $\mathbf{S}_{\text{rad}}$ takes the simple form [11]

$$\mathbf{S}_{\text{rad}} = \sum_{\mathbf{k}} \frac{\hbar \mathbf{k}}{k} (a_{\mathbf{k},1}^\dagger a_{\mathbf{k},1} - a_{\mathbf{k},-1}^\dagger a_{\mathbf{k},-1}). \quad (13)$$

As this operator contains only number operators, all its components mutually commute, so that

$$[S_i, S_j] = 0. \quad (14)$$

This simple, but somewhat surprising result seems to have gone unnoticed before. It implies that $\mathbf{S}_{\text{rad}}$ cannot generate rotations of the polarization of the field. Therefore, it cannot be considered as the spin angular-momentum operator. Hence the first objection to the separation (2) mentioned in the introduction does not apply. Furthermore, from the representation (13) of $\mathbf{S}_{\text{rad}}$ it follows that for an arbitrary unit vector $\mathbf{u}$, the operator

$$R(\alpha) = \exp[-i\alpha \mathbf{u} \cdot \mathbf{S}_{\text{rad}} / \hbar] \quad (15)$$

rotates the polarization of each $\mathbf{k}$ -component of the field around its wave vector $\mathbf{k}$, over an angle $\alpha \cos(\mathbf{u} \cdot \mathbf{k} / k)$. Hence the effective rotation vector is the projection of $\mathbf{u}$ along the wave vector $\mathbf{k}$. This rotation of the polarization preserves the transversality of the field.

In order to find the commutation rules for $\mathbf{L}_{\text{rad}}$, we first note that both $\mathbf{S}_{\text{rad}}$ and $\mathbf{L}_{\text{rad}}$ transform as vectors under rotations, so that

$$[J_i, L_j] = \sum_{\mathbf{k}} i\hbar \varepsilon_{ijk} L_k, \quad [J_i, S_j] = \sum_{\mathbf{k}} i\hbar \varepsilon_{ijk} S_k. \quad (16)$$

This can be derived from the general expressions (3). From the commutation relations (14) and (16), it follows immediately that the orbital angular-momentum operator obeys

$$[L_i, L_j] = \sum_{\mathbf{k}} i\hbar \varepsilon_{ijk} (L_k - S_k), \quad [L_i, S_j] = \sum_{\mathbf{k}} i\hbar \varepsilon_{ijk} S_k. \quad (17)$$

This shows explicitly that also $\mathbf{L}_{\text{rad}}$ is not an angular-momentum operator. It does not generate orbital rotations. Furthermore, $\mathbf{L}_{\text{rad}}$ and $\mathbf{S}_{\text{rad}}$ do not commute with each other, unlike $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$, and unlike the corresponding operators for internal and external angular momentum of matter particles.

Finally, the three commuting components of $\mathbf{S}_{\text{rad}}$ can be simultaneously diagonalized. Indeed, the form (13) of the spin operator makes it clear that in a circularly polarized plane-wave mode the spin vector has a well-defined value $\pm \hbar \mathbf{k} / k$. Since this very mode has also a well-defined linear momentum $\hbar \mathbf{k}$, $\mathbf{S}_{\text{rad}}$ commutes with the linear momentum operator $\mathbf{P}_{\text{rad}}$ of

the radiation field, which is in the same representation as (13) given by

$$\mathbf{P}_{\text{rad}} = \sum_{\mathbf{k}} \hbar \mathbf{k} (a_{\mathbf{k}, 1}^{\dagger} a_{\mathbf{k}, 1} + a_{\mathbf{k}, -1}^{\dagger} a_{\mathbf{k}, -1}). \quad (18)$$

In conclusion, we have shown that both «spin» $\mathbf{S}_{\text{rad}}$ and «orbital» angular momentum $\mathbf{L}_{\text{rad}}$ of a photon are well defined and separately measurable. However, these two quantities do not have the meaning of angular momentum. The commutation relations show that neither $\mathbf{S}_{\text{rad}}$ nor $\mathbf{L}_{\text{rad}}$ generate rotations. Instead, $\mathbf{S}_{\text{rad}}$ transforms the polarization of the field such that the transversality condition is preserved. In order to further investigate the physical significance of $\mathbf{L}_{\text{rad}}$ and $\mathbf{S}_{\text{rad}}$, one should consider the interaction of matter with the radiation field, in particular with a photon in an eigenstate of a given component of $\mathbf{L}_{\text{rad}}$ and $\mathbf{S}_{\text{rad}}$. This will be discussed elsewhere [4].

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Orbital angular momentum and nonparaxial light beams

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Abstract

The simple relationship between total angular momentum and energy and the seemingly natural separation of the angular momentum into spin and orbital components in the paraxial approximation, are investigated for a general nonparaxial form of monochromatic beam with near cylindrical symmetry.

1. Introduction

It has been argued [1] that the Laguerre–Gaussian modes [2] possess well-defined angular momentum that can be decomposed into an orbital component and a spin component associated with its polarisation. Such modes are not exact solutions of the full Maxwell theory but rather are solutions of the paraxial wave equation. The well-known analogy between paraxial theory and quantum mechanics [3] has lead to an eigenfunction description of the angular momentum of light [4] and a comparison between paraxial wave optics and quantum harmonic oscillators [5] which exploits the language of ladder operators. The fact that the Laguerre–Gaussian beams may be expressed in terms of, and experimentally produced from, the Hermite–Gaussian beams emitted from real lasers [6] permits experimental examination of the phenomenon.

Allen et al. [1] have noted that paraxial beams possessing cylindrical symmetry possess well defined total angular momentum per unit energy flux and that this angular momentum can be divided into orbital and spin components. It is possible that this is an artefact of the paraxial approximation. This paper seeks to address this question by examining the angular momentum properties of light without invoking the paraxial approximation. We obtain the general form for a beam propagating in the z -direction for which the x - and y -components of the electric field have azimuthal dependence $\exp(i\ell\phi)$. Beyond the paraxial approximation the total angular momentum per unit energy may still be calculated but it is no longer straightforward to identify the orbital and spin components. We also obtain expressions for the energy and angular momentum per unit length and identify a contribution not present in the paraxial approximation. We illustrate our results with an example that reduces to the Laguerre–Gaussian modes in the paraxial limit.

2. The paraxial approximation and angular momentum

The paraxial wave-equation provides an adequate description of nearly all optical resonator and beam propagation problems that arise in the study of real lasers [2]. It may readily be derived from the scalar wave-equation, which for a monochromatic wave

$$\psi(x, y, z, t) = \chi(x, y, z) \exp(-i\omega t) \quad (2.1)$$

reduces to the Helmholtz equation for the space-dependence

$$(\nabla^2 + k^2)\chi(x, y, z) = 0, \quad (2.2)$$

where $k = c/\omega$. We consider an optical beam propagating in the z -direction and introduce the form

$$\chi(x, y, z) = u(x, y, z) \exp(ikz). \quad (2.3)$$

The z -dependence of the wave amplitude $u(x, y, z)$ is essentially due to diffractive effects and is, in general, slow compared with the transverse variation in the width of the beam. The mathematical expression of this slow variation lies in the inequality

$$\left| \frac{\partial^2 u}{\partial z^2} \right| \ll \left| k \frac{\partial u}{\partial z} \right|, \left| \frac{\partial^2 u}{\partial x^2} \right|, \left| \frac{\partial^2 u}{\partial y^2} \right|. \quad (2.4)$$

Dropping the second derivative of u with respect to z leads to the paraxial wave equation

$$i \frac{\partial u}{\partial z} = -\frac{1}{2k} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u. \quad (2.5)$$

Lax et al. [7] have pointed out that the paraxial approximation makes the inconsistent assumption that one can have a plane polarised electromagnetic wave whose electric vector depends on transverse distance. They have, however, established that this result is a consistent zero-order solution of Maxwell's equations and identified the higher order correction terms. Davis [8] showed that if the vector potential of the field is assumed to be plane-polarised, the non-vanishing component of the vector potential obeys a scalar wave-equation and produces the same low-order components of the transverse and longitudinal field of a Gaussian beam as derived that by Lax et al. [7]. Pattanayak and Agrawal [9] have also obtained corrections to the paraxial theory and note that: "In the nonparaxial case, polarization is no longer a global property and changes on propagation." The approach of Davis et al., developed by Haus [10], forms the basis of the orbital angular momentum calculation presented by Allen et al. [1].

Poynting [11] inferred by recourse to mechanical analogy that circularly polarised light should exert a torque on a unit area of a quarter wave-plate equal to $\lambda/2\pi$ times the light energy per unit volume, as the plate converts the polarisation of the light from circular to linear polarisation. It follows that the ratio of angular momentum flux to energy flux becomes σ_z/ω where ω is the frequency of the light and σ_z is ± 1 for left- or right-handed light. The torque on a suspended birefringent plate was measured by Beth [12].

The calculation of the resulting torque assumed illumination of the plate by a plane wave and, although the concept of orbital angular momentum of light is well-known, the experiment takes cognisance only of the effect of spin. Rose [13] has considered the role of orbital angular momentum in multipole radiation and connects energy and total angular momentum with the free electromagnetic field. It is common, particularly in nuclear physics, to associate orbital angular momentum with radiative decay but Biedenhahn and Louck [14] warn that: "multipole fields for a given value of the total angular momentum ... do not have ... a definite value of the orbital angular momentum. It is, indeed, not possible to separate the total angular momentum of the photon field into an 'orbital' and a 'spin' part (this would contradict gauge invariance); the best that can be done is to define the helicity operator $\mathcal{S} \cdot \hat{\mathbf{P}}$ which is an observable (Beth)."

Nevertheless, Allen et al. [1] show for Laguerre–Gaussian beams *within the paraxial approximation* that the z -component of the total angular momentum density per unit power is given by

$$M_z = \frac{l}{\omega} |u|^2 + \frac{\sigma_z \rho}{2\omega} \frac{\partial |u|^2}{\partial \rho} \quad (2.6)$$

and that the ratio of angular momentum flux to energy flux is

$$\frac{J_z}{cP_z} = \frac{(l + \sigma_z)}{\omega}, \quad (2.7)$$

where P_z is the z -component of the momentum and σ_z is ± 1 for circularly polarised light and 0 for linearly polarised light. Here $u(\rho, \phi, z)$ is the complex scalar function, expressed in cylindrical polar coordinates, describing the distribution of the field amplitude of a Laguerre–Gaussian beam and is given by

$$u_{pl}(\rho, \phi, z) = \frac{C}{\sqrt{1 + z^2/z_R^2}} \left(\frac{\rho\sqrt{2}}{w(z)} \right)^l L_p^l \left(\frac{2\rho^2}{w^2(z)} \right) \exp \left(\frac{-\rho^2}{w^2(z)} \right) \exp \left(\frac{-ik\rho^2 z}{2(z^2 + z_R^2)} \right) \\ \times \exp(\pm i l \phi) \exp[i(2p + l + 1) \tan^{-1}(z/z_R)]. \quad (2.8)$$

where z_R is the Rayleigh range, $w(z)$ is the radius of the beam, L_p^l is the associated Laguerre polynomial, C is a constant and the beam waist is situated at $z=0$. As may be seen, the z -component of both the angular momentum/energy ratio and the angular momentum density naturally devolve into orbital and spin components associated with the values l and σ_z , respectively. The quantity l arises from the azimuthal dependence in the term $\exp(i l \phi)$ while σ_z , where $\sigma_z = i(\alpha\beta^* - \beta\alpha^*)$, is readily identifiable with spin when the electric field in the x - and y -directions is proportional to $(\alpha\hat{x} + \beta\hat{y})$. It may be noted that the spin term in the equation for M_z depends on the gradient of the field intensity. A plane wave propagating in the z -direction has no component of the angular momentum density $\epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B})$ in the z -direction. The correction terms for the field found by Davis [8] would lead to a small modification to expression (2.6) but their effect is not readily summed to all orders.

We conclude this section by examining some general properties of the electromagnetic field that will be useful in our discussion of angular momentum. We write the densities of linear and angular momentum for the field in terms of the electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) field as [15,16]

$$\mathbf{p} = \epsilon_0 \mathbf{E} \times \mathbf{B}, \quad \mathbf{j} = \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}). \quad (2.9, 2.10)$$

The total linear and angular momenta are found by integrating these densities over all space

$$\mathbf{P} = \int d^3r \mathbf{p}, \quad \mathbf{J} = \int d^3r \mathbf{j}. \quad (2.11, 2.12)$$

Specialising to monochromatic fields we introduce the complex notation

$$\mathbf{E} = (\mathbf{E} e^{-i\omega t} + \mathbf{E}^* e^{i\omega t})/2, \quad \mathbf{B} = (\mathbf{B} e^{-i\omega t} + \mathbf{B}^* e^{i\omega t})/2, \quad (2.13, 2.14)$$

where the asterisk denotes complex conjugation. Following van Enk and Nienhuis [4] we eliminate the magnetic field using the Maxwell equation

$$i\omega \mathbf{B} = \nabla \times \mathbf{E} \quad (2.15)$$

to write the field linear and angular momenta as

$$\mathbf{P} = \frac{\epsilon_0}{2i\omega} \int d^3r \mathbf{E}^* \times (\nabla \times \mathbf{E}), \quad \mathbf{J} = \frac{\epsilon_0}{2i\omega} \int d^3r \mathbf{r} \times [\mathbf{E}^* \times (\nabla \times \mathbf{E})]. \quad (2.16, 2.17)$$

On partial integration and using the transversality of $\mathbf{E}$ we find that for fields that vanish sufficiently quickly as $|\mathbf{r}| \rightarrow \infty$ we obtain the simpler expressions

$$\mathbf{P} = \frac{\epsilon_0}{2i\omega} \int d^3r \sum_{j=x,y,z} \mathbf{E}_j^* \nabla E_j, \quad (2.18)$$

$$\mathbf{J} = \frac{\epsilon_0}{2i\omega} \int d^3r \sum_{j=x,y,z} \mathbf{E}_j^* (\mathbf{r} \times \nabla) E_j + \frac{\epsilon_0}{2i\omega} \int d^3r \mathbf{E}^* \times \mathbf{E}. \quad (2.19)$$

It is tempting to associate the two terms making up the angular momentum with the orbital and spin components. Indeed, within the paraxial approximation these two terms give, for the Laguerre–Gaussian modes, contributions proportional to l and σ_z respectively. Yet as van Enk and Nienhuis note “the interpretation of the first and second terms on the right hand side of Eq. (2.19) as orbital angular momentum and spin is, although seemingly obvious, not without fundamental difficulties.”

In addition to the linear and angular momenta we also require the energy associated with the field. The energy density for the free electromagnetic field is [15]

$$U = \frac{1}{2} (\epsilon_0 \mathbf{E}^2 + \mu_0^{-1} \mathbf{B}^2) , \quad (2.20)$$

giving a total energy

$$U = \frac{1}{2} \int d^3r (\epsilon_0 \mathbf{E}^2 + \mu_0^{-1} \mathbf{B}^2) = \frac{\epsilon_0}{2} \int d^3r \mathbf{E}^* \cdot \mathbf{E} . \quad (2.21)$$

Finally we will examine the linear momentum, angular momentum and energy per unit length along a beam propagating in the z -direction. These quantities are found by integration of the respective densities over the x - y plane perpendicular to the direction of propagation. However, these quantities are not, in general, time-independent and therefore we will perform a cycle average over an optical period $2\pi/\omega$ to give the cycle averaged energy, linear momentum, angular momentum and energy per unit length which we denote $\mathcal{P}$, $\mathcal{J}$ and $\mathcal{E}$, respectively.

$$\mathcal{P} = \iint dx dy \mathbf{E}^* \times (\nabla \times \mathbf{E}) , \quad (2.22)$$

$$\mathcal{J} = \frac{\epsilon_0}{2i\omega} \iint dx dy \mathbf{r} \times (\mathbf{E}^* \times (\nabla \times \mathbf{E})) , \quad (2.23)$$

$$\mathcal{E} = \frac{\epsilon_0}{2} \iint dx dy \mathbf{E}^* \cdot \mathbf{E} . \quad (2.24)$$

We emphasise that expressions (2.16) to (2.24) have a validity that does not rely on the paraxial approximation and they form the basis for our analysis of nonparaxial beams.

3. Nonparaxial beams and angular momentum

The principal aim of this paper is to determine the extent to which some of the remarkable properties associated with the angular momentum carried by Laguerre–Gaussian beams are associated with the paraxial approximation. In order to address this question we develop a description of light beams based on the full Maxwell theory. There is, of course, a very wide variety of possible nonparaxial beams, but the considerations of the preceding section lead us to focus on monochromatic beams for which the x - and y -components of the electric field are proportional to $\exp(i l \phi)$. Accordingly we write the electric field as

$$\mathbf{E} = (\alpha \hat{x} + \beta \hat{y}) E(z, \rho) e^{i l \phi} + \hat{z} E_z , \quad (3.1)$$

where each component satisfies the Helmholtz equation and E_z has to be chosen to ensure the transversality of the electric field. We find that the general solution for $E(z, \rho)$ has the form

$$E(z, \rho) = \int_0^k d\kappa E(\kappa) J_l(\kappa \rho) \exp(i \sqrt{k^2 - \kappa^2} z) , \quad (3.2)$$

where $J_l(\kappa\rho)$ is the Bessel function of order l . As we have noted, E_z must also satisfy the Helmholtz equation and must have the correct form to give $\nabla \cdot \mathbf{E} = 0$. After lengthy but straight forward algebra we find that the electric field for a beam has the form

$$\mathbf{E} = \int_0^k d\kappa E(\kappa) e^{i\phi} \exp(i\sqrt{k^2 - \kappa^2}z) \times \left((\alpha\hat{x} + \beta\hat{y}) J_l(\kappa\rho) + \hat{z} \frac{\kappa}{2\sqrt{k^2 - \kappa^2}} [(\alpha - \beta) e^{-i\phi} J_{l-1}(\kappa\rho) - (\alpha + \beta) e^{i\phi} J_{l+1}(\kappa\rho)] \right). \quad (3.3)$$

A general transverse electric field of the form (3.1) includes an arbitrary superposition of z -polarised plane waves propagating in the x - y plane. The inclusion of such waves is not of interest here as they do not form a z -propagating beam. We note that the z -component of this field has a different dependence on the azimuthal angle to that which has been imposed on the x - and y -components. Our analysis of the properties of nonparaxial beams is based on the general form (3.3). We should note that this beam includes, as a special case, Durnin's non-diffracting Bessel beams [17].

The total energy, linear and angular momenta for the beam formally diverge. However, the corresponding quantities evaluated per unit length are better behaved. In particular, the energy per unit length is

$$\begin{aligned} \mathcal{E} = & \frac{\epsilon_0}{2} \int_0^{2\pi} d\phi \int_0^\infty \rho d\rho \mathbf{E}^* \cdot \mathbf{E} = \pi\epsilon_0 \int_0^k d\kappa \int_0^k d\kappa' \int_0^\infty \rho d\rho E^*(\kappa') E(\kappa) \exp[i(\sqrt{k^2 - \kappa^2} - \sqrt{k^2 - \kappa'^2})z] \\ & \times \left(J_l(\kappa'\rho) J_l(\kappa\rho) + \frac{\kappa'\kappa}{4\sqrt{k^2 - \kappa'^2} \sqrt{k^2 - \kappa^2}} [|\alpha - \beta|^2 J_{l-1}(\kappa'\rho) J_{l-1}(\kappa\rho) + |\alpha + \beta|^2 J_{l+1}(\kappa'\rho) J_{l+1}(\kappa\rho)] \right). \end{aligned} \quad (3.4)$$

Surprisingly, two of these integrals can be performed together using the Fourier-Bessel integral [18]

$$\int_p^q R \Phi(R) dR \int_0^\infty \lambda J_n(\lambda R) J_n(\lambda r) d\lambda = \Phi(r), \quad (3.5)$$

for $p < r < q$. The resulting energy per unit length acquires the simple form

$$\mathcal{E} = \frac{\pi\epsilon_0}{2} \int_0^k d\kappa \frac{|E(\kappa)|^2 (2k^2 - \kappa^2)}{\kappa(k^2 - \kappa^2)}. \quad (3.6)$$

The physical requirement that this quantity be finite, places constraints on the possible form of $E(\kappa)$; in particular it must tend to zero sufficiently quickly as $\kappa \rightarrow 0$, k . A similar calculation using, in addition, the Bessel function identity

$$\frac{d}{dx} J_n(x) \pm \frac{n}{x} J_n(x) = \pm J_{n\mp 1}(x) \quad (3.7)$$

gives equally simple exact forms for the linear and angular momentum per unit length associated with the beam

$$\mathcal{P}_z = \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 (2k^2 - \kappa^2)}{\kappa\sqrt{k^2 - \kappa^2}} \quad (3.8)$$

$$\mathcal{J}_z = [l + i(\alpha\beta^* - \beta\alpha^*)] \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 (2k^2 - \kappa^2)}{\kappa(k^2 - \kappa^2)} + i(\alpha\beta^* - \beta\alpha^*) \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 \kappa}{k^2 - \kappa^2}. \quad (3.9)$$

We note that the simple relationships

$$\mathcal{E} = \mathcal{P}_z c, \quad \mathcal{J}_z / \mathcal{E} = (l + \sigma_z) / \omega, \quad (3.10, 3.11)$$

which hold within the paraxial approximation, do not necessarily have general validity. The paraxial limit of the expressions (3.6), (3.8) and (3.9) is recovered, as are the simple relationships between $\mathcal{E}$, $\mathcal{P}_z$ and $\mathcal{J}_z$, if the function $E(\kappa)$ is sharply peaked at small values of κ so that we can neglect terms of order $(\kappa/k)^2$ in the integrands.

The expression for the total angular momentum per unit length, $\mathcal{J}_z$, bears closer examination. The first integral in Eq. (3.9) is simply $(l + \sigma_z)\mathcal{E}/\omega$, while the second term which depends on σ_z , but not on l , is the correction to the simple paraxial result. Thus, for ‘linearly polarised’ light, with $\sigma_z = 0$, the paraxial relation between $\mathcal{J}_z$ and $\mathcal{E}$ is *exact*. It is interesting to examine the origin of the components of $\mathcal{J}_z$. Recall from Eq. (2.23) that $\mathcal{J}_z$ of the angular momentum per unit length is obtained by integrating the z -component of $[\mathbf{r} \times \{\mathbf{E}^* \times (\nabla \times \mathbf{E})\}]$ over the x, y -plane. Application of vector calculus allows us to rewrite this expression in the form

$$[\mathbf{r} \times \{\mathbf{E}^* \times (\nabla \times \mathbf{E})\}]_z = \sum_{j=x,y,z} E_j^* \frac{\partial}{\partial \phi} E_j + (E_x^* E_y - E_y^* E_x) - \sum_{j=x,y,z} E_j^* \frac{\partial}{\partial x_j} (xE_y - yE_x), \quad (3.12)$$

where the last sum produces the final term in Eq. (3.9) on integration across the beam profile. It is tempting, following van Enk and Nienhuis [4] and the discussion in Sec. 2, to ascribe the orbital angular momentum and spin to the first and second terms in Eq. (3.12) respectively as is done in the paraxial limit. However, we find that these two contributions to the angular momentum per unit length are

$$\frac{\epsilon_0}{2i\omega} \int_0^{2\pi} d\phi \int_0^\infty \rho d\rho \sum_{j=x,y,z} E_j^* \frac{\partial}{\partial \phi} E_j = l \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 (2k^2 - \kappa^2)}{\kappa(k^2 - \kappa^2)} + \sigma_z \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 \kappa}{k^2 - \kappa^2}, \quad (3.13)$$

$$\frac{\epsilon_0}{2i\omega} \int_0^{2\pi} d\phi \int_0^\infty \rho d\rho (E_x^* E_y - E_y^* E_x) = \sigma_z \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 2}{\kappa}. \quad (3.14)$$

While the second of these terms depends on σ_z but not on l , the first depends on both. In the general nonparaxial case there is no simple separation into an l -dependent orbital and a σ_z -dependent spin component of angular momentum. It is worth noting, however, that the sum of these two contributions is precisely $(l + \sigma_z)\mathcal{E}/\omega$. The contribution to the z -component of the angular momentum per unit length of the final term in Eq. (3.12) is

$$- \frac{\epsilon_0}{2i\omega} \int_0^{2\pi} d\phi \int_0^\infty \rho d\rho \sum_{j=x,y,z} E_j^* \frac{\partial}{\partial x_j} (xE_y - yE_x) = \sigma_z \frac{\pi\epsilon_0}{2\omega} \int_0^k d\kappa \frac{|E(\kappa)|^2 \kappa}{k^2 - \kappa^2}. \quad (3.15)$$

To highlight the nonparaxial nature of this term we write it in a slightly different form. Reverting to Cartesian coordinates and performing the integrations containing derivatives with respect to x and y by parts we find

$$\begin{aligned} & - \frac{\epsilon_0}{2i\omega} \int_{-\infty}^\infty dx \int_{-\infty}^\infty dy \sum_{j=x,y,z} E_j^* \frac{\partial}{\partial x_j} (xE_y - yE_x) \\ & = - \frac{\epsilon_0}{2i\omega} \int_{-\infty}^\infty dy [E_x^* (xE_y - yE_x)]_{x=-\infty}^\infty - \frac{\epsilon_0}{2i\omega} \int_{-\infty}^\infty dx [E_y^* (xE_y - yE_x)]_{y=-\infty}^\infty \\ & + \frac{\epsilon_0}{2i\omega} \int_{-\infty}^\infty dx \int_{-\infty}^\infty dy (xE_y - yE_x) \left(\frac{\partial}{\partial x} E_x^* + \frac{\partial}{\partial y} E_y^* \right) - \frac{\epsilon_0}{2i\omega} \int_{-\infty}^\infty dx \int_{-\infty}^\infty dy E_z^* \frac{\partial}{\partial z} (xE_y - yE_x). \end{aligned} \quad (3.16)$$

The first two terms will be zero for physical fields (which fall off sufficiently quickly as $\rho \rightarrow \infty$). Writing the final term in Eq. (3.16) as

$$\frac{\epsilon_0}{2i\omega} \left(-\frac{\partial}{\partial z} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy E_z^* (xE_y - yE_x) + \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy (xE_y - yE_x) \frac{\partial}{\partial z} E_z^* \right) \quad (3.17)$$

and using the fact that $\nabla \cdot \mathbf{E}^* = 0$, the third term in Eq. (3.16) combines with Eq. (3.17) to give

$$-\frac{\epsilon_0}{2i\omega} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \sum_{j=x,y,z} E_j^* \frac{\partial}{\partial x_j} (xE_y - yE_x) = -\frac{\epsilon_0}{2i\omega} \frac{\partial}{\partial z} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy E_z^* (xE_y - yE_x). \quad (3.18)$$

The nonparaxial nature of this contribution is now clear; examining the general field expression (3.3) we see that in the paraxial limit the z -component of the field will be smaller than the x - and y -components by $\sim \kappa/k$. In addition the effect of the derivative with respect to z is to introduce a factor of order κ/k . The net effect is that the contribution (3.18) is of the order of magnitude of terms that are neglected within the paraxial approximation.

In concluding this section we summarise our main results. The general form for a monochromatic field with (near) cylindrical symmetry (3.3) leads to simple expressions for the energy, linear momentum and angular momentum per unit length, given in Eqs. (3.6), (3.8) and (3.9), respectively. These have a validity which does not rely on the paraxial approximation, but the simple relationships between these quantities that holds in the paraxial limit is lost. Moreover, the seemingly natural separation of the angular momentum into σ_z -dependent spin and l -dependent orbital components apparent in the paraxial approximation is no longer possible.

4. A nonparaxial generalisation of the Laguerre–Gaussian mode

The considerations of the preceding section have lead to a general description of the properties of monochromatic beams with (near) cylindrical symmetry. These are the natural, nonparaxial, extension of the familiar Laguerre–Gaussian modes [1,2]. We wish to emphasise that a laboratory laser mode is only accurately described by a solution of the full Maxwell equations. Its representation as a Hermite– or Laguerre–Gaussian mode is only an approximation. It seems likely, therefore, that careful experiments on sufficiently well controlled modes should be capable of detecting departures from the behaviour associated with the approximate form. In order to demonstrate the utility of our general formulae we provide a specific example of a nonparaxial mode and describe some of its properties, paying particular attention to the corrections to the simple paraxial results and to the paraxial limit itself.

The general form of the field given in Eq. (3.3) depends on the function $E(\kappa)$. We have a great deal of freedom in how we choose this function but, as we have noted above, it must tend to zero sufficiently rapidly as $\kappa \rightarrow 0$, k for the energy, momentum and angular momentum per unit length to be finite. As an example (which satisfies these requirements) we examine the field for which

$$E(\kappa) = u(l, p) \exp \left(-\frac{k\kappa^2 z_R}{2(k^2 - \kappa^2)} \right) \left(\frac{\kappa^2}{k^2 - \kappa^2} \right)^{(2p+l+1)/2} \left(\frac{k^2}{k^2 - \kappa^2} \right)^{1/2}, \quad (4.1)$$

where $u(l, p)$ is a constant for the beam and z_R is a characteristic length scale which, in the paraxial limit, we will associate with the Rayleigh range. This choice has been made to provide simple analytic expressions for the energy and angular momentum per unit length and to give the Laguerre–Gaussian modes directly upon taking the paraxial limit.

Inserting the example (4.1) into our general expressions for the energy and angular momentum per unit length and performing the integrations gives the simple results

$$\mathcal{E} = \frac{\pi\epsilon_0}{2} |u(l, p)|^2 (2p+l)! (kz_R)^{-(2p+l+1)} \left(1 + \frac{2p+l+1}{2kz_R}\right) \quad (4.2)$$

$$\mathcal{J}_z = (l + \sigma_z) \frac{\mathcal{E}}{\omega} + \frac{\sigma_z \pi \epsilon_0}{2\omega} \frac{|u(l, p)|^2}{2} (2p+l+1)! (kz_R)^{-(2p+l+2)}. \quad (4.3)$$

Unfortunately, there appears to be no correspondingly simple expression for the linear momentum per unit length. The ratio of the angular momentum per unit length to the energy per unit length is

$$\frac{\mathcal{J}_z}{\mathcal{E}} = \frac{(l + \sigma_z)}{\omega} + \frac{\sigma_z}{\omega} \left(\frac{2kz_R}{2p+l+1} + 1 \right)^{-1}. \quad (4.4)$$

In the paraxial limit, z_R is very much greater than the wavelength of the light. This implies that $kz_R \gg 1$ and the paraxial result (3.11) is readily regained.

We can obtain an explicit expression for the paraxial limit of the mode with $E(\kappa)$ given by Eq. (4.1) if the exponential is sharply peaked at small values of κ . This corresponds to the real part of z_R being large which, in turn, implies a large beam waist relative to the wavelength. In practice this means dropping terms outside of the exponential that are of order κ^2/k^2 and extending the upper limit of the integral in the general expression for the field (3.3) to infinity. Note, however, that it is necessary to keep terms in the z -dependent exponential of order $\kappa^2 z/k$. The resulting expression for the paraxial approximation to the x -component of the electric field is

$$E_x = \alpha u(l, p) e^{i\ell\phi} e^{ikz} \int_0^\infty d\kappa \exp\left(-\frac{\kappa^2}{2k}(z_R + iz)\right) \left(\frac{\kappa}{k}\right)^{2p+l+1} J_1(\kappa\rho). \quad (4.5)$$

This integral may be evaluated [19] with the result

$$E_x = \frac{\alpha u(l, p) p!}{2} e^{i\ell\phi} e^{ikz} k \left(\frac{kp}{2}\right)^l \left(\frac{2}{k(z_R + iz)}\right)^{p+l+1} \exp\left(\frac{kp^2}{2(z_R + iz)}\right) L_p^l\left(\frac{kp^2}{2(z_R + iz)}\right), \quad (4.6)$$

which we recognise as the ‘elegant’ form of the Laguerre–Gaussian mode [2]. This expression for the field is indeed a solution of the paraxial wave equation (2.5) as may be verified by direct substitution.

5. Conclusion

We have developed a general nonparaxial form of a monochromatic beam based on the full Maxwell theory. Beams with as near to cylindrical symmetry as allowed by the requirement that the electric field be transverse have a simple form parametrised by a mode function $E(\kappa)$ and a polarisation. We have shown that modes of this large class have simple forms for the energy, linear momentum and total angular momentum per unit length. Appropriate choice of $E(\kappa)$ allows the choice of a form for the modes which gives the Laguerre–Gaussian modes directly upon taking the paraxial limit. This approach allows the systematic investigation of mode effects arising from either paraxial or the nonparaxial form. It has allowed the specific problem of whether the spin and orbit components of the associated angular momentum of the beam may be legitimately separated in an obvious and natural way to be addressed here. We have also shown that corrections to specific paraxial calculations may be readily made within this framework.

The modes treated in this paper do not exhaust the family of near cylindrically symmetric beams. It might be interesting to examine beams for which the form of the field given in Eq. (3.3) is generalised by allowing α and β to be functions of κ . Finally, we note that the formulation of the modes is such as to allow their ready quantisation in order to permit a detailed investigation of atom–mode interactions. Naturally, a quantum description

of the full electromagnetic field requires the construction of a complete set of modes orthogonal to that introduced here.

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Optical angular-momentum flux*

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Abstract

We introduce the angular-momentum flux as the natural description of the angular momentum carried by light. We present four main results: (i) angular-momentum flux is the flow of angular momentum across a surface and, in conjunction with the more familiar angular-momentum density, expresses the conservation of angular momentum. (ii) The angular-momentum flux for a light beam about its axis (or propagation direction) can be separated into spin and orbital parts. This separation is gauge invariant and does not rely on the paraxial approximation. (iii) Angular-momentum flux can describe the propagation of angular momentum in other geometries, but the identification of spin and orbital parts is then more problematic. We calculate the flux for a component of angular momentum that is perpendicular to the axis of a light beam and for the field associated with an electric dipole. (iv) The theory can be extended to quantum electrodynamics.

Keywords: Angular momentum, electromagnetic theory

1. Introduction

Angular momentum has played an important part in electromagnetism since the early writings of Maxwell [1]. Indeed, Maxwell's crowning achievement of identifying light as electromagnetic waves was originally derived in terms of the rotational properties of the luminiferous aether, notably its *torsion* modulus¹. Maxwell also derived mechanical properties of light in the form of radiation pressure [3]. The mechanical property of momentum was identified by Poynting, who also associated angular momentum with circular polarization [4]. This phenomenon was later demonstrated experimentally [5].

The density of electromagnetic angular momentum is defined as the cross product of the radius vector from the axis of rotation and the Poynting vector [6]. This quantity has been calculated for the light emitted by a rotating dipole [7] and by a multipole [8]. The physical significance and form of the angular-momentum flux were identified by Humblet [9], who applied it to the study of optical angular momentum in the far field. It is also worth noting that the angular momentum of light made an early appearance in quantum theory [10].

Recent interest in optical angular momentum can be traced back to the seminal paper of Allen *et al* [11]. This work showed that the Laguerre–Gaussian modes, familiar from paraxial optics [12], carry a well-defined angular momentum. Moreover, this angular momentum can be decomposed into spin and orbital parts. The research activity stimulated by this paper is reviewed in [13] and a very readable introduction to optical angular momentum is given in [14].

Attempts to extend the ideas of optical angular momentum beyond the paraxial approximation seem to present a more complicated picture [15], in which there does not appear to be a natural or simple separation into spin and orbital parts. This is certainly plausible given the polarization (or spin) dependence of tightly focused, non-paraxial, light beams [16]. We shall show, however, that calculation of the correct quantity (the optical angular-momentum flux) does lead to well-defined fluxes for the spin and orbital angular momenta.

This paper presents the theory of the optical angular-momentum flux. We work with the Maxwell theory for the free electromagnetic field. No paraxial approximation is required and the theory should be considered to be exact within Maxwellian electromagnetism. We work throughout with the SI system of units in which Maxwell's equations for the free electromagnetic field have the form

* This paper is dedicated to Les Allen, physicist, artist and friend, who first introduced me to the beautiful subtleties of optical angular momentum.

¹ A modern account of this can be found in [2].

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= 0 \\
\nabla \cdot \mathbf{B} &= 0 \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \times \mathbf{B} &= \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}.
\end{aligned} \tag{1}$$

Important quantities are reproduced in the Gaussian system of units in appendix A.

The significance of a flux for a physical quantity is most readily appreciated in terms of conservation laws. We illustrate this principle in section 2 by reference to the conservation laws for charge, and for electromagnetic energy and momentum. We derive, in section 3, the form of the electromagnetic angular-momentum flux by appealing to the conservation of angular momentum. The main part of the paper (section 4) is concerned with the angular-momentum flux for light beams and its separation into spin and orbital parts. We show, in sections 5 and 6, that the theory of optical angular momentum can also be applied to the field radiated by an electric dipole and can be extended into the quantum realm.

2. Electromagnetic conservation laws

The physical quantities energy, momentum, angular momentum and, to some extent, charge are important because they are conserved. This conservation can be expressed as a continuity equation relating a density and a flux density, or current, of the conserved quantity. The best known example is the conservation of charge as expressed by the continuity equation [6]

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0, \tag{2}$$

where ρ is the charge density and $\mathbf{j}$ is the current density or ‘charge flux density’. Integration of this equation over a volume of space and application of Gauss’ theorem leads to

$$\frac{\partial}{\partial t} \int_V \rho \, dV = - \int_S \mathbf{j} \cdot d\mathbf{S}, \tag{3}$$

which connects the change in charge in the volume V to the flow of current through its surface S .

It is instructive to express the conservation of other quantities in the same way. The conservation of vector quantities such as momentum then requires a flux in the form of a two-index object. The representation of such quantities in vector notation can be awkward and so we shall work with the index notation commonly used to describe tensors. In this notation, a Roman letter index will take any of the values 1, 2 and 3 corresponding to the x , y and z directions. Repeated indices are summed over all three values so that equation (2) becomes

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} j_i = 0. \tag{4}$$

A more complete introduction to this notation can be found in [17].

The energy density for the free electromagnetic field is

$$W = \frac{1}{2}(\epsilon_0 E^2 + \mu_0^{-1} B^2). \tag{5}$$

It follows from Maxwell’s equations (1) that the continuity equation expressing conservation of energy is

$$\frac{\partial W}{\partial t} + \frac{\partial}{\partial x_i} [\mu_0^{-1} (\mathbf{E} \times \mathbf{B})_i] = 0, \tag{6}$$

where $\mu_0^{-1} (\mathbf{E} \times \mathbf{B})_i$ is the energy flux density. We recognize that this quantity is also the momentum density, in the form of Poynting’s vector, multiplied by the square of the speed of light. The conservation of momentum can be expressed as three continuity equations, one for each Cartesian component. The conservation of the i -component of momentum, $\epsilon_0 (\mathbf{E} \times \mathbf{B})_i$, is associated with the continuity equation

$$\frac{\partial}{\partial t} [\epsilon_0 (\mathbf{E} \times \mathbf{B})_i] + \frac{\partial}{\partial x_j} T_{ij} = 0. \tag{7}$$

Here T_{ij} is the momentum flux density

$$T_{ij} = \frac{1}{2} \delta_{ij} (\epsilon_0 E^2 + \mu_0^{-1} B^2) - \epsilon_0 E_i E_j - \mu_0^{-1} B_i B_j, \tag{8}$$

where δ_{ij} is the usual Kronecker delta. It is inevitable that the conservation of a vector quantity (like the momentum) should be associated with a two-index flux, just as the conservation of a scalar quantity (like the energy) is associated with a one-index (or vector) flux. The change in the energy in a given volume is associated with the flux of energy across its surface and the integral form of (6) expresses this:

$$\frac{\partial}{\partial t} \int_V W \, dV = - \int_S \mu_0^{-1} (\mathbf{E} \times \mathbf{B})_i \, dS_i. \tag{9}$$

Similarly, the conservation of the i -component of momentum can be expressed in integral form in terms of the flux of momentum across the surface bounding the volume of interest:

$$\frac{\partial}{\partial t} \int_V \epsilon_0 (\mathbf{E} \times \mathbf{B})_i \, dV = - \int_S T_{ij} \, dS_j. \tag{10}$$

It is natural to express the conservation of angular momentum in the same way, in terms of an angular-momentum density and a flux density. We shall use the idea of angular-momentum conservation to obtain the correct flux density in the next section.

It will be helpful to be able to express cross-products without explicit use of vector notation. To this end we shall employ the permutation symbol ϵ_{ijk} [17]. This takes the value +1 if $ijk = 123, 312, 231$, the value -1 if $ijk = 321, 132, 213$ and is zero otherwise. With this notation, the i -component of the momentum density becomes $\epsilon_0 (\mathbf{E} \times \mathbf{B})_i = \epsilon_0 \epsilon_{ijk} E_j B_k$.

3. Electromagnetic angular-momentum flux

The angular-momentum density for the electromagnetic field is obtained, in analogy with mechanics, by forming the cross product of the position vector with the momentum density [6]. We can write the angular-momentum density as a vector in the form²

$$\mathbf{j} = \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}). \tag{11}$$

The total electromagnetic angular momentum associated with a volume V is obtained by integration of this quantity over the volume to give

$$\mathbf{J} = \int_V \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) \, dV. \tag{12}$$

We shall find it convenient to work with the tensor notation in which the i -component of the angular-momentum density is

$$j_i = \epsilon_0 \epsilon_{ijk} x_j \epsilon_{klm} E_l B_m = \epsilon_0 (E_i x_j B_j - B_i x_j E_j). \tag{13}$$

² This should not be confused with the $\mathbf{j}$ that was used to denote current density in section 2. The current makes no further appearances in this paper and henceforth $\mathbf{j}$ will always denote the angular-momentum density.

The angular momentum is a conserved quantity and hence we should be able to express its conservation by means of a continuity equation analogous to that for linear momentum (7). Hence, we seek an angular-momentum flux density, which we write as M_{li} , associated with the flux of the i -component of angular momentum through a surface oriented in the direction l . The conservation of the i -component of the electromagnetic angular momentum could then be expressed in the form

$$\frac{\partial}{\partial t} j_i + \frac{\partial}{\partial x_l} M_{li} = 0. \quad (14)$$

A natural quantity to propose for the angular-momentum flux density is the ‘cross product’ of the position with the momentum-flux density (8). This gives

$$M_{li} = \epsilon_{ijk} x_j \left[\frac{1}{2} \delta_{kl} (\epsilon_0 E^2 + \mu_0^{-1} B^2) - \epsilon_0 E_k E_l - \mu_0^{-1} B_k B_l \right]. \quad (15)$$

It is straightforward to show (using Maxwell’s equations (1)) that this quantity does satisfy the continuity equation (14) and therefore is the required angular-momentum flux density. It is possible to derive this quantity rather more elegantly and directly by means of the relativistic theory. A brief demonstration of this is presented in appendix B.

It is worth noting that the dimensions of the angular-momentum flux density are angular momentum per unit area per unit time. This suggests that we can interpret it in terms of a *flow* of angular momentum through a surface. This idea is supported by the integral form of the continuity equation for the i -component of angular momentum:

$$\frac{\partial}{\partial t} J_i = \frac{\partial}{\partial t} \int_V j_i dV = - \int_S M_{li} dS_l, \quad (16)$$

where S is the closed surface bounding the volume V . More generally, we shall associate the quantity $\int_S M_{li} dS_l$ with the flux of angular momentum through the surface S even if this surface is not closed.

It is often useful to separate angular momentum into internal and external or spin and orbital parts. The form of this separation for the electromagnetic field has been debated at length and remains unclear [18, 19]. There appear to be fundamental reasons why such a separation is not ‘physically observable’ [19], although it has been suggested that a physical separation is possible in which neither parts are true angular momenta [20, 21]. Nevertheless, at least for light beams within the paraxial approximation [12], the separation does seem to have a physical and experimentally demonstrable meaning [13, 22, 23]. We shall see in the following section, for the special case of the angular momentum of a light beam about its axis, that it is meaningful to identify spin and orbital parts of the angular momentum. The correct identification of these parts requires the use of the angular-momentum flux.

4. Light beams

The recent surge of interest in optical angular momentum has been driven by studies of suitably prepared laser beams [13]. Indeed the paper that kindled this interest dealt with the orbital angular momentum of Laguerre–Gaussian laser modes [11]. This paper and many subsequent ones considered the angular

momentum of a monochromatic, cylindrically symmetric beam about its own axis. For the Laguerre–Gaussian beams, the azimuthal angular dependence of the complex electric field is $\exp(il\phi)$ and the angular momentum of the beam contains a contribution that is proportional to l . Allen *et al.*, citing Marcuse [24], exploited the analogy between quantum mechanics and paraxial optics to identify this contribution with the orbital angular momentum of the light beam. This has become the accepted meaning of the expression ‘orbital angular momentum’ for a light beam. We should note, however, that the angular momentum of the beam has the same value about any axis that is parallel to the axis of the beam [25]. This means that we cannot separate our spin and orbital angular momenta on the basis of dependence on the axis (as is done with mechanical internal and external angular momenta).

Attempts to go beyond the paraxial approximation suggested that the separation of the angular momentum into spin and orbital contributions was an artefact of the paraxial approximation [15]. We shall see that this is not the case and that the problems encountered previously in working outside the paraxial approximation are removed by the correct use of the optical angular-momentum flux.

It is clear from our earlier discussion that the important quantity in discussing the angular momentum for light beams is the optical angular-momentum flux. This is the quantity of angular momentum crossing a surface in unit time. Hence the angular momentum transported by a light beam across a surface will be an optical angular-momentum flux. Earlier studies have concentrated on the angular momentum per unit power or per unit length [11, 15, 18].

4.1. Angular-momentum flux for a beam about its axis

The most studied example of optical angular momentum is that of a cylindrically symmetric (or near symmetric) light beam about its axis. Suppose that we have such a beam, the axis of which defines the z -axis. The angular-momentum flux density for this beam through a plane oriented in the z -direction is

$$\begin{aligned} M_{zz} &= M_{33} \\ &= y(\epsilon_0 E_x E_z + \mu_0^{-1} B_x B_z) - x(\epsilon_0 E_y E_z + \mu_0^{-1} B_y B_z). \end{aligned} \quad (17)$$

The total angular-momentum flux through a plane of constant z is $\iint M_{zz} dx dy$. Consider the effect of displacing the axis about which the angular-momentum flux is calculated to $x = -a$, $y = -b$. This changes the total angular-momentum flux density to

$$\begin{aligned} \iint M_{zz} dx dy &\rightarrow \iint M_{zz} dx dy + a \iint T_{yz} dx dy \\ &\quad - b \iint T_{xz} dx dy, \end{aligned} \quad (18)$$

where T_{xz} and T_{yz} are the x - and y -components of the linear-momentum flux through the surface. The total angular-momentum flux is independent of the displacements a and b if $\iint T_{xz} dx dy = 0$ and $\iint T_{yz} dx dy = 0$. This means that all beams, for which the total linear momentum carried through the surface is perpendicular to that surface, will have an angular-momentum flux density that is the same about any axis that is parallel to the axis of the beam. This generalizes

Berry's result for angular momentum to angular-momentum flux [25].

In order to proceed, it is helpful to introduce complex electric and magnetic fields and to limit our attention to monochromatic beams with angular frequency ω . We express the real electric and magnetic field components E_i and B_j in terms of complex field amplitudes $\mathcal{E}_i$ and B_j as

$$\begin{aligned} E_i &= \text{Re} [\mathcal{E}_i \exp(-i\omega t)] \\ B_j &= \text{Re} [B_j \exp(-i\omega t)]. \end{aligned} \quad (19)$$

These complex field amplitudes are related by Maxwell's equations (1) in the form

$$\begin{aligned} B_j &= \frac{1}{i\omega} \epsilon_{jkl} \frac{\partial}{\partial x_k} \mathcal{E}_l \\ \mathcal{E}_j &= -\frac{c^2}{i\omega} \epsilon_{jkl} \frac{\partial}{\partial x_k} B_l. \end{aligned} \quad (20)$$

We are now in a position to derive expressions for the spin and orbital parts of the total angular-momentum flux. The manipulations are a little involved and the reader who is primarily interested in the form and consequences of these contributions may wish to bypass these details by going straight to equations (25) and (26). The cycle-averaged angular-momentum flux density obtained from (17) is

$$\bar{M}_{zz} = \frac{1}{2} \text{Re} [y(\epsilon_0 \mathcal{E}_x \mathcal{E}_z^* + \mu_0^{-1} B_x^* B_z) - x(\epsilon_0 \mathcal{E}_y \mathcal{E}_z^* + \mu_0^{-1} B_y^* B_z)]. \quad (21)$$

It is instructive to use the Maxwell equations (20) to eliminate the z -components of the electric and magnetic fields to give

$$\begin{aligned} \bar{M}_{zz} &= \frac{1}{2} \text{Re} \left[\epsilon_0 (y \mathcal{E}_x - x \mathcal{E}_y) \frac{c^2}{i\omega} \left(\frac{\partial}{\partial x} B_y^* - \frac{\partial}{\partial y} B_x^* \right) \right. \\ &\quad \left. + \mu_0^{-1} (y B_x^* - x B_y^*) \frac{1}{i\omega} \left(\frac{\partial}{\partial x} \mathcal{E}_y - \frac{\partial}{\partial y} \mathcal{E}_x \right) \right] \\ &= \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left\{ -i \left[-y \frac{\partial}{\partial y} (\mathcal{E}_x B_x^*) - x \frac{\partial}{\partial x} (\mathcal{E}_y B_y^*) \right. \right. \\ &\quad \left. \left. + x \mathcal{E}_y \frac{\partial}{\partial y} B_x^* + y B_x^* \frac{\partial}{\partial x} \mathcal{E}_y + y \mathcal{E}_x \frac{\partial}{\partial x} B_y^* + x B_y^* \frac{\partial}{\partial y} \mathcal{E}_x \right] \right\}. \end{aligned} \quad (22)$$

The total angular-momentum flux is obtained by integrating this quantity over the whole xy -plane:

$$\begin{aligned} \mathcal{M}_{zz} &= \iint \bar{M}_{zz} dx dy = \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left\{ -i \iint dx dy \right. \\ &\quad \left. \times \left[-y \frac{\partial}{\partial y} (\mathcal{E}_x B_x^*) + x \mathcal{E}_y \frac{\partial}{\partial y} B_x^* + x B_y^* \frac{\partial}{\partial y} \mathcal{E}_x + x \leftrightarrow y \right] \right\} \\ &= \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left\{ -i \iint dx dy \left[\mathcal{E}_x B_x^* - B_x^* \frac{\partial}{\partial y} \mathcal{E}_y \right. \right. \\ &\quad \left. \left. - \mathcal{E}_x \frac{\partial}{\partial y} B_y^* + x \leftrightarrow y \right] \right\}, \end{aligned} \quad (23)$$

where we have integrated by parts to obtain the second line and $x \leftrightarrow y$ denotes an exchange of all explicit occurrences of x and y . Each of the terms in the second line of (23) arises from the corresponding term in the first line. Hence we can combine the two expressions by taking the first term from the second line together with one half of the second and third terms from

both lines and for the same combinations for the terms with x and y interchanged. This procedure gives the expression

$$\begin{aligned} \mathcal{M}_{zz} &= \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left\{ -i \iint dx dy \right. \\ &\quad \times \left[\mathcal{E}_x B_x^* + \frac{1}{2} B_x^* \left(y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y} \right) \mathcal{E}_y \right. \\ &\quad \left. + \frac{1}{2} \mathcal{E}_x \left(y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y} \right) B_y^* + x \leftrightarrow y \right] \Big\} \\ &= \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left\{ -i \iint \rho d\rho d\phi \left[(\mathcal{E}_x B_x^* + \mathcal{E}_y B_y^*) \right. \right. \\ &\quad \left. \left. + \frac{1}{2} \left(-B_x^* \frac{\partial}{\partial \phi} \mathcal{E}_y + \mathcal{E}_y \frac{\partial}{\partial \phi} B_x^* - \mathcal{E}_x \frac{\partial}{\partial \phi} B_y^* + B_y^* \frac{\partial}{\partial \phi} \mathcal{E}_x \right) \right] \right\}, \end{aligned} \quad (24)$$

where we have introduced the cylindrical polar coordinates (ρ, ϕ, z) .

It is tempting to express the total angular-momentum flux $\mathcal{M}_{zz}$ as the sum of two contributions

$$\mathcal{M}_{zz}^{\text{spin}} = \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left[-i \iint \rho d\rho d\phi (\mathcal{E}_x B_x^* + \mathcal{E}_y B_y^*) \right] \quad (25)$$

$$\begin{aligned} \mathcal{M}_{zz}^{\text{orbit}} &= \frac{\epsilon_0 c^2}{4\omega} \text{Re} \left[-i \iint \rho d\rho d\phi \right. \\ &\quad \left. \times \left(-B_x^* \frac{\partial}{\partial \phi} \mathcal{E}_y + \mathcal{E}_y \frac{\partial}{\partial \phi} B_x^* - \mathcal{E}_x \frac{\partial}{\partial \phi} B_y^* + B_y^* \frac{\partial}{\partial \phi} \mathcal{E}_x \right) \right] \end{aligned} \quad (26)$$

and to associate these with the spin and orbital angular-momentum fluxes. These quantities contain only the electric and magnetic fields and hence each is manifestly gauge invariant. This separation is also physically meaningful as we now show. We start by considering paraxial light before turning to the more general non-paraxial case. We consider the effects of a birefringent element and of an element that imparts an azimuthal phase dependence on the beam. Such devices have been employed respectively to modify the polarization (by means of waveplates [26]) and the azimuthal phase dependence of light propagating through them (see [13] and references therein). A birefringent element has a fast and a slow axis for which the refractive indices are n_f and n_s . For simplicity we shall take these axes to be the x and y axes respectively. The effect on the fields of propagation through the birefringent element is to introduce the phase shifts

$$\begin{aligned} \mathcal{E}_x &\rightarrow \mathcal{E}_x \exp(in_f k L) & B_y &\rightarrow B_y \exp(in_f k L) \\ \mathcal{E}_y &\rightarrow \mathcal{E}_y \exp(in_s k L) & B_x &\rightarrow B_x \exp(in_s k L), \end{aligned} \quad (27)$$

where k is the wavenumber for the beam and L is the propagation length. This transformation changes the spin part of the total angular-momentum flux (25) to

$$\begin{aligned} \mathcal{M}_{zz}^{\text{spin}} &\rightarrow \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left[-i \iint \rho d\rho d\phi (\mathcal{E}_x B_x^* \exp[ikL(n_f - n_s)] \right. \\ &\quad \left. + \mathcal{E}_y B_y^* \exp[-ikL(n_f - n_s)]) \right] \end{aligned} \quad (28)$$

but leaves the orbital part (26) *unchanged*. This supports our contention that $\mathcal{M}_{zz}^{\text{spin}}$ should be identified with the spin, or polarization, part of the angular-momentum flux. If, however,

we consider an element that imparts an azimuthal phase shift then our fields will all experience the same shift:

$$\begin{aligned} \mathcal{E}_x &\rightarrow \mathcal{E}_x \exp(im\phi) & \mathcal{B}_y &\rightarrow \mathcal{B}_y \exp(im\phi) \\ \mathcal{E}_y &\rightarrow \mathcal{E}_y \exp(im\phi) & \mathcal{B}_x &\rightarrow \mathcal{B}_x \exp(im\phi). \end{aligned} \quad (29)$$

This transformation changes the orbital part of the total angular-momentum flux (26) to

$$\mathcal{M}_{zz}^{\text{orbit}} \rightarrow \mathcal{M}_{zz}^{\text{orbit}} + \frac{\epsilon_0 c^2}{2\omega} \text{Re} \left[m \iint \rho \, d\rho \, d\phi (\mathcal{B}_x^* \mathcal{E}_y + \mathcal{E}_x \mathcal{B}_y^*) \right] \quad (30)$$

but leaves the spin part (25) *unchanged*. This supports our contention that $\mathcal{M}_{zz}^{\text{orbit}}$ should be identified with the orbital rather than the spin part of the angular-momentum flux. This analysis holds only within the paraxial approximation and in order to demonstrate a more general validity we need to consider the behaviour of non-paraxial beams.

We can investigate the spin and orbit parts of the total angular-momentum flux, beyond the paraxial approximation, by means of the rather general beam introduced in [15]. This describes an electromagnetic field satisfying the full Maxwell equations (1) and corresponds to a beam propagating in the $+z$ -direction with azimuthal phase dependence $\exp(il\phi)$. The components of this electric field are

$$\begin{aligned} \mathcal{E}_x &= \alpha \int_0^k d\kappa \mathcal{E}(\kappa) \exp(il\phi) \exp(i\sqrt{k^2 - \kappa^2}z) J_l(\kappa\rho) \\ \mathcal{E}_y &= \beta \int_0^k d\kappa \mathcal{E}(\kappa) \exp(il\phi) \exp(i\sqrt{k^2 - \kappa^2}z) J_l(\kappa\rho) \\ \mathcal{E}_z &= \int_0^k d\kappa \mathcal{E}(\kappa) \exp(il\phi) \exp(i\sqrt{k^2 - \kappa^2}z) \frac{\kappa}{2\sqrt{k^2 - \kappa^2}} \\ &\quad \times [(\alpha - \beta) \exp(-i\phi) J_{l-1}(\kappa\rho) \\ &\quad - (\alpha + \beta) \exp(i\phi) J_{l+1}(\kappa\rho)] \end{aligned} \quad (31)$$

where $J_l(\kappa\rho)$ is the Bessel function of order l and $k = \omega/c$. The complex numbers α and β are chosen so as to satisfy the equation $|\alpha|^2 + |\beta|^2 = 1$. We also require the magnetic field and can find this using the first of the Maxwell equations (20):

$$\begin{aligned} \mathcal{B}_x &= \int_0^k d\kappa \frac{\mathcal{E}(\kappa) \exp(i\sqrt{k^2 - \kappa^2}z)}{\omega\sqrt{k^2 - \kappa^2}} \left[-\beta \frac{2k^2 - \kappa^2}{2} \right. \\ &\quad \times \exp(il\phi) J_l(\kappa\rho) + \frac{\kappa^2}{4} (\exp(i(l-2)\phi)(\alpha - \beta) J_{l-2}(\kappa\rho) \\ &\quad \left. - \exp(i(l+2)\phi)(\alpha + \beta) J_{l+2}(\kappa\rho)) \right] \\ \mathcal{B}_y &= \int_0^k d\kappa \frac{\mathcal{E}(\kappa) \exp(i\sqrt{k^2 - \kappa^2}z)}{\omega\sqrt{k^2 - \kappa^2}} \left[\alpha \frac{2k^2 - \kappa^2}{2} \right. \\ &\quad \times \exp(il\phi) J_l(\kappa\rho) + \frac{\kappa^2}{4} (\exp(i(l-2)\phi)(-\beta - \alpha) \\ &\quad \times J_{l-2}(\kappa\rho) - \exp(i(l+2)\phi)(-\beta + \alpha) J_{l+2}(\kappa\rho)) \left. \right] \\ \mathcal{B}_z &= \int_0^k d\kappa \frac{\kappa \mathcal{E}(\kappa) \exp(i\sqrt{k^2 - \kappa^2}z)}{2\omega} [(\beta - \alpha) \\ &\quad \times \exp(i(l+1)\phi) J_{l+1}(\kappa\rho) - (i\beta + \alpha) \\ &\quad \times \exp(i(l-1)\phi) J_{l-1}(\kappa\rho)]. \end{aligned} \quad (32)$$

We should note that for these fields $\iint T_{xz} \, dx \, dy = 0$ and $\iint T_{yz} \, dx \, dy = 0$, so that the total linear momentum carried through a plane oriented in the z -direction is parallel to the beam axis. This means that the total angular-momentum flux will be the same about any axis parallel to the axis of the beam.

From the form of the electric field (31) we expect the spin angular-momentum flux to depend on α and β but not on l . The orbital angular-momentum flux, however, should depend on l but not on α and β . It is straightforward to show that this is indeed the case. If we insert our expressions for the electric and magnetic field amplitudes into our formulae for the spin and orbit parts of the angular momentum flux, (25) and (26), we find

$$\mathcal{M}_{zz}^{\text{spin}} = i(\alpha\beta^* - \alpha^*\beta) \frac{\pi\epsilon_0 c^2}{2\omega^2} \int_0^k d\kappa |\mathcal{E}(\kappa)|^2 \frac{2k^2 - \kappa^2}{\kappa\sqrt{k^2 - \kappa^2}} \quad (33)$$

$$\mathcal{M}_{zz}^{\text{orbit}} = l \frac{\pi\epsilon_0 c^2}{2\omega^2} \int_0^k d\kappa |\mathcal{E}(\kappa)|^2 \frac{2k^2 - \kappa^2}{\kappa\sqrt{k^2 - \kappa^2}}. \quad (34)$$

In obtaining these expressions we have made use of the Fourier-Bessel integral [27], which we can write in the form

$$\int_p^q d\kappa' \kappa' \Phi(\kappa') \int_0^\infty d\rho \rho J_l(\kappa'\rho) J_l(\kappa\rho) = \Phi(\kappa), \quad (35)$$

for $p < \kappa < q$. It is clear that these quantities have the expected dependences and this justifies making the separation of the angular-momentum flux into spin and orbital parts, even if we are operating outside the paraxial approximation. It is convenient to combine the α and β -dependent factor into a single term in the form

$$\sigma_z = i(\alpha\beta^* - \alpha^*\beta). \quad (36)$$

This term takes the values ± 1 for circularly polarized light and zero for linear polarization.

In their original paper, Allen *et al* [11] established, within the paraxial approximation, a proportionality between the angular momentum and energy fluxes. This led them to identify $\sigma_z \hbar$ and $l \hbar$ respectively as the spin and orbital angular momentum ‘per photon’. We are now in a position to show that this relationship also holds beyond the paraxial approximation. In order to do so, we require the energy flux through the surface. This quantity is the integral over the surface of the cycle-averaged energy flux density, $(2\mu_0)^{-1} \text{Re} (\mathcal{E}_x \mathcal{B}_y^* - \mathcal{E}_y \mathcal{B}_x^*)$. Hence the energy flux is [15]

$$\begin{aligned} \mathcal{F} &= \frac{1}{2\mu_0\omega} \text{Re} \iint \rho \, d\rho \, d\phi (\mathcal{E}_x \mathcal{B}_y^* - \mathcal{E}_y \mathcal{B}_x^*) \\ &= \frac{\pi}{2\omega\mu_0} \int_0^k d\kappa |\mathcal{E}(\kappa)|^2 \frac{2k^2 - \kappa^2}{\kappa\sqrt{k^2 - \kappa^2}}. \end{aligned} \quad (37)$$

The integral in this equation is the same as that occurring in (33) and (34). Hence we can write

$$\begin{aligned} \frac{\mathcal{M}_{zz}^{\text{spin}}}{\mathcal{F}} &= \frac{\sigma_z}{\omega} \\ \frac{\mathcal{M}_{zz}^{\text{orbit}}}{\mathcal{F}} &= \frac{l}{\omega} \\ \frac{\mathcal{M}_{zz}}{\mathcal{F}} &= \frac{l + \sigma_z}{\omega}, \end{aligned} \quad (38)$$

which justifies our assertion that $\mathcal{M}_{zz}^{\text{spin}}$ and $\mathcal{M}_{zz}^{\text{orbit}}$ are indeed the spin and orbit parts of the angular-momentum flux for a paraxial or non-paraxial beam about its axis and through a surface that is perpendicular to the direction of propagation. If we multiply both the numerator and denominator, in each of these expressions, by $\hbar$ then we recover the idea of spin and orbital angular momenta $\sigma_z \hbar$ and $l \hbar$ for each quantum of energy $\hbar \omega$. It should be stressed, however, that these expressions have been derived within classical electromagnetism. Inserting $\hbar$ may do no more than indicate a possibly helpful physical picture.

4.2. Paraxial approximation

The angular-momentum flux, within the paraxial limit, has been calculated by integrating the angular-momentum density, rather than the angular-momentum flux density, over the plane perpendicular to the direction of propagation [11]. An attempt to extend the calculation beyond the paraxial limit, again using the angular-momentum density [15], led to the erroneous conclusion that a separation of the angular-momentum flux into spin and orbit parts was not possible outside the paraxial approximation. It is now clear that the correct quantity is the angular-momentum flux density. It is interesting, however, to ask why an analysis based on the angular-momentum density gives the correct result within the paraxial approximation. To see why this is so, we consider the cycle-averaged angular-momentum flux density given in (21). The paraxial approximation applies to beams that have a small transverse wavevector. This corresponds to neglecting terms that are of second order in κ/k [15]. Inspection of the electric and magnetic field amplitudes (31) and (32) shows that the z -components of both fields are first order in κ/k and hence the paraxial approximation to $\bar{M}_{zz}$ requires only the x - and y -components of the fields to order zero in κ/k . At this order of approximation we have

$$\mathcal{E}_x \simeq cB_y, \quad \mathcal{E}_y \simeq -cB_x. \quad (39)$$

If we make these replacements in (21) then we find

$$\bar{M}_{zz} \simeq \frac{1}{2} c \epsilon_0 \text{Re} [\mathcal{E}_z^* (yB_y + xB_x) - B_z (\mathcal{E}_y^* + x\mathcal{E}_x^*)]. \quad (40)$$

This quantity is simply the product of the cycle-average of the z -component of the angular-momentum density (13) multiplied by the speed of light. Hence, within the paraxial approximation, this quantity gives the same result as that obtained by integration over the (more strictly correct) quantity given in (21).

We can also make the approximations (39) in our expressions (25) and (26) for the spin and orbit angular-momentum fluxes. This gives

$$\mathcal{M}_{zz}^{\text{spin}} \simeq \frac{\epsilon_0 c}{i2\omega} \iint \rho \, d\rho \, d\phi (\mathcal{E}_x^* \mathcal{E}_y - \mathcal{E}_y^* \mathcal{E}_x) \quad (41)$$

$$\mathcal{M}_{zz}^{\text{orbit}} \simeq \frac{\epsilon_0 c}{i2\omega} \iint \rho \, d\rho \, d\phi \left(\mathcal{E}_x^* \frac{\partial}{\partial \phi} \mathcal{E}_x + \mathcal{E}_y^* \frac{\partial}{\partial \phi} \mathcal{E}_y \right). \quad (42)$$

These are the expressions derived by van Enk and Nienhuis [18], within the paraxial approximation, for the spin and orbital angular momenta per unit length, multiplied by the speed of light.

4.3. Flux for other components of angular momentum

We are not restricted to calculating the angular-momentum flux for the component of angular momentum that is parallel to the beam axis. The fluxes of the other components of angular momentum also have physical significance. We shall illustrate this significance by calculating the flux through the xy -plane of the y -component of angular momentum for the fields (31) and (32). The y -component of the angular-momentum flux density through a surface oriented in the z -direction is (from (15))

$$M_{zy} = -z(\epsilon_0 E_x E_z + \mu_0^{-1} B_x B_z) - \frac{x}{2} [\epsilon_0 (E_x^2 + E_y^2 - E_z^2) + \mu_0^{-1} (B_x^2 + B_y^2 - B_z^2)]. \quad (43)$$

If we integrate this quantity over the entire xy -plane then we find the value zero. This is reasonable as the y -axis is a symmetry axis for the beam and we would not expect to find a net flux of angular momentum. Other axes, parallel to the y -axis, will not be symmetry axes for the beam and we can expect a non-zero angular-momentum flux about these.

Consider the flux of the y -component of angular momentum about the axis $x = -a$. For this axis the angular-momentum flux density has the form (43) with $x \rightarrow x + a$ and the total flux of the y -component of angular momentum is

$$\mathcal{M}_{zy}(a) = \iint \rho \, d\rho \, d\phi M_{zy} = \frac{a}{2} \iint \rho \, d\rho \, d\phi \times [\epsilon_0 (E_x^2 + E_y^2 - E_z^2) + \mu_0^{-1} (B_x^2 + B_y^2 - B_z^2)]. \quad (44)$$

We can interpret this expression very simply. It is the product of the distance a between the axis of rotation and the centre of the beam with the flux of the z -component of the linear momentum through the surface. Hence it expresses the usual angular momentum induced by a linear momentum off-set from the axis of rotation by the distance a . In this sense, this component of angular-momentum flux associated with the beam is an exterior angular momentum. It is of a different character to the spin and orbit angular-momentum fluxes associated with the beam about its axis. A separation of this angular-momentum flux into spin and orbital components, in the sense discussed above in section 4.1, does not appear to be meaningful.

5. Dipole field

At a fundamental level, light is emitted by atoms (or similar microscopic sources). For this reason it is of interest to ask how angular momentum is transferred from excited atoms to the emitted light. This problem has received considerable attention [8, 9, 21, 28, 29]. Here we shall calculate the rate of radiation of angular momentum by the dipole using the angular-momentum flux. Our approach is similar to that adopted by Humblet [9] except that we do not require the surface, through which the angular-momentum flux is calculated, to be far from the source. Our calculation is entirely classical, but an extension to a fully quantum calculation should be possible using the methods developed in [28].

We consider the complex electric and magnetic fields associated with a point dipole positioned at the origin with complex dipole moment $\mu_j \exp(-i\omega t)$. The complex electric and magnetic fields at position $\mathbf{r}$, generated by the dipole, are [6, 28, 30, 31]

$$\mathcal{E}_j(\mathbf{r}) = -\frac{\exp(i\omega r/c)}{4\pi\epsilon_0 r^3} \left[[\mu_j - 3(\mu_l \hat{r}_l) \hat{r}_j] \left(1 - \frac{i\omega r}{c}\right) - [\mu_j - (\mu_l \hat{r}_l) \hat{r}_j] \left(\frac{\omega r}{c}\right)^2 \right] \quad (45)$$

$$\mathcal{B}_j(\mathbf{r}) = \frac{i\omega\mu_0 \exp(i\omega r/c)}{4\pi r^2} \epsilon_{jkl} \hat{r}_k \mu_l \left(1 - \frac{i\omega r}{c}\right), \quad (46)$$

where $\hat{r}$ is the unit vector in the radial direction and the exponential factor takes account of retardation due to the finite speed of light. We can use these fields to calculate the cycle-averaged flux of angular momentum through a sphere of radius r centred on the emitting dipole. The cycle-averaged flux of the z -component of angular momentum through a (radially oriented) element of the surface of the sphere is

$$\begin{aligned} \bar{M}_{rz} &= \frac{1}{2} \text{Re} \left[\frac{1}{2} (x \hat{y} \cdot \hat{r} - y \hat{x} \cdot \hat{r}) (\epsilon_0 \mathcal{E}_k^* \mathcal{E}_k + \mu_0^{-1} \mathcal{B}_m^* \mathcal{B}_m) \right. \\ &\quad \left. - \epsilon_0 (x \mathcal{E}_y^* - y \mathcal{E}_x^*) \mathcal{E}_r - \mu_0^{-1} (x \mathcal{B}_y^* - y \mathcal{B}_x^*) \mathcal{B}_r \right] \\ &= -\frac{1}{2} \text{Re} [\epsilon_0 (x \mathcal{E}_y^* - y \mathcal{E}_x^*) \mathcal{E}_r], \end{aligned} \quad (47)$$

where the hats again denote unit vectors. Integrating this quantity over the surface of the sphere gives the total flux of the z -component of angular momentum through the surface of the sphere, which we find to be

$$\begin{aligned} \mathcal{M}_{rz} &= \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} d\phi \, \bar{M}_{rz} \\ &= \sigma_z \frac{|\mu|^2 \omega^3}{12\pi \epsilon_0 c^3}, \end{aligned} \quad (48)$$

where we associate $\sigma_z = i(\mu_x \mu_y^* - \mu_y \mu_x^*)/|\mu|^2$ with the z -component of angular momentum for the dipole. It is interesting to compare this with the flux of energy across the surface of the sphere:

$$\begin{aligned} \mathcal{F} &= \frac{1}{2\mu_0} \int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} d\phi \, \text{Re} (\epsilon_{ijk} \hat{r}_j \mathcal{E}_k \mathcal{B}_i^*) \\ &= \frac{|\mu|^2 \omega^4}{12\pi \epsilon_0 c^3}. \end{aligned} \quad (49)$$

The ratio of the total angular-momentum flux to the energy flux is

$$\frac{\mathcal{M}_{rz}}{\mathcal{F}} = \frac{\sigma_z}{\omega}, \quad (50)$$

which has the same form as we found for light beams (38). It has been suggested that half of this can be attributed to spin and half to orbital angular momentum [29]. We have not been able to find, however, any natural or convincing way to separate the angular-momentum flux into spin and orbital components.

6. Quantum theory

The classical theory presented in this paper has a natural extension to quantum electrodynamics. The electric and magnetic fields become operators in the quantum theory, satisfying the equal-time canonical commutation relation [32]

$$[\hat{E}_j(\mathbf{r}, t), \hat{B}_k(\mathbf{r}', t)] = i \frac{\hbar}{\epsilon_0} \epsilon_{jkl} \frac{\partial}{\partial x'_l} \delta(\mathbf{r} - \mathbf{r}'), \quad (51)$$

where the carets denote quantum operators. Maxwell's equations (1) remain valid but with the classical electric and magnetic fields replaced by these operators. It is helpful

to separate the field operators into positive- and negative-frequency parts corresponding to time evolutions of the form $\exp(-i\omega t)$ and $\exp(i\omega t)$ respectively:

$$\hat{E}_j^{(+)}(\mathbf{r}, t) = \int_0^\infty \hat{E}_j(\mathbf{r}, \omega) \exp(-i\omega t) d\omega \quad (52)$$

$$\hat{E}_j^{(-)}(\mathbf{r}, t) = \int_0^\infty \hat{E}_j^\dagger(\mathbf{r}, \omega) \exp(i\omega t) d\omega, \quad (53)$$

with similar expressions for the magnetic fields. These positive- and negative-frequency parts of the fields are associated with photon annihilation and creation operators respectively. The commutator (51) implies the existence of a divergent zero-point energy associated with the vacuum state of the field [30, 32]. A number of physical quantities discussed in this paper will be ill defined in the quantum theory because of this zero-point energy. These include the energy density (5), the momentum-flux density (8) and the angular-momentum flux density (15).

We can obtain well-behaved quantities by subtracting the vacuum contributions from these quantities. This corresponds to the familiar approach in which field quantities are expressed in normal order [30]. The normal-ordered form of operator products has annihilation operators to the right and creation operators to the left so that the normal-ordered square of the electric field is

$$\begin{aligned} : \hat{E}^2 : &= : (\hat{E}_j^{(-)} + \hat{E}_j^{(+)}) (\hat{E}_j^{(-)} + \hat{E}_j^{(+)}) : \\ &= 2\hat{E}_j^{(-)} \hat{E}_j^{(+)} + \hat{E}_j^{(-)} \hat{E}_j^{(-)} + \hat{E}_j^{(+)} \hat{E}_j^{(+)}. \end{aligned} \quad (54)$$

Normal-ordered products are denoted by a pair of colons so our normal-ordered angular-momentum flux density operator is

$$\begin{aligned} : \hat{M}_{li} : &= \epsilon_{ijk} x_j \left[\frac{1}{2} \delta_{kl} (\epsilon_0 : \hat{E}^2 : + \mu_0^{-1} : \hat{B}^2 :) \right. \\ &\quad \left. - \epsilon_0 : \hat{E}_k \hat{E}_l : - \mu_0^{-1} : \hat{B}_k \hat{B}_l : \right]. \end{aligned} \quad (55)$$

This normal-ordered operator satisfies the operator analogue of the conservation of angular momentum (14) with the angular-momentum density operator also expressed in normal order in the form

$$: \hat{j}_i : = \epsilon_0 : (\hat{E}_i x_j \hat{B}_j - \hat{B}_i x_j \hat{E}_j) :. \quad (56)$$

Hence these normal-ordered quantities do represent a conservation law for the normal-ordered angular momentum and the choice of normal-ordered quantities is physically meaningful.

The quantities introduced in this section allow us to extend our treatments of the angular momentum of light beams and dipole fields into the quantum regime and to discuss problems such as the transfer of angular momentum in elementary absorption and emission by atoms [28]. We shall develop these ideas further elsewhere.

7. Conclusion

In this paper, we have presented the theory of optical angular-momentum flux. This quantity can be derived from consideration of the conservation of angular momentum for the free electromagnetic field. Expression (15) for the angular-momentum flux density is exact in that it does not depend

on any paraxial (or related) approximation. We have applied this quantity to calculate the flux of a light beam about its axis (or propagation direction). In this case, we found that it is possible to separate the angular momentum into well-defined spin and orbital contributions. This separation is gauge invariant and the fluxes of spin and orbital angular momentum reduce to the known paraxial forms [18] in the appropriate limit. We have also shown that the optical angular-momentum flux can be applied to the study of other components of optical angular momentum and to the radiation associated with an electric dipole. The theory of optical angular-momentum flux follows from Maxwell's equations and these are valid as operator equations in quantum electrodynamics. It is no surprise, therefore, that the theory presented here can also be applied to the quantum theory of light.

The theory presented in this paper opens new possibilities for the study of the angular momentum of light. We conclude with a brief indication of three such possibilities: (i) the form of the linear electromagnetic momentum inside material media has long been debated (for a review see [33]). It is possible that the study of electromagnetic angular momentum may provide a clearer picture of the momentum and momentum transfer inside such media. (ii) The separation of angular-momentum flux into spin and orbital parts remains a problem for all but the simplest geometries. It is possible, however, that a study of angular-momentum flux for more complicated light sources may provide insight into this problem. In particular we envisage extending the analysis of section 5 to multipole sources [8, 34]. (iii) The quantum theory of optical angular-momentum flux remains to be explored. It is likely that recent ideas relating to the entanglement of orbital angular momentum [35, 36] can be extended to obtain more exotic states in which spin and orbital angular momenta are entangled. We intend to pursue these ideas elsewhere.

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Appendix A. Gaussian units

In this paper we have worked with the SI system of units. The Gaussian system is also widely used and so we present here the main formulae in Gaussian units. A more complete discussion of electromagnetic units can be found in [6].

Maxwell's equations for the free electromagnetic field (1):

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0 \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}\quad (\text{A.1})$$

Energy density for the free electromagnetic field (5):

$$W = \frac{1}{8\pi} (E^2 + B^2). \quad (\text{A.2})$$

The energy-flux density for the free electromagnetic field is $(c/4\pi)(\mathbf{E} \times \mathbf{B})_i$ and the density of the i -component of the momentum density is $(4\pi c)^{-1}(\mathbf{E} \times \mathbf{B})_i$. The momentum-flux density (8) becomes

$$T_{ij} = \frac{1}{4\pi} \left[\frac{1}{2} \delta_{ij} (E^2 + B^2) - E_i E_j - B_i B_j \right]. \quad (\text{A.3})$$

Angular-momentum density (11):

$$\mathbf{j} = \frac{1}{4\pi c} \mathbf{r} \times (\mathbf{E} \times \mathbf{B}). \quad (\text{A.4})$$

Angular-momentum flux density (15):

$$M_{lij} = \frac{1}{4\pi} \epsilon_{ijk} x_j \left[\frac{1}{2} \delta_{kl} (E^2 + B^2) - E_k E_l - B_k B_l \right]. \quad (\text{A.5})$$

The ratios of the total angular-momentum flux to the energy flux calculated in this paper are unaffected by the change of units.

Finally, the electric and magnetic fields associated with an electric dipole (45) and (46), become

$$\begin{aligned}\mathcal{E}_j(\mathbf{r}) &= -\frac{\exp(i\omega r/c)}{r^3} \left[[\mu_j - 3(\mu_l \hat{r}_l) \hat{r}_j] \left(1 - \frac{i\omega r}{c} \right) \right. \\ &\quad \left. - [\mu_j - (\mu_l \hat{r}_l) \hat{r}_j] \left(\frac{\omega r}{c} \right)^2 \right] \\ \mathcal{B}_j(\mathbf{r}) &= \frac{i\omega \exp(i\omega r/c)}{cr^2} \epsilon_{jkl} \hat{r}_k \mu_l \left(1 - \frac{i\omega r}{c} \right).\end{aligned}\quad (\text{A.6})$$

$$\mathcal{B}_j(\mathbf{r}) = \frac{i\omega \exp(i\omega r/c)}{cr^2} \epsilon_{jkl} \hat{r}_k \mu_l \left(1 - \frac{i\omega r}{c} \right). \quad (\text{A.7})$$

Appendix B. Relativistic construction

The most elegant and natural theory of the free electromagnetic field is that provided by the special theory of relativity. In this appendix we construct the theory of electromagnetic angular momentum using the tools of relativity. Physical quantities are represented as tensors with indices (written as Greek letters) ranging over four values 0, 1, 2 and 3 corresponding to the t -, x -, y - and z -components. As is usual in relativity, we shall retain the summation convention with repeated Greek indices implying summation over the four dimensions of spacetime. We shall continue to use Roman indices to denote the three spatial dimensions. We shall also work with a rationalized system of units in which $c = \epsilon_0 = \mu_0 = 1$. The analysis presented in this appendix follows from that given by Weinberg [37]. Similar analyses can be found in a number of texts [6, 38, 39].

The symmetric energy-momentum tensor for the electromagnetic field is [37]

$$T^{\alpha\beta} = F^{\alpha}_{\gamma} F^{\beta\gamma} - \frac{1}{4} \eta^{\alpha\beta} F_{\gamma\delta} F^{\gamma\delta}. \quad (\text{B.1})$$

This quantity is denoted $\Theta^{\alpha\beta}$ in Jackson [6]. Here $\eta^{\alpha\beta}$ is the Minkowski metric, which takes the value +1 for $\alpha = \beta = 1, 2, 3$, the value -1 for $\alpha = \beta = 0$ and is zero otherwise. The antisymmetric field tensor $F^{\alpha\beta} (= -F^{\beta\alpha})$ has the non-zero components

$$\begin{aligned}
F^{12} = B_3 = B_z & & F^{23} = B_1 = B_x & & F^{31} = B_2 = B_y \\
F^{01} = E_1 = E_x & & F^{02} = E_2 = E_y & & F^{03} = E_3 = E_z.
\end{aligned}
\tag{B.2}$$

The energy and momentum densities are

$$T^{00} = \frac{1}{2}(E^2 + B^2) \tag{B.3}$$

$$T^{i0} = T^{0i} = (\mathbf{E} \times \mathbf{B})_i = \epsilon_{ijk} E_j B_k \tag{B.4}$$

and the remaining components are

$$T^{ij} = T^{ji} = [\frac{1}{2}\delta_{ij}(E^2 + B^2) - E_i E_j - B_i B_j]. \tag{B.5}$$

The conservation of energy and momentum becomes, in the relativistic theory, the conservation of energy–momentum and the mathematical expression of this conservation is as the simple equation

$$\frac{\partial}{\partial x^\alpha} T^{\alpha\beta} = 0. \tag{B.6}$$

This leads us to associate T^{i0} with the energy–flux density (as well as with the momentum density) and T^{ij} with the flux of the j -component of the momentum through a surface oriented in the direction i .

The angular momentum is introduced in terms of the rank three tensor

$$M^{\gamma\alpha\beta} = x^\alpha T^{\beta\gamma} - x^\beta T^{\alpha\gamma}. \tag{B.7}$$

This tensor, like the energy–momentum tensor, is a conserved quantity in that it obeys the continuity equation

$$\frac{\partial}{\partial x^\gamma} M^{\gamma\alpha\beta} = 0. \tag{B.8}$$

The angular momentum tensor may be defined as

$$J^{\alpha\beta} = \int d^3x M^{0\alpha\beta}, \tag{B.9}$$

the spatial components of which are the familiar angular momenta. The i -component of the angular momentum is $J_i = \frac{1}{2}\epsilon_{ijk} J^{jk}$. In particular the z -component is

$$\begin{aligned}
J_z = J^{12} &= \int d^3x (x^1 T^{20} - x^2 T^{10}) \\
&= \int d^3x [x(E \times B)_y - y(E \times B)_x].
\end{aligned}
\tag{B.10}$$

The x - and y -components are J^{23} and J^{31} respectively. The quantity

$$j_i = \frac{1}{2}\epsilon_{ijk} M^{0jk} = \epsilon_{ijk} x_j (E \times B)_k = [\mathbf{r} \times (\mathbf{E} \times \mathbf{B})]_i \tag{B.11}$$

is the i -component of the angular-momentum density with M^{023} , M^{031} and M^{012} being its x -, y - and z -components respectively. If j_i is the i -component of the angular momentum density, then it follows from the conservation law (B.8) that $M^l_i = \frac{1}{2}\epsilon_{ijk} M^{ijk}$ is the angular-momentum flux density for a surface pointing in the direction l . Hence the total flux for the i -component of angular momentum through a surface S is $\int_S M^l_i dS^l = \frac{1}{2}\epsilon_{ijk} \int_S M^{ijk} dS_l$. The i -component of the angular-momentum flux density is

$$M^l_i = \epsilon_{ijk} x^j T^{kl} = \epsilon_{ijk} x_j [\frac{1}{2}\delta_{kl}(E^2 + B^2) - E_k E_l - B_k B_l]. \tag{B.12}$$

If we rewrite this using SI units then we recover the form (15) given earlier.

We see that the form of the angular-momentum flux density appears naturally in the relativistic formulation as a consequence of angular-momentum conservation as expressed in (B.8).

Note added in proof. The significance of optical angular-momentum flux has also been noted by Alexeyev *et al* [40], who applied the idea to study angular momentum in optical fibres. These authors also suggested a possible form for the flux of the spin part of the angular-momentum flux [41]. I am grateful to Michael Berry for bringing this work to my attention.

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